Remainder Terms of the Riemann Zeta Function

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Abstract

Starting from the Riemann–Siegel decomposition of $\zeta(s)$ given in Siegel's 1932 paper, we introduce a single change of variable, $t = I(T)$, that replaces Siegel's coupled pair "imaginary part t / summation index m" with one real index T. Under this reparameterization the formula is *identically* Riemann–Siegel. We then split Siegel's single remainder integral R into two exact pieces, R_{1ps} and R_{2ps}, and prove that $R = R_{1ps} + R_{2ps}$, so that

$$\zeta \;=\; \Sigma_1 + R_{1ps} + \Sigma_2 + R_{2ps}.$$

Our central observation is that each of these remainders is nothing more than *one additional, fractional partial summand* appended to its Dirichlet sum:

$$\zeta(s) = \sum_{n=1}^{m} n^{-s} + \hat{d}_1\,(m+1)^{-s} + \chi(s) \sum_{n=1}^{m} n^{s-1} + \hat{d}_2\,\chi(s)\,(m+1)^{s-1},$$

with $\hat{d}_1, \hat{d}_2$ real numbers (always positive on the critical line), the fractions of those two summands that are used. As a corollary, when $\sigma = \frac{1}{2}$ one has $d_1 = d_2$ (equivalently $\hat{d}_1 = \hat{d}_2$); the fact is formally verified in Lean. The decomposition was discovered through experimental mathematics using a spiral visualization of the partial sums, described later in the paper.

Contents

1 Introduction

This paper makes three claims, in increasing order of importance.

1. **A reparameterization.** We introduce a change of variable $t = I(T)$ that collapses Siegel's two roles for t (the imaginary part of the input and the index of summation) into a single real "index" T. With this substitution the Riemann–Siegel formula is unchanged in value; it is the *same* formula, viewed through a different variable, T, instead of t.

2. **A decomposition of the remainder.** Siegel's formula writes $\zeta = \Sigma_1 + \Sigma_2 + R$, where Σ_1, Σ_2 are finite Dirichlet sums and R is a remainder integral. We define two exact remainder terms, R_{1ps} and R_{2ps}, and prove

$$R \;=\; R_{1ps} + R_{2ps}, \qquad \text{hence} \qquad \zeta \;=\; \Sigma_1 + R_{1ps} + \Sigma_2 + R_{2ps}. \tag{1}$$

   Equation (1) is the main result of the paper.

3. **An interpretation.** The two new remainders are not opaque integrals: each is exactly *one more summand* of its Dirichlet sum, namely the $(m+1)$st term, scaled by a real "fractional" weight $\hat{d}_1$ or $\hat{d}_2$, always positive on the critical line and positive elsewhere outside narrow windows around the poles of §8.4 (§5 distinguishes these dimensionless *fractional amounts* from the closely related absolute lengths d_1, d_2). In this sense the remainder is simply the partial sum carried one fractional step further.

A corollary of the interpretation is that on the critical line $\sigma = \frac{1}{2}$ the two weights coincide, $d_1 = d_2$; we return to this at the end, where we also record that the fact has been formally verified in Lean. We emphasize that this critical-line symmetry, while pleasing, is *not* the focus of the paper. The decomposition (1) and its "one more partial summand" reading are.

Origin of the result. The decomposition was not found analytically. It emerged from experimental mathematics: plotting the partial sums of ζ as a Euler-like spiral in the complex plane and studying its geometry with a visualization tool we have developed (and published) over several years. The spiral itself has precedent. Erickson [7] plots the partial sums, identifies the shapes as Cornu spirals, labels their centers C_n, finds $C_0 = \zeta(s)$, and traces the symmetry he sees among the C_n back to the approximate functional equation. Nickel [14, 15] studies the same Argand diagram of the steps n^{-s} geometrically, pairing each early step with a conjugate region of linked Euler (Cornu) spirals, recovering the functional equation from that symmetry and locating zeros where two vectors point in opposite directions with equal magnitudes, which happens exactly when $\sigma = \frac{1}{2}$. Kapitonets [8] names the polyline of steps the Riemann spiral and works with its signed radius of curvature, its reverse points and its inflection points. That geometric story (the spiral, a local coordinate frame, and the "yin" and "yang" curves that actually produce R_{1ps} and R_{2ps}) is deferred to §9 and §11, so that the reader sees the clean algebraic result first and its experimental origin second.

Roadmap. §2 recalls Siegel's 1932 decomposition. §3 introduces the mapping $I(T)$ and shows the reparameterized formula is identical to Riemann–Siegel. §4 defines R_{1ps}, R_{2ps} and proves (1). §5 establishes the "one more partial summand" interpretation, and §6 recasts each partial sum together with its remainder as a product of homogeneous transformation matrices, one per link, the remainder entering as one extra fractional link. §7 discusses other remainders: §7.1 contrasts the Kuznetsov / Siegel f remainders, and §7.2 names the three splits R_{rs}, R_{ps}, and R_{ak} side by side, and §7.3 records how the remainders scale when $\{T\}$ is held fixed. §8 studies the weight d_1 as a function of the index: its handoff jumps, two exactly continuous coordinates built from it, and zeta-free closed-form approximations on and off the critical line. §9 tells the experimental/geometric story behind the result. §10 collects the $I(T)$ functions, including the bisector-frame origin of $I(T)$ and variants for L-functions. §11 develops the yin and yang curves from which the weights d_1 and d_2 were derived. §12 collects further observations, including the critical-line corollary for zeta zeros and the link-number connection. §13 sets the construction beside the prior literature on the geometry of these spirals.

2 Siegel's 1932 decomposition

We begin with the Riemann–Siegel decomposition of $\zeta(s)$ into two finite Dirichlet sums plus a remainder integral, as it appears in Siegel's 1932 paper [18] (his equation 13):

$$\zeta(s) = \sum_{n=1}^{m} n^{-s} \;+\; \frac{(2\pi)^s}{2\,\Gamma(s)\,\cos\!\left(\frac{\pi s}{2}\right)} \sum_{n=1}^{m} n^{s-1} \;+\; \frac{(2\pi)^s e^{\frac{\pi i s}{2}}}{\Gamma(s)\left(e^{2\pi i s}-1\right)} \int_{C_2} g(x)\,\mathrm{d}x, \tag{2}$$

where the path C_2 consists of two half-lines meeting at the point $\eta\left(1-\frac{\epsilon}{2}\right)$ for a small fixed $\epsilon > 0$, one half-line passing through the saddle point $\eta = +\sqrt{t/2\pi}$ and the other through $-(m+\frac{1}{2})$, and g is defined below.

It is convenient to name the three pieces. Recall the completion factor

$$\chi(s) \;:=\; 2^s \pi^{s-1} \sin\!\left(\tfrac{\pi s}{2}\right)\Gamma(1-s) \;=\; \frac{(2\pi)^s}{2\,\Gamma(s)\,\cos\!\left(\frac{\pi s}{2}\right)}, \tag{3}$$

so that, writing $s = \sigma + it$ and taking the first m terms,

$$\Sigma_1(s) = \sum_{n=1}^{m} n^{-s}, \tag{4}$$

$$\Sigma_2(s) = \frac{(2\pi)^s}{2\,\Gamma(s)\,\cos\!\left(\frac{\pi s}{2}\right)} \sum_{n=1}^{m} n^{s-1} = \chi(s)\sum_{n=1}^{m} n^{s-1}, \tag{5}$$

$$R(s) = \frac{(2\pi)^s e^{\frac{\pi i s}{2}}}{\Gamma(s)\left(e^{2\pi i s}-1\right)} \int_{C_2} g(x)\,\mathrm{d}x, \tag{6}$$

where

$$g(x) = x^{s-1}\,\frac{e^{-2\pi i m x}}{e^{2\pi i x}-1}. \tag{7}$$

With these names, (2) reads

$$\zeta(s) = \Sigma_1 + \Sigma_2 + R. \tag{8}$$

In Siegel's formulation m is the index of summation, and to minimize the remainder he takes

$$m = \left\lfloor \sqrt{\tfrac{t}{2\pi}} \right\rfloor. \tag{9}$$

Thus t plays two roles at once: it is the imaginary part of the input s, and, through (9), it also fixes the number of terms m. In this way m depends on t. The next section reverses the dependency.

3 The same formula, reparameterized: The $I(T)$ mapping

We introduce a change-in-variable mapping for t:

$$t \;=\; \frac{\pi\,\big(2\,T+1\big)}{\log\!\big(\frac{1}{T}+1\big)} \;=\; I(T), \qquad m \;=\; \lfloor T \rfloor. \tag{10}$$

Here T is a real number, while $m = \lfloor T \rfloor$ is an integer. So now t depends on m, the reverse of Siegel's (9). The single function I therefore determines both the imaginary part of the input and the integer index of summation; for this reason we call T "the index" even though it is real. Both t and m are functions of T.

For the remainder of the paper we replace Siegel's $s = \sigma + it$ by

$$s \;=\; \sigma + i\,I(T), \tag{11}$$

and we carry σ and T as two separate arguments of ζ. Nothing about the *value* of ζ changes; we have only renamed the variable that generates t and m. In this notation Siegel's decomposition (8) becomes

$$\zeta(\sigma, T) = \Sigma_1 + \Sigma_2 + R, \tag{12}$$

where, with $m = \lfloor T \rfloor$,

$$\Sigma_1(\sigma, T) = \sum_{n=1}^{m} \frac{1}{n^{\,\sigma + i\,I(T)}}, \tag{13}$$

$$\Sigma_2(\sigma, T) = \chi\big(\sigma + i\,I(T)\big) \sum_{n=1}^{m} \frac{1}{n^{\,1-\sigma-i\,I(T)}}. \tag{14}$$

Using the identity

$$\frac{(2\pi)^s e^{\frac{\pi i s}{2}}}{\Gamma(s)\big(e^{2\pi i s}-1\big)} \;=\; \frac{\chi(s)}{e^{i\pi s}-1}, \tag{15}$$

the remainder becomes

$$R(\sigma, T) = \frac{\chi\big(\sigma + i\,I(T)\big)}{e^{\,i\pi(\sigma + i\,I(T))} - 1} \int_{C_2} g(x)\,\mathrm{d}x. \tag{16}$$

The point of this section is only that (12) *is* (8): the reparameterization is an exact identity, not an approximation. Everything that follows takes place inside this reindexed but otherwise unchanged Riemann–Siegel formula.

Remark 3.1. The derivation of $I(T)$ and its approximations and inverse are developed in §10.1. The context *behind* the mapping is not needed to state or use (12).

4 Decomposing the remainder: $R = R_{1ps} + R_{2ps}$

We now split Siegel's single remainder R into two exact pieces, named R_{1ps} and R_{2ps} (the subscript "*ps*" distinguishes them from the approximate remainders of other authors). Define

$$R_{1ps}(\sigma, T) = d_1(\sigma, T)\,\cdot\,\lceil T \rceil^{-i\,I(T)}, \tag{17}$$

$$R_{2ps}(\sigma, T) = d_2(\sigma, T)\,\cdot\,\lceil T \rceil^{\,i\,I(T)} \cdot \frac{\chi(\sigma + i\,I(T))}{|\chi(\sigma + i\,I(T))|}, \tag{18}$$

with real weights

$$d_1(\sigma, T) = |R| \, \frac{\sin\bigl(\omega - \arg R + \arg \chi\bigr)}{\sin\bigl(2\omega + \arg \chi\bigr)}, \tag{19}$$

$$d_2(\sigma, T) = |R| \, \frac{\sin\bigl(\omega + \arg R\bigr)}{\sin\bigl(2\omega + \arg \chi\bigr)}, \tag{20}$$

where $\arg R = \arg R(\sigma, T)$, $\arg \chi = \arg \chi(\sigma + iI(T))$, and

$$\omega(T) \;=\; I(T)\, \log\bigl(\lfloor T+1 \rfloor\bigr). \tag{21}$$

Geometrically, ω is the *absolute* direction angle, measured from the positive real axis, of the next summand of the first Dirichlet sum: with $t = I(T)$ and $m = \lfloor T \rfloor$, the summand $n^{-s} = n^{-\sigma} e^{-it\ln n}$ rotates by $\theta_n = -t \ln \frac{n+1}{n}$ relative to its predecessor, and these relative rotations telescope,

$$\theta_1 + \theta_2 + \cdots + \theta_m \;=\; -t\ln\tfrac{2}{1} - t\ln\tfrac{3}{2} - \cdots - t\ln\tfrac{m+1}{m} \;=\; -t\ln(m+1) \;=\; -\omega, \tag{22}$$

so the $(m+1)$st summand points along $e^{-i\omega}$.

The two quotients are one intersection seen from two sides. In the frame that lays the bisector link of §9.1 along the real axis, the segment joining the two ends of the reverse bisector link, the yin and yang points, crosses that axis at distance d_1 from the joint, and the same crossing read in the reverse frame is d_2; §11.2–§11.3 derive (19)–(20) that way, which is how they were found.

Note: both d_1 and d_2 are real, and on the critical line they are always positive. Writing $\psi = \arg \chi(\sigma + iI(T))$, the definitions (17)–(18) express R as a combination of two fixed *unit* directions,

$$R \;=\; d_1\, e^{-i\omega} \;+\; d_2\, e^{\,i(\omega+\psi)}, \tag{23}$$

namely the phases of the next summand of each Dirichlet sum. Requiring $d_1, d_2 \in \mathbb{R}$ turns (23) into a 2×2 real linear system (its real and imaginary parts), whose solution by Cramer's rule is exactly the pair of sine expressions (19)–(20); hence d_1, d_2 are real. They are positive exactly when R lies in the cone spanned by the two directions $e^{-i\omega}$ and $e^{i(\omega+\psi)}$, so that both numerators carry the same sign as the common denominator $\sin(2\omega+\psi)$. On the critical line R never leaves the cone: there $d_1 = d_2$ (Corollary 8.1) and the common value stays strictly inside the link, the normalized fraction $\lceil T \rceil^{\sigma} d_1$ sweeping $\approx [0.23,\, 0.78]$ in every unit interval (§8); we verify $d_1 = d_2 > 0$ at every one of 30,562 samples of a dense grid over $1 \le T \le 30$, including magnified windows around the near-parallel instants at fractional parts $\approx \frac{1}{4}, \frac{3}{4}$, and in unit windows at $T = 100$, 300, and 1000 (`check_d1_positive_critical.py`). Off the critical line the weights are positive outside a narrow window around each pole of §8.4; inside such a window R exits the cone and one of the two weights turns negative (Remark 12.2, where $d_1 = -d_2$ exactly at a pole). Geometrically, positivity is the statement that each fractional summand points *forward* along its partial sum, a positive fraction of the next term.

Theorem 4.1. *For all σ and T,*

$$R_{1ps} + R_{2ps} = R.$$

Consequently,

$$\zeta = \Sigma_1 + R_{1ps} + \Sigma_2 + R_{2ps}.$$

Proof. Write $\phi = \arg R$, $\psi = \arg \chi$, and $\omega = \omega(T)$, so that $R = |R| e^{i\phi}$ and, for non-integer T, $\lceil T \rceil^{\,iI(T)} = e^{i\omega}$ and $\chi/|\chi| = e^{i\psi}$.

Step 1 (substitute). From (17)–(20),

$$R_{1ps} = |R|\,\frac{\sin(\omega-\phi+\psi)}{\sin(2\omega+\psi)}\,e^{-i\omega}, \qquad R_{2ps} = |R|\,\frac{\sin(\omega+\phi)}{\sin(2\omega+\psi)}\,e^{i\omega}\,e^{i\psi}.$$

Step 2 (add).

$$R_{1ps}+R_{2ps} = \frac{|R|}{\sin(2\omega+\psi)}\Big[\,e^{-i\omega}\sin(\omega-\phi+\psi)+e^{i(\omega+\psi)}\sin(\omega+\phi)\,\Big].$$

Step 3 (expand via $\sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i}$).

$$e^{-i\omega}\sin(\omega-\phi+\psi) = \frac{e^{i(\psi-\phi)}-e^{i(-2\omega+\phi-\psi)}}{2i},$$
$$e^{i(\omega+\psi)}\sin(\omega+\phi) = \frac{e^{i(2\omega+\psi+\phi)}-e^{i(\psi-\phi)}}{2i}.$$

Step 4 (add the expansions). The $e^{i(\psi-\phi)}$ terms cancel, leaving

$$\frac{1}{2i}\Big[\,e^{i(2\omega+\psi+\phi)}-e^{i(\phi-\psi-2\omega)}\,\Big].$$

Step 5 (factor $e^{i\phi}$).

$$=\frac{e^{i\phi}}{2i}\Big[\,e^{i(2\omega+\psi)}-e^{-i(2\omega+\psi)}\,\Big] = e^{i\phi}\sin(2\omega+\psi).$$

Step 6 (substitute back).

$$R_{1ps}+R_{2ps} = \frac{|R|}{\sin(2\omega+\psi)}\cdot e^{i\phi}\sin(2\omega+\psi) = |R|e^{i\phi} = R.$$

Integer T. At integer T one has $\lceil T\rceil^{\,iI(T)} = -e^{i\omega}$, and the factor $(-1)^{\lfloor T\rfloor}$ in R shifts $\phi=\arg R$ by π. That π shift flips the signs of both sine terms in d_1, d_2, while the -1 contributes a further sign; the two sign flips cancel, so $R = R_{1ps}+R_{2ps}$ holds for all T. □

A formalization of the algebra of Theorem 4.1 in Lean is given in Appendix A. It assumes no axioms of its own; the appendix records what its three hypotheses are and what is left outside its scope. Since $R = R_{1ps}+R_{2ps}$, we have in particular proved the exact formula

$$\zeta(\sigma,T) = \Sigma_1 + R_{1ps} + R_{2ps} + \Sigma_2. \tag{24}$$

5 The remainders are "one more partial summand"

Theorem 4.1 is more than an algebraic identity. Unlike Siegel's integral R (or the approximate remainders R_{ak} of Kuznetsov), which bear no resemblance to the terms of the Dirichlet sums, each of R_{1ps} and R_{2ps} is *exactly the next term* of its partial sum, scaled by a real number (positive in the sense of the note after (23)). We call such a term a *fractional summand.*

Concretely, the last term actually included in Σ_1 is the $\lfloor T\rfloor$-th term, and the very next term would be the $\lceil T\rceil$-th. The remainder R_{1ps} lies along that next term, shortened by a real factor; and similarly for R_{2ps} along the next term of Σ_2. Table 1 makes this precise.

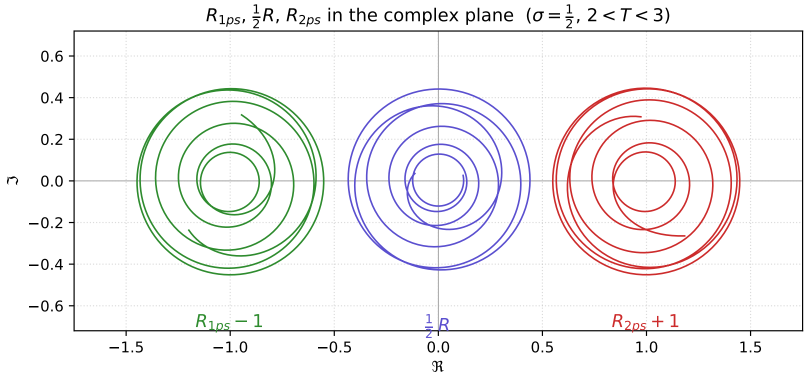

Figure 1: $R_{1ps}(\sigma, T) - 1$ (green, left), $\frac{1}{2}R(\sigma, T)$ (violet, middle), $R_{2ps}(\sigma, T) + 1$ (red, right), for $\sigma = \frac{1}{2}$ and $2 < T < 3$. The middle trajectory is the average of the left and right trajectories (the unit shifts cancel), since $R_{1ps} + R_{2ps} = R$. Generated by `fig_remainder_average.py`.

	Last summand	Next summand	Fractional amount	Partial summand
R_{1ps}	$\dfrac{1}{\lfloor T \rfloor^{\sigma + iI(T)}}$	$\dfrac{1}{\lceil T \rceil^{\sigma + iI(T)}}$	$\lceil T \rceil^{\sigma} d_1$	$d_1 \lceil T \rceil^{-iI(T)}$
R_{2ps}	$\dfrac{\chi(\sigma + iI(T))}{\lfloor T \rfloor^{1-\sigma-iI(T)}}$	$\dfrac{\chi(\sigma + iI(T))}{\lceil T \rceil^{1-\sigma-iI(T)}}$	$\dfrac{\lceil T \rceil^{1-\sigma}}{\lvert\chi(\sigma + iI(T))\rvert} d_2$	$d_2 \lceil T \rceil^{iI(T)} \dfrac{\chi(\sigma + iI(T))}{\lvert\chi(\sigma + iI(T))\rvert}$

Table 1: Each remainder is the $(m+1)$st ("next") summand of its Dirichlet sum, scaled by a real fractional amount. The final column is the resulting *partial summand* = (fractional amount) $\times$ (next summand), which simplifies to R_{1ps} and R_{2ps} respectively.

Reading (24) in this light, and writing $s = \sigma + iI(T)$ and $m = \lfloor T \rfloor$, the zeta function takes the clean "one more summand" form

$$\boxed{\zeta(s) = \sum_{n=1}^{m} n^{-s} + \hat{d}_1\,(m+1)^{-s} + \chi(s)\sum_{n=1}^{m} n^{s-1} + \hat{d}_2\,\chi(s)\,(m+1)^{s-1}} \tag{25}$$

with $\hat{d}_1, \hat{d}_2$ reals, positive on the critical line (and off it outside the pole windows of §8.4). In words: each finite Dirichlet sum, carried exactly one fractional term past its cutoff, reproduces ζ with no remainder terms.

The coefficients here are the *fractional amounts* of Table 1, not the weights d_1, d_2 of (17)–(18). The hat is what distinguishes them, and the two differ by the length of the link they sit on:

$$\hat{d}_1 = \lceil T \rceil^{\sigma}\, d_1, \qquad \hat{d}_2 = \frac{\lceil T \rceil^{\,1-\sigma}}{|\chi|}\, d_2. \tag{26}$$

The unhatted weight multiplies a *unit* vector, so $|R_{1ps}| = d_1$ is an absolute length, whereas $\hat{d}_1$ multiplies the full summand $(m+1)^{-s}$ and is therefore the dimensionless fraction of that summand which is used. At $\sigma = \frac{1}{2}$, $T = 6.18$ for instance, $d_1 = 0.0875$ while $\hat{d}_1 = \sqrt{7}\, d_1 = 0.2315$.

6 Summands as links and joints

Equation (24) presents ζ as the sum of the four terms $\Sigma_1 + R_{1ps} + \Sigma_2 + R_{2ps}$, two of which are partial sums. Read geometrically, each summand of a partial sum is one *link* of a planar chain, and each remainder R_{1ps}, R_{2ps} is one additional *fractional* link. One can think of $\hat{d}_1$ (and likewise $\hat{d}_2$) as the percentage of the way along that additional link at which the chain stops: the link taken at its full length would be the whole $(m+1)$st summand, and the chain walks the fraction $\hat{d}_1$ of it. Write $s = \sigma + iI(T)$, $t = I(T)$, $m = \lfloor T \rfloor$, and $\omega = t\ln(m+1)$, and denote two complex numbers by

$$B_1 = \Sigma_1 + R_{1ps}, \qquad B_2 = \Sigma_2 + R_{2ps}, \qquad \zeta = B_1 + B_2. \tag{27}$$

Each remainder is a single extra link appended to its partial sum, pointing in the direction of the $(m+1)$-th summand with a real length d_1 or d_2 (the fractional summands of §5):

$$R_{1ps} = d_1\,\lceil T \rceil^{-iI(T)} = d_1\, e^{-i\omega}, \qquad R_{2ps} = d_2\,\lceil T \rceil^{\,iI(T)}\,\frac{\chi}{|\chi|} = d_2\, e^{\,i(\omega + \arg\chi)}, \tag{28}$$

with d_1, d_2 as in (19)–(20). The first equality in each is the definition (17)–(18); the second uses that, for non-integer T, $\lceil T \rceil = \lfloor T+1 \rfloor$, so $\lceil T \rceil^{\pm iI(T)} = e^{\pm iI(T)\log\lceil T \rceil} = e^{\pm i\omega}$ by the definition $\omega = I(T)\log\lfloor T+1 \rfloor$, together with $\chi/|\chi| = e^{\,i\arg\chi}$.

6.1 Sum form

Splitting ζ into its partial sum plus this last fractional link,

$$B_1 = \qquad \sum_{n=1}^{m} n^{-s} \;\; + d_1\, e^{-i\omega}, \tag{29}$$

$$B_2 = \chi(s)\sum_{n=1}^{m} n^{\,s-1} + d_2\, e^{\,i(\omega+\arg\chi)}. \tag{30}$$

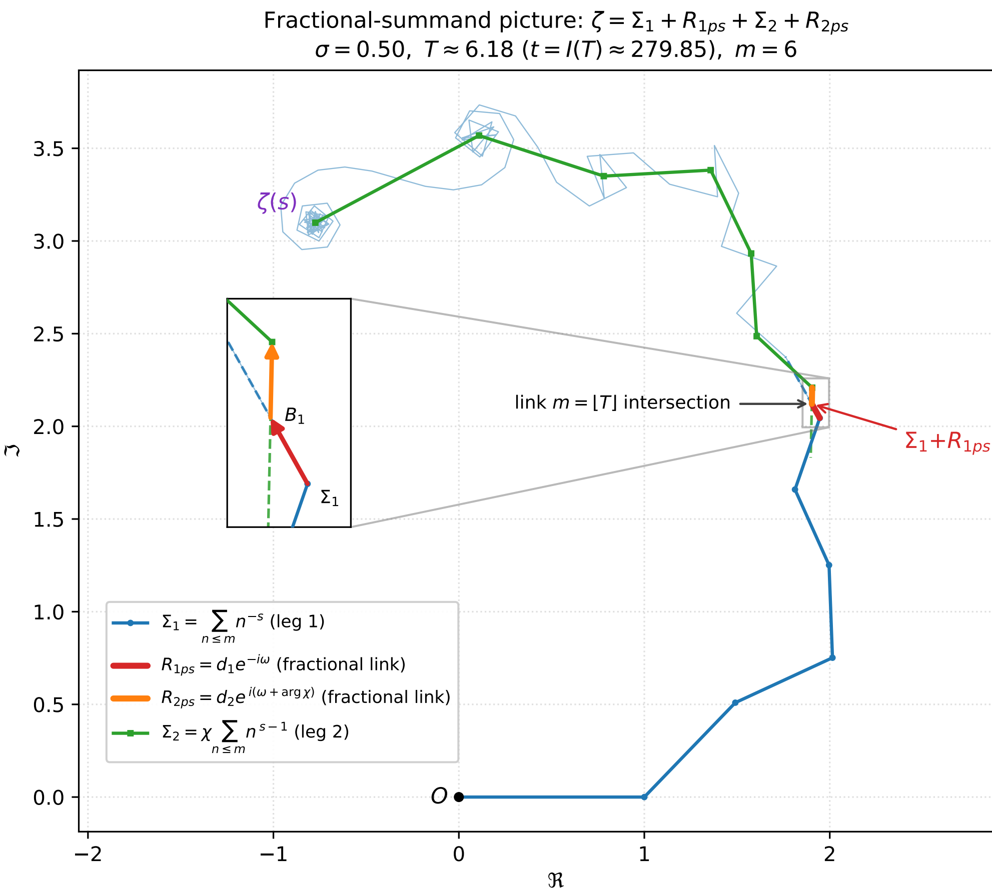

Figure 2: The fractional-summand picture: pictorial (spiral) representation of the four pieces of (25) in the complex plane for $\sigma = 0.5$ and $T \approx 6.18$ (equivalently $t = I(T) \approx 279.85$, $m = 6$). The blue path is the partial-sum spiral $\Sigma_1 = \sum_{n \le m} n^{-s}$; the green path, laid tip-to-tail from the point $\Sigma_1 + R_{1ps}$, is $\Sigma_2 = \chi \sum_{n \le m} n^{s-1}$. The short red and orange segments $R_{1ps} = d_1 e^{-i\omega}$ and $R_{2ps} = d_2 e^{i(\omega + \arg\chi)}$ lie along link m of each spiral (the faint dashed vectors show the full next summand, of which each remainder is a "fractional" part), so that the four pieces add up to the point $\zeta(s)$. Underneath, the thin faint blue path continues the Σ_1 chain past m out to link $\lfloor I(T)/\pi \rfloor = 89$, winding into its spiral near ζ. On the critical line $d_1 = d_2$, so the two remainders have equal length. The fractional links are short here ($d_1 = d_2 \approx 0.087$), so the inset zooms in on them. Generated by `fig1_spiral_summands.py`.

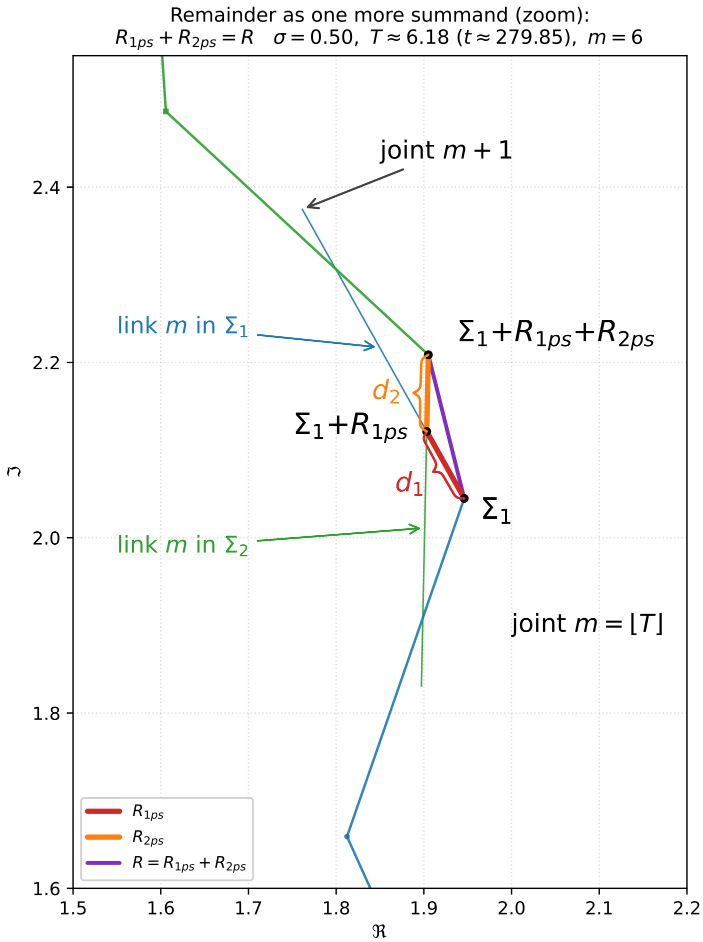

Figure 3: Zoom into the joint region of Figure 2 (imaginary part ≈ 1.5–2.6), showing the theorem $R_{1ps} + R_{2ps} = R$ as a small triangle. Starting from the joint Σ_1 (end of the sum-1 partial sum), the red link R_{1ps} reaches $\Sigma_1 + R_{1ps}$ and the orange link R_{2ps} reaches $\Sigma_1 + R_{1ps} + R_{2ps}$; the straight purple vector from Σ_1 to that last joint is Siegel's remainder $R = R_{1ps} + R_{2ps}$ (see the legend). The curly braces labelled d_1 and d_2 span the two fractional links, marking their lengths, i.e. how far each reaches from its own joint. The thin blue and green lines show the entire extent each fractional summand would have if it were not shortened to the lengths d_1 and d_2: each extends its fractional link to the full $(m+1)$st summand. The faint blue and green chains are the neighboring links of the two partial sums, arriving at and departing from the joint region. Generated by `fig2_remainder_zoom.py`.

Componentwise. Writing out real and imaginary parts,

$$\Re B_1 = \sum_{n=1}^{m} \frac{\cos(t\ln n)}{n^{\sigma}} + d_1 \cos\bigl(t\ln(m+1)\bigr), \tag{31}$$

$$\Im B_1 = -\sum_{n=1}^{m} \frac{\sin(t\ln n)}{n^{\sigma}} - d_1 \sin\bigl(t\ln(m+1)\bigr), \tag{32}$$

$$\Re B_2 = \sum_{n=1}^{m} \frac{|\chi|\cos(\arg\chi + t\ln n)}{n^{1-\sigma}} + d_2 \cos\bigl(\arg\chi + t\ln(m+1)\bigr), \tag{33}$$

$$\Im B_2 = \sum_{n=1}^{m} \frac{|\chi|\sin(\arg\chi + t\ln n)}{n^{1-\sigma}} + d_2 \sin\bigl(\arg\chi + t\ln(m+1)\bigr). \tag{34}$$

6.2 Product form

The goal of this subsection is to write each link as a homogeneous coordinate transformation matrix, and then to group the links into a single transformation matrix, so that ζ takes the form of a matrix product with four components: the two sums Σ_1, Σ_2 and the two partial-summand remainders R_{1ps}, R_{2ps}.

Each summand is a link in the plane. We number the links of each chain from zero: link n runs from joint n to joint $n+1$ (joint n being the value of the partial sum after n summands), so link n represents the $(n+1)$st summand. The m summands of a partial sum are thus links $0, \ldots, m-1$, and each remainder is the *fractional link* m. We encode one link by the 3×3 matrix with link length l and rotation θ from the previous link,[1]

$$M(\theta, l) = \begin{pmatrix} \cos\theta & -\sin\theta & l\cos\theta \\ \sin\theta & \cos\theta & l\sin\theta \\ 0 & 0 & 1 \end{pmatrix}, \tag{35}$$

where the accumulated position is the top two entries of the last column. The link matrix rotates by θ and then translates forward l along the new heading. Its inverse is again a link-shaped matrix, of a second type that translates *back* l along the current heading and then undoes the rotation:

$$M(\theta, l)^{-1} = \begin{pmatrix} \cos\theta & \sin\theta & -l \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{36}$$

Forward links build chain 1; inverses are what let chain 2 be walked backwards.

Chain 1. Collect the m summand links of Σ_1 into a single product and give the fractional link its own name:

$$P_{\Sigma_1} = \prod_{n=0}^{m-1} M\bigl(\theta_n,\ (n+1)^{-\sigma}\bigr), \qquad M_{R_{1ps}} = M\bigl(-t\ln\tfrac{m+1}{m},\ d_1\bigr), \tag{37}$$

[1] This is the planar case of the Denavit–Hartenberg convention of robot kinematics [6], in which each link of a serial chain is encoded by the 4×4 homogeneous transformation $\mathrm{Rot}_z(\theta_i)\,\mathrm{Trans}_z(d_i)\,\mathrm{Trans}_x(a_i)\,\mathrm{Rot}_x(\alpha_i)$ built from four parameters: the joint angle θ_i, the joint offset d_i, the link length a_i and the link twist α_i. A planar chain has $d_i = \alpha_i = 0$, which leaves the two parameters (θ_i, a_i) and the 3×3 matrix (35); (42) is then the forward-kinematic product of the chain. One collision of names to watch: the Denavit–Hartenberg offset d_i has nothing to do with the weights d_1, d_2 of (19)–(20).

where link n carries the two parameters θ_n, its rotation from the previous link, and $l_n = (n+1)^{-\sigma}$, its length: as in (22), $\theta_0 = 0$ and $\theta_n = -t \ln \frac{n+1}{n}$ for $n \geq 1$. Written out, with the first two factors and the last two shown, the chain that reaches B_1 is

$$P_{\Sigma_1} M_{R_{1ps}} = \underbrace{\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } 0} \underbrace{\begin{pmatrix} \cos\theta_1 & -\sin\theta_1 & \frac{\cos\theta_1}{2^\sigma} \\ \sin\theta_1 & \cos\theta_1 & \frac{\sin\theta_1}{2^\sigma} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } 1} \cdots \underbrace{\begin{pmatrix} \cos\theta_{m-1} & -\sin\theta_{m-1} & \frac{\cos\theta_{m-1}}{m^\sigma} \\ \sin\theta_{m-1} & \cos\theta_{m-1} & \frac{\sin\theta_{m-1}}{m^\sigma} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } m-1} \underbrace{\begin{pmatrix} \cos\theta_m & -\sin\theta_m & d_1\cos\theta_m \\ \sin\theta_m & \cos\theta_m & d_1 \sin\theta_m \\ 0 & 0 & 1 \end{pmatrix}}_{\text{fractional link } m}, \tag{38}$$

the leading factor being the unit step $1^{-s} = 1$ along the real axis, where $\theta_0 = 0$ and $l_0 = 1$ collapse (35) to a pure translation, and the trailing factor being $M_{R_{1ps}}$, whose rotation $\theta_m = -t \ln \frac{m+1}{m}$ continues the same pattern while its length is d_1 rather than a power of $m+1$. The product carries the frame from the origin to the bisector point $B_1 = \Sigma_1 + R_{1ps}$, whose coordinates are the top two entries of the last column, and it arrives with accumulated rotation $-t\ln(m+1) = -\omega$ by the telescoping (22).

Chain 2. The second chain is traversed tip-first, exactly as Figure 2 draws leg 2: the fractional link R_{2ps} leaves B_1, and the summand links follow in descending order. Walking a link backwards is exactly its inverse (36); regrouping each rotation with the backward translation that follows it returns everything to the forward type (35), with negated parameters and each rotation paired with the preceding link's length. Two bookkeeping factors appear at the ends of the backward walk: a leading pure translation $M(0, -d_2)$, and a trailing pure rotation $M(-\arg\chi, 0)$, the functional-equation factor χ peeled off on its own. The leading translation absorbs the zero-length joint $M(2\omega + \arg\chi + \pi,\ 0)$ that must sit between the two chains (a rotation followed by a translation is a single link), defining the backward fractional link

$$M_{R_{2ps}} \;:=\; M\big(2\omega + \arg\chi + \pi,\ 0\big)\, M(0,\ -d_2) \;=\; M\big(2\omega + \arg\chi + \pi,\ -d_2\big); \tag{39}$$

the backward summand links collect into

$$P_{\Sigma_2} \;:=\; \prod_{n=m}^{1} M\big(\theta_n,\ -|\chi|\, n^{\sigma-1}\big) \quad (n \text{ descending}, \ \theta_n = -t\ln \tfrac{n+1}{n}); \tag{40}$$

and the trailing rotation is simply dropped, since a post-multiplied zero-length joint never moves the position column, only the final orientation.

Written out as (38) was, with the first two factors and the last two shown and with $\psi = \arg\chi$ as before, the second chain is

$$\begin{aligned} M_{R_{2ps}} P_{\Sigma_2} = {} & \underbrace{\begin{pmatrix} \cos(2\omega+\psi+\pi) & -\sin(2\omega+\psi+\pi) & -d_2\cos(2\omega+\psi+\pi) \\ \sin(2\omega+\psi+\pi) & \cos(2\omega+\psi+\pi) & -d_2\sin(2\omega+\psi+\pi) \\ 0 & 0 & 1 \end{pmatrix}}_{\text{fractional link } m} \underbrace{\begin{pmatrix} \cos\theta_m & -\sin\theta_m & -\frac{|\chi|\cos\theta_m}{m^{1-\sigma}} \\ \sin\theta_m & \cos\theta_m & -\frac{|\chi|\sin\theta_m}{m^{1-\sigma}} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } m-1} \cdots \\ & \cdots \underbrace{\begin{pmatrix} \cos\theta_2 & -\sin\theta_2 & -\frac{|\chi|\cos\theta_2}{2^{1-\sigma}} \\ \sin\theta_2 & \cos\theta_2 & -\frac{|\chi|\sin\theta_2}{2^{1-\sigma}} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } 1} \underbrace{\begin{pmatrix} \cos\theta_1 & -\sin\theta_1 & -|\chi|\cos\theta_1 \\ \sin\theta_1 & \cos\theta_1 & -|\chi|\sin\theta_1 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } 0}, \end{aligned} \tag{41}$$

the leading rotation being the merged joint of (39). The links run backwards, from the fractional link m down to link 0, each brace naming the link whose length that factor carries; the rotation it carries is the θ_n of the link after it, which is the regrouping. Its last column is not B_2 but $e^{i\omega}B_2$, leg 2 read in the frame chain 1 leaves behind, whose orientation is $-\omega$; multiplying on the left by $P_{\Sigma_1} M_{R_{1ps}}$ turns that back onto world axes and adds B_1, which is (42). All three columns agree to 28 digits at $\sigma = \frac{1}{2}$, $T = 6.18$ (`check_chain2_matrix.py`).

The full 3×3 equation for ζ. With these four names the two chains combine into one serial product, in a single link type:

$$Z \;=\; P_{\Sigma_1}\, M_{R_{1ps}}\, M_{R_{2ps}}\, P_{\Sigma_2} \;=\; \begin{pmatrix} \cos\Phi & -\sin\Phi & \Re\,\zeta \\ \sin\Phi & \cos\Phi & \Im\,\zeta \\ 0 & 0 & 1 \end{pmatrix}, \qquad \Phi = \pi + \arg\chi. \tag{42}$$

The factors line up one to one with the terms of (24), repeated here for comparison:

$$\zeta(\sigma, T) = \Sigma_1 + R_{1ps} + R_{2ps} + \Sigma_2. \tag{24}$$

Figure 4 draws (42) with one arrow per factor.

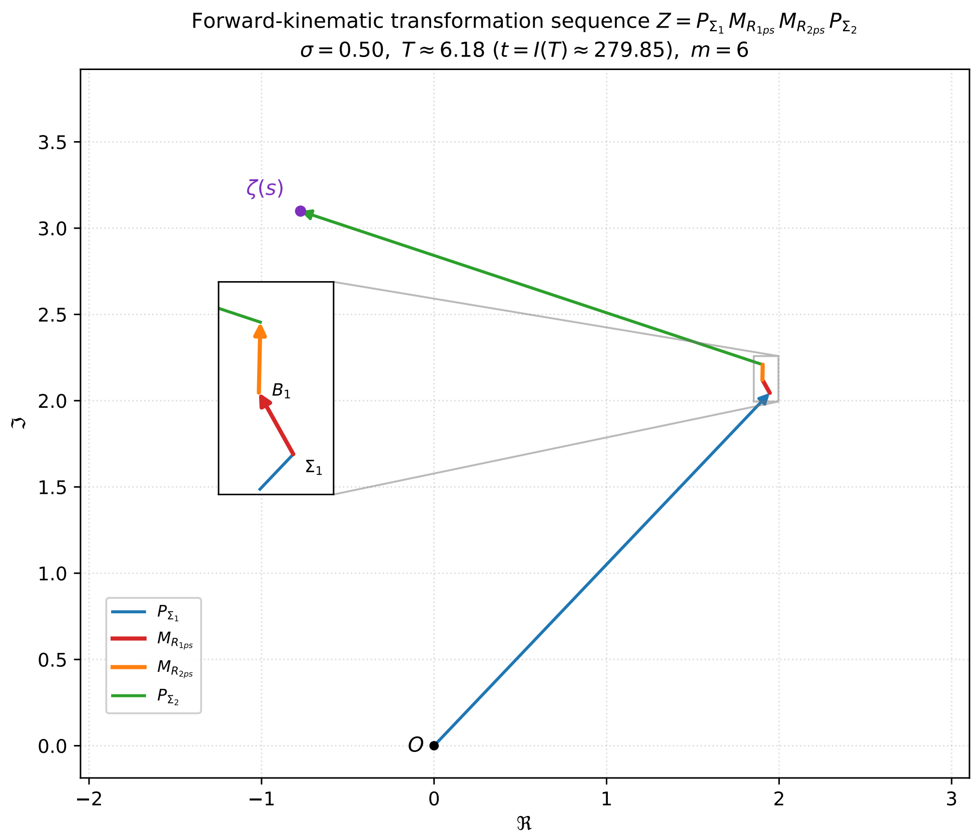

Figure 4: The serial chain of (42) at $\sigma = \frac{1}{2}$, $T = 6.18$ ($t = I(T) \approx 279.85$, $m = 6$), one arrow for the net displacement of each matrix factor: P_{Σ_1} (blue, O to Σ_1), $M_{R_{1ps}}$ (red, to B_1), $M_{R_{2ps}}$ (orange, to $B_1 + R_{2ps}$; its rotation parameter carries the merged joint $2\omega + \arg\chi + \pi$), and P_{Σ_2} (green, to ζ). The four arrows are the four terms of $\zeta = \Sigma_1 + R_{1ps} + R_{2ps} + \Sigma_2$; the two fractional links are short here ($d_1 = d_2 \approx 0.087$), so the inset zooms in on them. Colors match Figure 2. Generated by `fig_zeta_chain.py`.

Three features of (42) deserve comment. First, the rotation $2\omega + \arg\chi + \pi$ carried by $M_{R_{2ps}}$ has three jobs at once: $+\omega$ cancels chain 1's accumulated rotation $-t\ln(m+1) = -\omega$, a further $+(\omega + \arg\chi)$ pre-rotates against the counter-rotation of the backward-walked chain 2, and the final $+\pi$ rotates the backward translations to point forward again; together $\omega + (\omega + \arg\chi) + \pi =$

$2\omega + \arg\chi + \pi$. Second, the links of P_{Σ_2} carry chain 1's very rotation parameters θ_n: in this form the two chains use identical rotations and differ only in their lengths, $(n+1)^{-\sigma}$ forward versus $-|\chi|\, n^{\sigma-1}$ backward. Third, the net displacements of the four factors P_{Σ_1}, $M_{R_{1ps}}$, $M_{R_{2ps}}$, P_{Σ_2} are exactly the four terms Σ_1, $R_{1ps} = d_1 e^{-i\omega}$, $R_{2ps} = d_2 e^{i(\omega+\arg\chi)}$, Σ_2 of (24), and the final orientation $\Phi = \pi + \arg\chi$ is the direction of $-\chi/|\chi|$: the trailing rotation $M(-\arg\chi, 0)$ (the functional-equation factor χ, peeled off automatically when the base link of chain 2 is regrouped) was discarded, so χ's direction is left showing in the orientation block, and Z packages the pair $(\zeta, \arg\chi)$ in one matrix. (Appending the discarded $M(-\arg\chi, 0)$ would leave the position column untouched and make the orientation the constant π.)

7 Other remainders

7.1 Kuznetsov's remainders

Our remainders R_{1ps}, R_{2ps} are not the only way to split Siegel's R. It is worth contrasting them with the remainder of Alexey Kuznetsov [9], whose approximation to the zeta function we also use for numerical work. The essential difference is one of *kind*: Kuznetsov's remainder is an *approximation* to R, whereas $R_{1ps} + R_{2ps} = R$ is an *exact* identity (Theorem 4.1).

Writing Kuznetsov's approximate remainder as R_{ak} (the subscript "ak" for Alexey Kuznetsov, and to distinguish it from Siegel's exact, unsubscripted R), his method gives

$$R \;\approx\; R_{ak} \;=\; -\tfrac{1}{2}(-1)^{\lfloor T \rfloor}\Big(I_1(\sigma, T) + \chi\big(\sigma + iI(T)\big)\, I_2(\sigma, T)\Big), \tag{43}$$

where I_1, I_2 are the two integrals of Kuznetsov's construction, each evaluated as a short finite series in precomputed coefficients $\omega_0, \omega_1[n], \lambda[n]$ (his notation; this ω_0 is unrelated to the direction angle $\omega(T)$ of §4); in some of our calculations we use 8 such coefficients, which already gives several digits of accuracy across the range of interest. Just as we split R into two pieces, the two terms of (43) define Kuznetsov's two half-remainders

$$R_{ak} = R_{1ak} + R_{2ak}, \qquad R_{1ak} = -\tfrac{1}{2}(-1)^{\lfloor T \rfloor} I_1, \qquad R_{2ak} = -\tfrac{1}{2}(-1)^{\lfloor T \rfloor} \chi\, I_2, \tag{44}$$

so that, in parallel with (24),

$$\zeta \;\approx\; \Sigma_1 + R_{1ak} + \Sigma_2 + R_{2ak}.$$

The split (44) mirrors the functional equation: R_{1ak} pairs with the first Dirichlet sum Σ_1 and R_{2ak} carries the functional-equation factor χ alongside Σ_2. Figure 5 compares the two routes from Σ_1 to $\Sigma_1 + R$: the exact fractional links on one side and Kuznetsov's dashed pair on the other.

This pairing is exactly Siegel's: in his 1932 paper [18] Siegel writes ζ as $f(s)$ plus its reflected counterpart, and the first side is

$$f(s) \;\approx\; \Sigma_1 + R_{1ak}, \tag{45}$$

i.e. the first partial sum completed by an approximation to the first half of the remainder. The sign is $\approx$ rather than $=$ because R_{1ak} is the quantity $-\frac{1}{2}(-1)^{\lfloor T \rfloor} I_1$ of (44), with I_1 evaluated as a truncated series; read the other way round, as the *definition* $R_{1ak} := f(s) - \Sigma_1$, the relation is an identity and the approximation moves into the numerical evaluation of R_{1ak}.

The form of (45) is Siegel's own, and it is worth recording where in his paper to look for it. He defines f in his §3, equation 58, as the contour integral

$$f(s) \;=\; \int_{0 \swarrow 1} \frac{x^{-s}\, e^{\pi i x^2}}{e^{\pi i x} - e^{-\pi i x}}\, \mathrm{d}x, \tag{46}$$

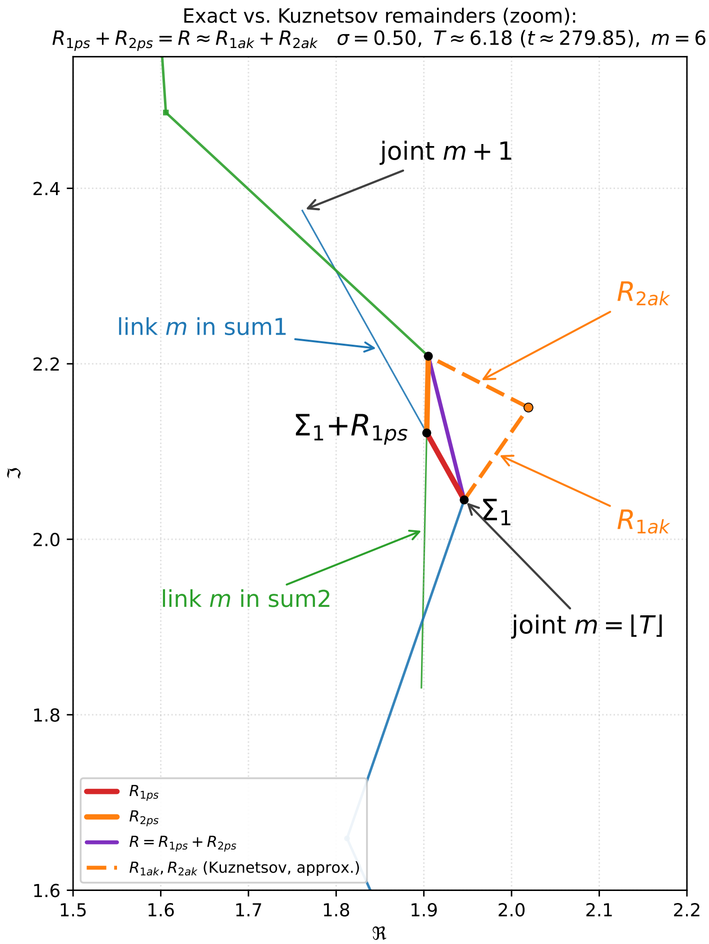

Figure 5: Exact versus Kuznetsov remainders, in the same zoom region as Figure 3. The exact fractional summands R_{1ps} (red) and R_{2ps} (orange) run from Σ_1 to $\Sigma_1 + R$ on one side of the resultant R (purple); Kuznetsov's approximate half-remainders R_{1ak}, R_{2ak} are drawn as the orange dashed pair. Since $R_{1ak} + R_{2ak} \approx R$ (here to $\sim 10^{-14}$), the dashed path lands on $\Sigma_1 + R$; unlike the exact links, the ak links do not lie along a summand direction. Computed with 8 Kuznetsov coefficients and generated by `fig4_kuznetsov_zoom.py`.

where his symbol $0 \swarrow 1$ marks a straight path crossing the real axis between 0 and 1, running from upper right to lower left parallel to the bisector of the first and third quadrants; it is the reflection in the real axis of the path $0 \nwarrow 1$ carrying the integral Φ of his §1. The Gaussian factor $e^{\pi i x^2}$ decays in both directions along that line, so (46) converges. He then develops f asymptotically in his §4, where his equation 84 reads

$$f(s) \;=\; \sum_{n=1}^{m_1} n^{-s} \;+\; O\big((|s|/2\pi e)^{-\sigma/2}\big) \qquad (\sigma \geq 0,\ t > 0), \tag{47}$$

with $m_1 = [\Re\eta_1 - \Im\eta_1]$ and $\eta_1 = \sqrt{s/2\pi i}$, which to leading order is Siegel's cutoff $\lfloor\sqrt{t/2\pi}\rfloor$ of (9). Earlier in the same section he states it in words as well: developing f asymptotically yields "as a principal term a sum of $[\sqrt{t/2\pi}\,]$ summands, that is $\sum_{n=1}^{m} n^{-s}$". So "first partial sum plus a remainder" is Siegel's own description of f, and it confirms that $\approx$ is the right sign in (45): at

$\sigma = \frac{1}{2}$ the error term of (47) is only $O(t^{-1/4})$. One caution on the cutoff: because $\sqrt{t/2\pi} \approx T + \frac{1}{2}$ (§10.1), Siegel's m_1 is one more than our $m = \lfloor T \rfloor$ whenever $\{T\} > \frac{1}{2}$. This does not disturb (45), since R_{1ak} carries the cutoff as a parameter and re-centers with it; it is Siegel's choice, not ours, that makes the remainder smallest. Equation numbers cited here are those of Siegel's paper as reprinted in his collected works, the numbering carried over by the Barkan–Sklar translation [19].[2]

Neither Siegel nor Kuznetsov names the other side, but it is $\Sigma_2 + R_{2ak}$, and adding the two recovers ζ to the same accuracy. The contrast with our result is that $\Sigma_1 + R_{1ps}$ and $\Sigma_2 + R_{2ps}$ are exact however they are read, and, moreover, each remainder there is a single fractional summand (§5).

7.2 The three remainders naming convention summary

There are a number of remainders in play, so to keep them straight we have named them with subscripts. Siegel's remainder integral is written R when unsplit; each named split below recovers that same R (exactly, or approximately for the Kuznetsov form):

1. **Riemann–Siegel half-split.**

   $$R \;=\; R_{rs} \;=\; R_{1rs} + R_{2rs}, \qquad R_{1rs} = R_{2rs} = \tfrac{1}{2}R.$$

   Here R_{1rs} is simply half the remainder integral.

2. **Partial-summand split** (§4–§5).

   $$R \;=\; R_{ps} \;=\; R_{1ps} + R_{2ps}.$$

   Here R_{1ps} is a partial next summand: the $(\lceil T \rceil)$-th Dirichlet term of Σ_1, shortened by the real weight d_1.

3. **Kuznetsov / Siegel f split** (§7.1).

   $$R \;\approx\; R_{ak} \;=\; R_{1ak} + R_{2ak}.$$

   Here R_{1ak} is Siegel's $f(s)$ minus the first partial sum: $R_{1ak} = f(s) - \Sigma_1$ (approximately, once R_{1ak} is evaluated numerically).

All three are measured from the same pair of joints, both at index $m = \lfloor T \rfloor$: each $R_{1\bullet}$ leaves joint m of the Σ_1 chain, which is the partial sum Σ_1 itself, and each $R_{2\bullet}$ arrives at joint m of the Σ_2 chain, which is the point $\Sigma_1 + R$. A split therefore chooses nothing but the point $\Sigma_1 + R_{1\bullet}$ at which the two halves meet, its own bisector point in the sense of §9.1; the anchoring joints are common to all three.

7.3 Scaling the remainders with the fractional part of T unchanged

Write $T = \lfloor T \rfloor + \{T\}$ for the integer and fractional parts of the index. Empirically, the *shape* of the remainder figure depends almost entirely on the fractional part $\{T\}$. By shape we mean the relative angles among R, R_{1ps}, R_{2ps}, R_{1ak}, and R_{2ak} once the configuration is rotated into the frame in which R lies along the positive real axis. The integer part $\lfloor T \rfloor$ mainly retunes the overall

[2] One incidental finding while checking (45) against the source: the translation's equation 83 prints the sum as $\sum n^{s-1}$, while equation 84, derived from it, has $\sum n^{-s}$. The numerics say n^{-s} is correct, so equation 83 carries a typo in the Barkan–Sklar translation.

size: with $n = \lceil T \rceil$ the lengths scale like $n^{-\sigma}$ times a slowly varying amplitude. Figure 6 makes this visible. Each panel shows only the remainders, drawn in the R-frame (rotated so R lies on the real axis and translated so the midpoint of R sits at the origin), with a common axis scale. The top row holds $\{T\} = 0.18$ at $T = 6.18$, 50.18, and 100.18; the bottom row holds $\{T\} = 0.72$ at $T = 6.72$, 50.72, and 100.72. Across a row the figure shrinks but keeps its shape; across a column (same $\lfloor T \rfloor$, different $\{T\}$) the shape changes.

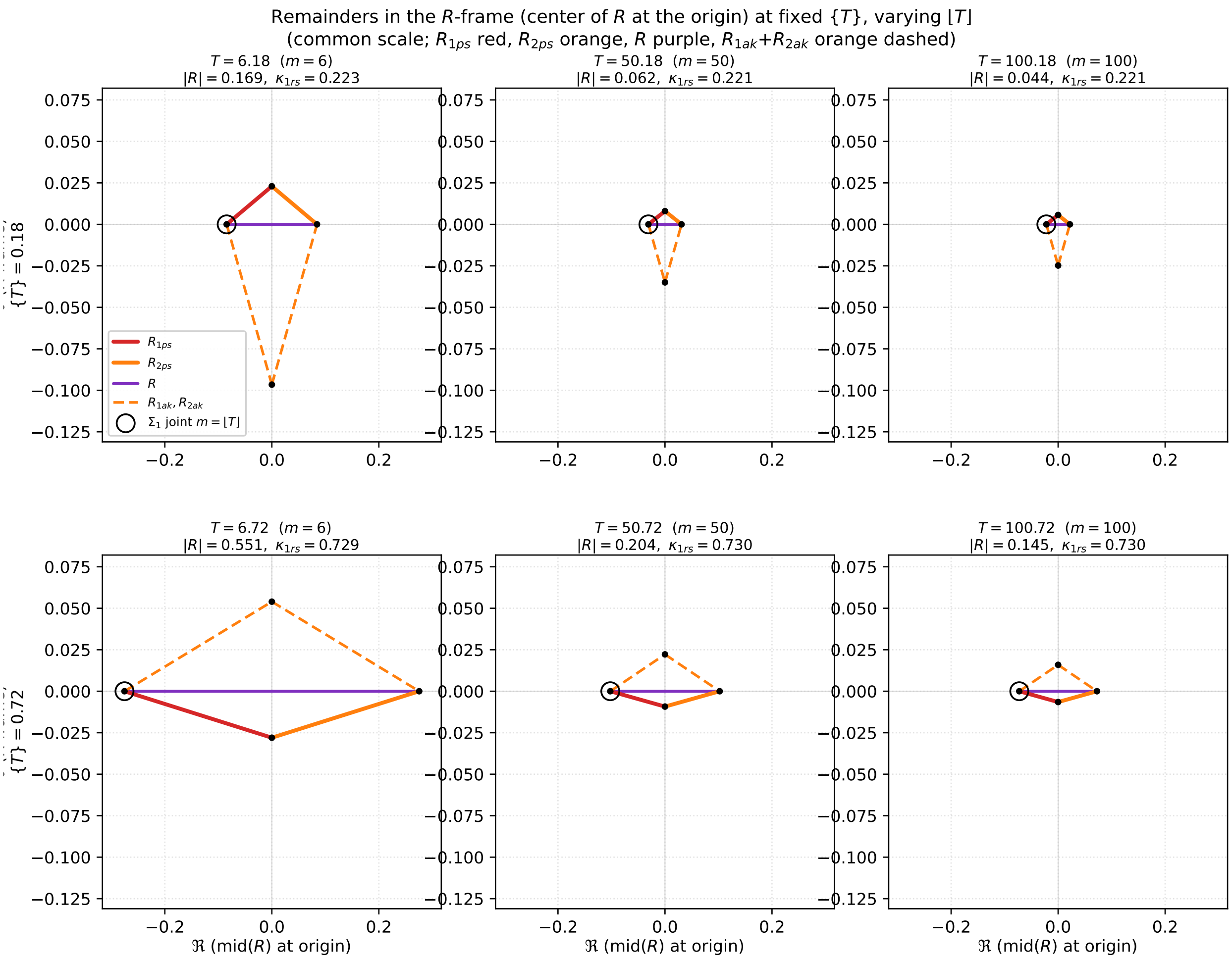

Figure 6: Remainders in the R-frame at fixed fractional part $\{T\}$, varying $\lfloor T \rfloor$. The frame is centered on the midpoint of R (so R runs from $-|R|/2$ to $+|R|/2$ on the real axis). Top row: $\{T\} = 0.18$ at $T = 6.18, 50.18, 100.18$. Bottom row: $\{T\} = 0.72$ at $T = 6.72, 50.72, 100.72$. Colors match Figure 5: R_{1ps} red, R_{2ps} orange, R purple, R_{1ak} and R_{2ak} orange dashed. The large open circle marks the left endpoint of R, which is the anchoring joint Σ_1 at $m = \lfloor T \rfloor$ that all three splits leave from. All six panels share the same axis scale. Generated by `fig_remainder_scale_panels.py`.

To make the scaling precise, factor out the leading $n^{-\sigma}$ from each half-remainder length by writing

$$|R_{1\bullet}(\sigma, T)| \;=\; \kappa_{1\bullet}(\sigma, T)\, n^{-\sigma}, \qquad n = \lceil T \rceil,$$

for the three first halves $R_{1\bullet} \in \{R_{1rs}, R_{1ps}, R_{1ak}\}$ of §7.2. Each split is then on the same footing: $R = R_{1rs} + R_{2rs} = R_{1ps} + R_{2ps} \approx R_{1ak} + R_{2ak}$, and for the rs half-split one has $R_{1rs} = R_{2rs} = \frac{1}{2}R$

so $\kappa_{2rs} = \kappa_{1rs}$. The three amplitudes admit the closed forms

$$\begin{aligned}
\kappa_{1rs} &= \tfrac{1}{2}\, n^{\sigma}\, |R| \;\approx\; \tfrac{1}{4}\, n^{\sigma}\, \big|I_1(s) + \chi(s)\, I_2(s)\big|, \\
\kappa_{1ps} &= d_1\, n^{\sigma} = 2\kappa_{1rs}\, \frac{\sin(\omega - \arg R + \psi)}{\sin(2\omega + \psi)}, \\
\kappa_{1ak} &= \tfrac{1}{2}\, n^{\sigma}\, |I_1(s)|,
\end{aligned} \tag{48}$$

with $s = \sigma + iI(T)$, $\omega = I(T)\log n$, $\psi = \arg\chi(s)$, and I_1, I_2 the Kuznetsov integrals of §7.1; the "$\approx$" in the first line holds when R is evaluated by Kuznetsov's method (43), and the second equality in the κ_{1ps} line is exact with κ_{1rs} read as $\frac{1}{2}n^{\sigma}|R|$. Consequently, at fixed σ and fixed $\{T\} = f$, the passage from $T_1 = N_1 + f$ to $T_2 = N_2 + f$ multiplies every half-remainder length by

$$\frac{|R_{1\bullet}(\sigma, T_2)|}{|R_{1\bullet}(\sigma, T_1)|} = \Big(\frac{N_1 + 1}{N_2 + 1}\Big)^{\sigma} \frac{\kappa_{1\bullet}(\sigma, T_2)}{\kappa_{1\bullet}(\sigma, T_1)}. \tag{49}$$

When the fractional part is held fixed the ratio of κ's is nearly 1, so the leading factor $\big((N_1 + 1)/(N_2 + 1)\big)^{\sigma}$ already accounts for almost all of the size change. Figure 7 checks this in the R-frame: the left panel is $T = 6.18$; the right panel is $T = 50.18$ with all remainder vectors scaled by the exact factor $\lambda = |R|_{6.18}/|R|_{50.18}$ from (49) (equivalently $\kappa_{1rs,1}/\kappa_{1rs,2}$ times the $n^{-\sigma}$ ratio). The two panels nearly coincide.

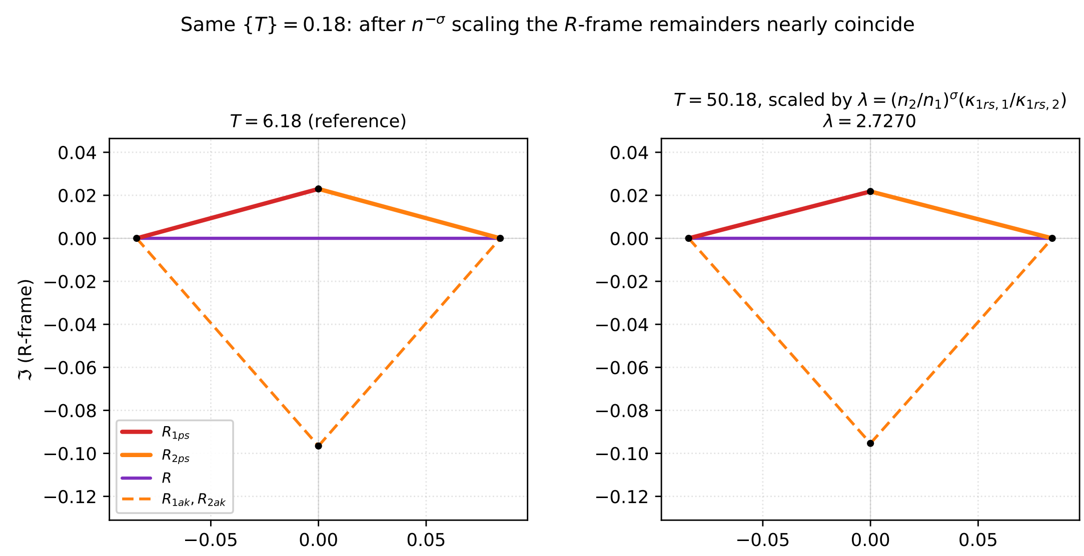

Figure 7: Same fractional part $\{T\} = 0.18$: R-frame remainders at $T = 6.18$ (left) and at $T = 50.18$ after scaling by $\lambda = \big(n_2/n_1\big)^{\sigma}(\kappa_{1rs,1}/\kappa_{1rs,2})$ (right). The scaled figure matches the reference to high accuracy ($|R_{1ps}|$ agrees to about 0.3%). Generated by `fig_remainder_scale_panels.py`.

8 The positive real function d_1 and its periodicity

The weights d_1 and d_2 of §4 are the quantitative heart of the two remainders: $R_{1ps} = d_1 e^{-i\omega}$ and $R_{2ps} = d_2 e^{i(\omega + \arg\chi)}$ are fixed in direction, so everything the remainders do is carried by the two real functions $d_1(\sigma, T)$ and $d_2(\sigma, T)$, positive on the critical line (§4). This section studies d_1 as a function of the index T: its critical-line symmetry $d_1 = d_2$, its behavior across the integer-T handoffs, and an asymptotic closed form.

8.1 The exact formulas

Corollary 8.1. *When $\sigma = \frac{1}{2}$ the two fractional weights coincide, $d_1 = d_2$, and hence $|R_{1ps}| = |R_{2ps}|$: the two fractional summands are always equal in length on the critical line.*

This follows from the definitions (19)–(20) together with the functional symmetry of χ on the critical line. Explicitly, when $\sigma = \frac{1}{2}$ the functional equation forces $|\chi| = 1$ and $\arg R = \frac{1}{2}\arg\chi$ modulo π (proved in Appendix B). Writing $\psi = \arg\chi$ and substituting $\arg R = \frac{\psi}{2}$ into (19)–(20), the two numerators collapse to the *same* value:

$$d_1 = |R|\,\frac{\sin\!\big(\omega - \frac{\psi}{2} + \psi\big)}{\sin(2\omega + \psi)} = |R|\,\frac{\sin\!\big(\omega + \frac{\psi}{2}\big)}{\sin(2\omega + \psi)}, \qquad d_2 = |R|\,\frac{\sin\!\big(\omega + \frac{\psi}{2}\big)}{\sin(2\omega + \psi)}, \tag{50}$$

so $d_1 = d_2$. Equivalently, setting $d_1 = d_2$ in (23) gives $R = d_1\big(e^{-i\omega} + e^{i(\omega+\psi)}\big) = 2d_1\cos\!\big(\omega + \frac{\psi}{2}\big)\,e^{i\psi/2}$, which indeed has argument $\frac{\psi}{2}$ (or $\frac{\psi}{2} + \pi$, where the cosine is negative), consistent with $\arg R = \frac{1}{2}\arg\chi$ modulo π. We have verified this in Lean; the proof is recorded in Appendix B. We stress that this critical-line equality is a pleasant consequence of the decomposition, not its motivation.

Since $d_1 = d_2$ on the critical line, a single curve there tells the whole story of both remainders, and it is worth drawing. Figure 8 plots $d_1(\frac{1}{2}, T)$ for $1 \le T \le 7$, and the graph is unexpectedly tame: as a function of the height t the zeta landscape is famously erratic, but viewed through the index T the weight d_1 oscillates once per unit of T, every unit interval the same shape, with only two blemishes: the amplitude decays slowly, and the curve jumps at each integer T, where $m = \lfloor T \rfloor$ increments and the fractional summands hand off to the next link. The bottom panel shows that a single normalization (by the length of the link that carries d_1) removes the decay and most of the jumps at once, leaving a curve that looks nearly *periodic* in T. That is a strong hint that d_1 is, at heart, a fixed waveform in the fractional part of T dressed up by the link geometry. The rest of this section chases that hint down, first by removing the jumps exactly, then by identifying the waveform in closed form.

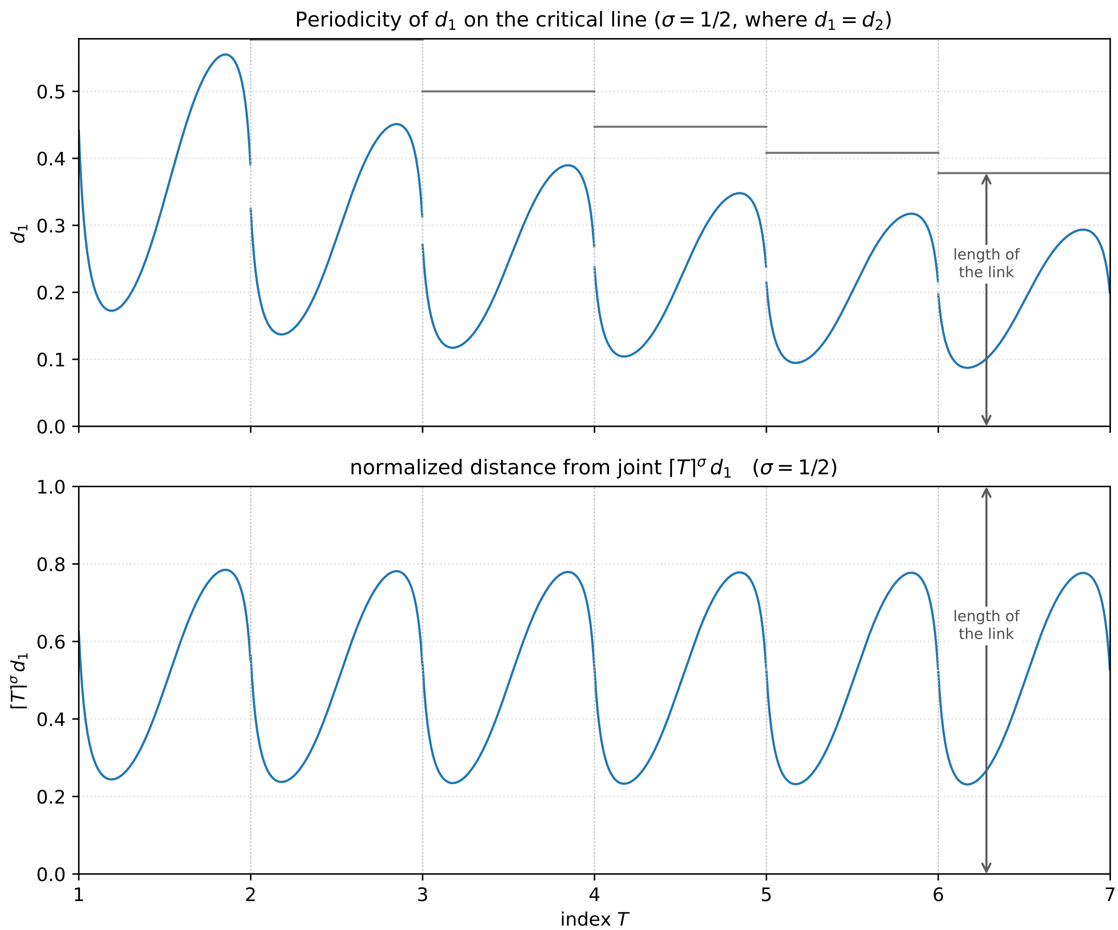

Figure 8: Top: the common fractional weight $d_1 = d_2$ on the critical line, plotted against the index T for $1 \leq T \leq 7$. The curve oscillates once per unit of T and jumps at each integer T, where $m = \lfloor T \rfloor$ increments and the fractional summands hand off to the next link; the amplitude decays slowly as T grows. The grey step is the length $\lceil T\rceil^{-\sigma}$ of the link that carries the fractional summand, which d_1 never exceeds; the panel keeps its original top, so the steps for $1 \leq T < 3$ lie at or above the frame. The double-headed arrow on $6 < T < 7$ spans that link length. Bottom: the same formula for d_1 normalized by the length $\lceil T\rceil^{-\sigma}$ of link m, which carries it: the *normalized distance from joint* $\lceil T\rceil^{\sigma} d_1$, the fraction of the way along link m at which the point $B_1 = \Sigma_1 + R_{1ps}$ sits (cf. §11). The panel is drawn on the full scale 0 to 1 of the normalized link, so 0 is joint m and 1 is joint $m+1$; the double-headed arrow, at the same T as the one above, marks that same link seen at unit length. The normalization removes both the decay and (nearly) the jumps: the fraction sweeps the same range $\approx [0.23, 0.78]$ in every unit interval, staying clear of both ends of the link. So the normalized curve converges to a periodic waveform: sampled at matched fractional parts of T, its values agree to $\approx 2 \times 10^{-3}$ by $T = 20$, and the residual discrepancy halves with each doubling of T, the rate set by the link-length mismatch $(1 + 1/\lfloor T \rfloor)^{\sigma} \to 1$ of (51) below. Computed exactly (mpmath) via the Cramer solution of (23). Generated by `fig_d1_critical.py`.

Removing the jumps exactly. The normalization in the bottom panel of Figure 8 nearly removes the jumps, and the residue can be traced to a single exact relation. At a handoff $T = n$ the outgoing link ($m = n - 1$) and the incoming link ($m = n$) fold back onto one another while the point B_1 moves continuously in the plane, which forces

$$d_1^+ \;=\; n^{-\sigma} \;-\; d_1^-, \tag{51}$$

where d_1^- and d_1^+ are the limits from the left and from the right. In terms of the plotted fraction $f = \lceil T\rceil^\sigma d_1$ this reads $f^+ = (1+\frac{1}{n})^\sigma(1-f^-)$: an orientation flip $f \mapsto 1-f$ (after the fold the fraction is measured from the other end of the link) composed with a unit mismatch (the new link is shorter than the old by the factor $(1+\frac{1}{n})^{-\sigma}$). The mismatch factor tends to 1, which is why the jumps in the bottom panel shrink as T grows; the replacement $f \mapsto 1-f$ also contributes less and less, because the curve happens to cross the integers near $f \approx \frac{1}{2}$, where $1-f \approx f$.

Because (51) is exact, the unit intervals can be glued into a coordinate with no jumps at all: measure the position of B_1 along the folded chain in one *fixed* unit (the first link, of length 1), flipping the orientation and absorbing the constant $n^{-\sigma}$ at each handoff of the bisector point from one link to the next (see §10.1 for more on these handoffs). Unrolling the recursion gives

$$h(T) \;=\; \sum_{k=2}^{\lfloor T\rfloor} (-1)^k\, k^{-\sigma} \;+\; (-1)^{\lfloor T\rfloor+1}\, d_1(T), \tag{52}$$

and at every integer the term entering the sum cancels the jump of d_1 identically, so h is continuous everywhere. The price is that h *flattens*: in a fixed unit the links themselves shrink, so the oscillating part $d_1 = O(T^{-\sigma})$ decays, while the accumulated constants are the partial sums of the convergent alternating series $2^{-\sigma} - 3^{-\sigma} + 4^{-\sigma} - \cdots = 1-\eta(\sigma)$, with η the Dirichlet eta function. Hence $h(T) \to 1-\eta(\sigma)$, which at $\sigma = \frac{1}{2}$ is ≈ 0.3951.

Continuity and a non-decaying amplitude can be had at once by re-zooming *smoothly*, with T^σ, instead of with the step function $\lceil T\rceil^\sigma$ (whose abrupt changes at the integers are exactly what created the residual jumps):

$$p(T) \;=\; T^\sigma\Big(h(T) - \big(1-\eta(\sigma)\big)\Big). \tag{53}$$

As a product of continuous functions p is exactly continuous, and since $h(T)-(1-\eta(\sigma)) = O(T^{-\sigma})$ (the decaying d_1 plus the tail of the alternating series), the factor T^σ restores a bounded oscillation. Figure 9 compares the two coordinates: both are continuous, with only kinks (no jumps) at the integers, while h contracts toward its limit $1-\eta(\frac{1}{2})$ and p keeps a steady amplitude. The red curve is computed from the simplified local form (58), derived below, which agrees with (53) to machine precision.

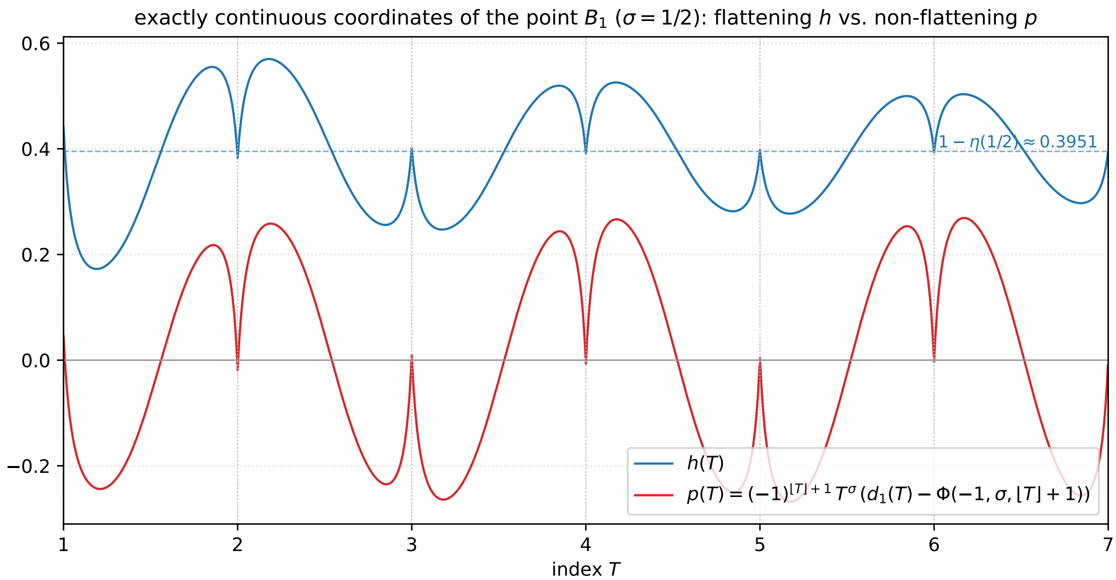

Figure 9: The two exactly continuous coordinates of the point B_1 on the critical line, $1 \le T \le 7$, in a single panel for comparison. Blue: $h(T)$ of (52), the position along the folded chain in a fixed unit; it has no jumps at the integers, only kinks where the links fold, and its oscillation contracts toward the limit $1 - \eta(\frac{1}{2}) \approx 0.3951$ (dashed line): the swing shrinks from ≈ 0.40 on the first interval to ≈ 0.24 by $T = 7$. Red: $p(T)$, the same coordinate re-zoomed smoothly by T^σ, computed from the simplified local form (58); it is equally continuous but its amplitude holds steady (≈ 0.53 peak to peak) instead of flattening. Continuity was verified numerically at each integer (mismatch $\sim 10^{-7}$ at offsets of 10^{-6}). Generated by `fig_h_p_continuous.py`.

A pinned waveform. Each arch of p in Figure 9 is nearly the vertical flip of its neighbours, but the arches do not begin and end at zero, so flipping alternate intervals would not make them line up. Taking one-sided limits at an integer n (where $\lfloor T \rfloor = n - 1$ on the left) gives the endpoint value exactly:

$$p(n) = (-1)^n n^\sigma \big(d_1(n^-) - \Phi(-1,\sigma,n)\big) = (-1)^{n+1}\varepsilon_n, \qquad \varepsilon_n := n^\sigma \big(\Phi(-1,\sigma,n) - d_1(n^-)\big) > 0, \tag{54}$$

so ε_n is the shortfall of the fold fraction from the alternating-tail center

$$n^\sigma \Phi(-1,\sigma,n) = \tfrac{1}{2} + \frac{\sigma}{4n} - \frac{\sigma(\sigma+1)(\sigma+2)}{48\,n^3} + O(n^{-5}) = \tfrac{1}{2} + \frac{1}{8n} - \frac{5}{128\,n^3} + O(n^{-5}) \quad \text{at } \sigma = \tfrac{1}{2}, \tag{55}$$

by Boole summation, $\sum_{j\ge 0}(-1)^j f(n+j) = \frac{1}{2}f(n) - \frac{1}{4}f'(n) + \frac{1}{48}f'''(n) - \cdots$, whose even derivatives drop out, so there is no n^{-2} term. On the critical line the measured fold fractions $n^\sigma d_1(n^-)$ are 0.54073, 0.53086, 0.52479 at $n = 2, 3, 4$ and 0.501237 at $n = 100$, giving $\varepsilon_n = 1.80 \times 10^{-2}, 9.55 \times 10^{-3}, 5.90 \times 10^{-3}$ and 1.26×10^{-5}, that is $\varepsilon_n \approx 0.127\, n^{-2}$ (fitted exponent -1.99 by $n = 100$): the ends approach zero on the order of T^{-2}, without ever reaching it at finite T. There is no zeta-free closed form for ε_n, because the fold relation (51) cancels identically for *any* value of $d_1(n^-)$ and so leaves the fold fraction free; the correction must use the endpoint values themselves.

Flipping alternate intervals, on the other hand, is free: it simply deletes the alternating sign from (58), since with $m = \lfloor T \rfloor$

$$P(T) \;:=\; (-1)^m p(T) \;=\; T^\sigma\big(\Phi(-1,\sigma,m+1) - d_1(T)\big), \tag{56}$$

whose two ends on $[m, m+1]$ are exactly $-\varepsilon_m$ and $+\varepsilon_{m+1}$ by (54), so P still jumps by $2\varepsilon_n$ at each integer. Subtracting the chord between those two values pins both ends: with $x = T - m$,

$$\mathcal{W}(T) \;=\; T^{\sigma}\big(\Phi(-1,\sigma,m+1) - d_1(T)\big) \;+\; (1-x)\,\varepsilon_m \;-\; x\,\varepsilon_{m+1}, \tag{57}$$

a single formula covering the intervals $[1,2], [3,4], \ldots$ and $[2,3], [4,5], \ldots$ with the second family already flipped. Figure 10 draws it: $\mathcal{W}$ vanishes at every integer, is continuous on $[1,\infty)$ with no sign alternation left, and converges like $1/T$ to the tangent waveform $\mathcal{W}_\infty(q) = \frac{1}{2}\tan(2\pi q)\tan\big(2\pi(q-\frac{1}{4})(q-\frac{3}{4})\big)$, $q = \{T\}$, which is the $T \to \infty$ form of the Riemann–Siegel first term (59) below and does vanish at $q = 0, 1$ exactly. Continuity is had exactly; C^1 only in the limit. The one-sided slopes at $T = n$ are 7.515 and 6.920 at $n = 2$ and 7.482 and 7.309 at $n = 8$, both approaching $\mathcal{W}_\infty'(0) = \mathcal{W}_\infty'(1) = \pi\tan\frac{3\pi}{8} = 7.5845$, so a corner of size $O(1/T)$ survives at each integer. Replacing the chord in (57) by the cubic Hermite that also matches the end slopes removes the corner exactly, at the cost of inflating each arch by about 0.03; and since $\mathcal{W}_\infty''(0) = -\mathcal{W}_\infty''(1)$, no pinning can do better than C^1 in any case.

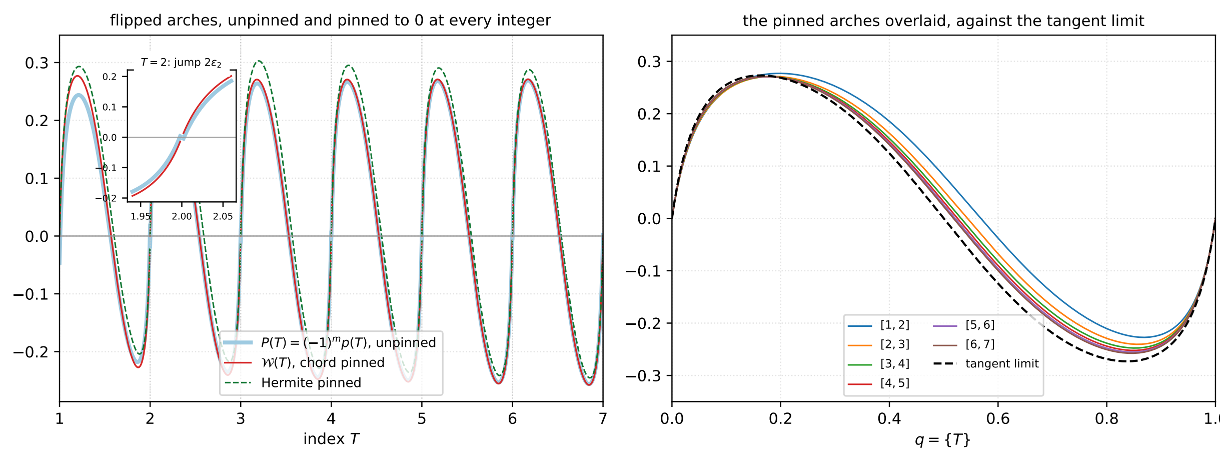

Figure 10: The flipped arches of p on the critical line, $1 \le T \le 7$. *Left:* the unpinned flip P of (56) (pale blue), which jumps by $2\varepsilon_n$ at each integer (inset, at $T = 2$); the chord-pinned $\mathcal{W}$ of (57) (red), which is zero there exactly; and the Hermite-pinned variant (dashed green), which also matches the end slopes but inflates the arch. *Right:* the six pinned arches drawn against the fractional part $q = \{T\}$, with the tangent limit $\mathcal{W}_\infty$ dashed; the deviation falls from 0.087 on $[1,2]$ to 0.025 on $[6,7]$. Generated by `fig_pinned_waveform.py`.

8.2 Approximation of d_1 when $\sigma = \frac{1}{2}$

Distributing T^σ and substituting (52) simplifies p considerably: the partial sum inside h cancels the head of the series $1 - \eta(\sigma) = \sum_{k=2}^{\infty}(-1)^k k^{-\sigma}$ term by term, leaving only its tail, and the common sign $(-1)^{\lfloor T\rfloor+1}$ factors out:

$$p(T) \;=\; (-1)^{\lfloor T\rfloor+1}\, T^{\sigma}\Big[\, d_1(T) \;-\; \Phi\big(-1,\sigma,\lfloor T\rfloor+1\big)\Big], \qquad \Phi(-1,\sigma,a) = \sum_{j=0}^{\infty}\frac{(-1)^j}{(a+j)^{\sigma}}, \tag{58}$$

with Φ the Lerch transcendent. In this form all the accumulated history of (52) has disappeared and p is *local*: it is just the deviation of d_1 from a reference value, the alternating sum of all the remaining link lengths $(\lfloor T\rfloor + 1)^{-\sigma} - (\lfloor T\rfloor + 2)^{-\sigma} + \cdots$. An alternating series is roughly half its first term, $\Phi(-1, \sigma, \lfloor T\rfloor + 1) \approx \frac{1}{2}(\lfloor T\rfloor + 1)^{-\sigma}$, which is half the length of link m, so asymptotically

$$p(T) \;\approx\; (-1)^{\lfloor T\rfloor+1}\Big(\lceil T\rceil^{\sigma} d_1 - \tfrac{1}{2}\Big),$$

which is the normalized fraction of Figure 8, recentered at $\frac{1}{2}$ and given alternating orientation. This is also why that fraction crosses the integers near $\frac{1}{2}$: continuity of p pins the fraction, at each integer, near the alternating-tail center, which is close to $\frac{1}{2}$.

On the critical line the substitution can be pushed all the way: d_1 itself acquires a zeta-free closed form. When $\sigma = \frac{1}{2}$ we have $\arg R = \frac{1}{2}\arg\chi$, so the Cramer solution (19) reduces to

$$d_1 \;=\; \frac{\widetilde{R}}{2\cos\big(\omega + \frac{\psi}{2}\big)}, \qquad \widetilde{R} = e^{-i\psi/2} R \;\;\text{(real)},$$

the two-equal-vector form of R seen in the proof of Corollary 8.1. For $\widetilde{R}$ we substitute the saddle-point evaluation of Siegel's integral, which is the first correction term of the Riemann–Siegel formula,

$$\widetilde{R} = (-1)^{N-1}\Big(\tfrac{2\pi}{t}\Big)^{1/4} C_0(\hat{p}) + O\big(t^{-3/4}\big), \qquad C_0(\hat{p}) = \frac{\cos\big(2\pi(\hat{p}^2 - \hat{p} - \frac{1}{16})\big)}{\cos(2\pi\hat{p})}, \qquad \sqrt{t/2\pi} = N + \hat{p}.$$

Every quantity on the right is an elementary function of $t = I(T)$, kept exact throughout:

$$z = \sqrt{t/2\pi}, \qquad N = \lfloor z\rfloor, \qquad \hat{p} = z - N, \qquad \omega = t\ln(m+1), \qquad m = \lfloor T\rfloor.$$

On the line $\chi = e^{-2i\vartheta(t)}$ with ϑ the Riemann–Siegel theta function, so $\widetilde{R}$ is $e^{i\vartheta}R$ up to a sign that cancels against the cosine, and $d_1 = e^{i\vartheta}R \,/\, 2\cos(\omega - \vartheta)$ *exactly*. Substituting the Riemann–Siegel first term for $e^{i\vartheta}R$ gives

$$d_1\big(\tfrac{1}{2}, T\big) \;\approx\; \big[N = m{+}1\big]\,(m{+}1)^{-1/2} \;+\; (-1)^{N-1}\Big(\tfrac{2\pi}{t}\Big)^{1/4} \frac{C_0(\hat{p})}{2\cos\big(\omega - \vartheta(t)\big)}, \tag{59}$$

where the Iverson bracket adds one full link length when $N = m + 1$, that is, when the Riemann–Siegel main sum carries one more summand pair than our Σ_1, Σ_2, which happens (roughly) when $\operatorname{frac}(T) > \frac{1}{2}$. The formula is elementary and zeta-free: ϑ may be replaced by its asymptotic $\vartheta(t) = \frac{t}{2}\ln\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t}$ at no cost (the two agree here to 4×10^{-7} even at $T = 1$). Its maximum error is 0.006 on $[1, 2)$, 0.002 on $[3, 4)$, 0.0009 on $[6, 7)$ and 0.0002 on $[20, 21)$. That is about 1% of the swing of d_1 at the start, decaying like $T^{-3/2}$ and spread flat across each unit interval (Figure 11). The error is purely the Riemann–Siegel truncation, $O(t^{-3/4})$; the next correction term of that formula would push further. And since $R = 2d_1\cos(\omega + \frac{\psi}{2})\, e^{i\psi/2}$ on the line, (59) is effectively a closed-form approximation of Siegel's remainder there as well.

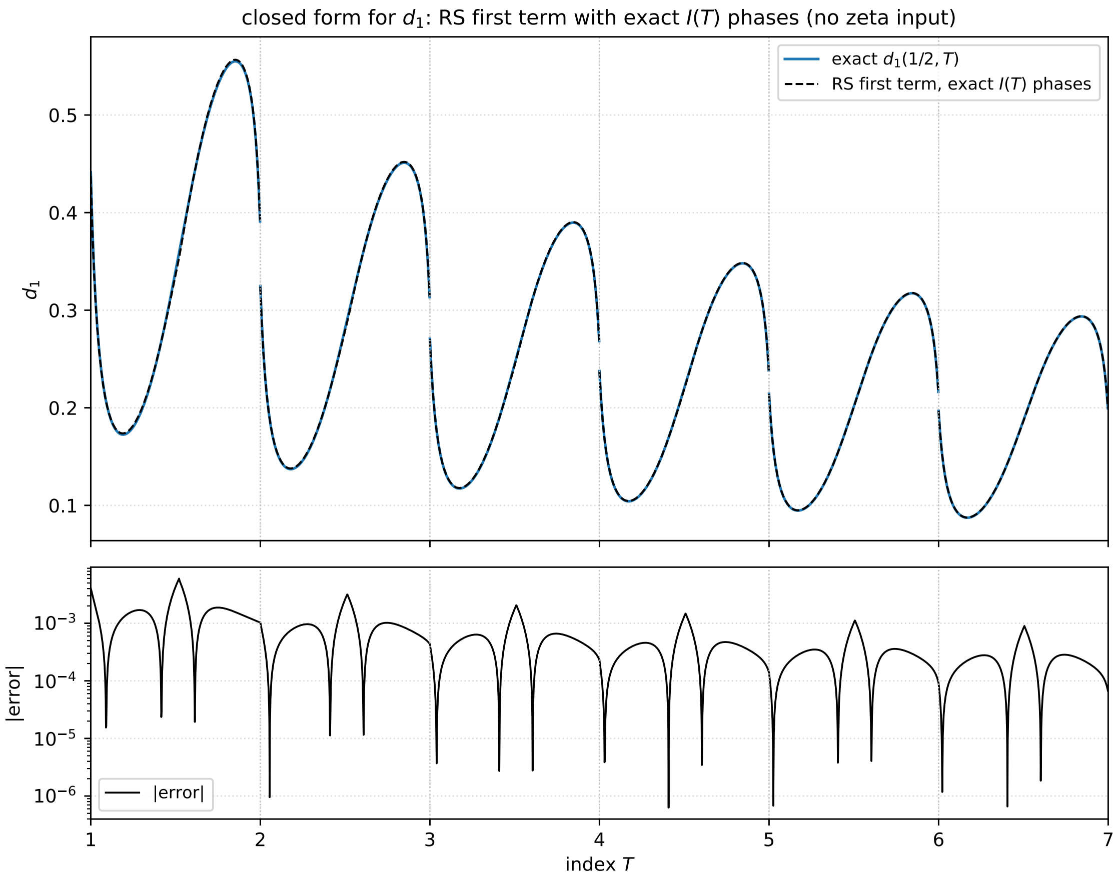

Figure 11: The closed form (59): the Riemann–Siegel first term with the phases kept exact through $t = I(T)$, no zeta input. Top: exact d_1 (blue) against (59) (black dashed), visually indistinguishable, $1 \le T \le 7$. Bottom: |error| on a log scale, with maximum 0.006 on $[1, 2)$, decaying like $T^{-3/2}$, spread flat in the fractional part. Generated by `fig_d1_rs_phases.py`.

8.3 Approximation of d_1 and d_2 when $\sigma \neq \frac{1}{2}$

There is a general-σ approximation as well, and it needs no new machinery beyond what is already in the paper. Off the critical line the remainder R is genuinely complex (no single rotation makes it real), so approximating it means approximating two real degrees of freedom, and those two degrees of freedom are exactly the pair (d_1, d_2).

Siegel's saddle-point analysis was never restricted to the critical line; the first term for the remainder at general $s = \sigma + it$ splits into one piece per chain:

$$R(s) \;\approx\; (-1)^{N-1}\,\frac{C_0(\hat{p})}{2}\Big[a^{-\sigma}\,e^{-i\tilde{\vartheta}(t)} \;+\; \chi(s)\,a^{\sigma-1}\,e^{+i\tilde{\vartheta}(t)}\Big], \qquad a = \sqrt{t/2\pi}, \tag{60}$$

with $t = I(T)$ exact, $N = \lfloor a \rfloor$, $\hat{p} = a - N$, the same universal $C_0(\hat{p})$ of §8.2, and the elementary phase $\tilde{\vartheta}(t) = \frac{t}{2}\ln\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8}$. When $N = m+1$ the extra summand pair $(m+1)^{-s} + \chi\,(m+1)^{s-1}$ is added, exactly as in (59). Feeding this approximate R to the Cramer solution (19)–(20) with the exact $\omega = t\ln(m+1)$ and $\psi = \arg\chi$ then yields *both* d_1 and d_2 at once, still with no zeta input (χ is Gamma factors). The two bracket pieces are the per-chain remainders, with the same structure as the Kuznetsov split of §7.1, and at $\sigma = \frac{1}{2}$, where $\chi = e^{-2i\vartheta}$, the bracket collapses to $2e^{-i\vartheta}$: (60)

reduces exactly to (59). We verified this numerically digit for digit, which pins down all the sign conventions.

Because the Cramer solution is linear in R, the solve can be carried out once and for all: substituting (60) into (19)–(20) gives the general-σ analogue of (59) in closed form,

$$\begin{aligned} d_1(\sigma,T) \approx{}& \big[N=m{+}1\big]\,(m{+}1)^{-\sigma} \\ &+(-1)^{N-1}\,\frac{C_0(\hat p)}{2\sin(2\omega+\psi)}\Big[a^{-\sigma}\sin\big(\omega+\psi+\tilde\vartheta\big)+|\chi|\,a^{\sigma-1}\sin\big(\omega-\tilde\vartheta\big)\Big], \end{aligned} \tag{61}$$

$$\begin{aligned} d_2(\sigma,T) \approx{}& \big[N=m{+}1\big]\,|\chi|\,(m{+}1)^{\sigma-1} \\ &+(-1)^{N-1}\,\frac{C_0(\hat p)}{2\sin(2\omega+\psi)}\Big[a^{-\sigma}\sin\big(\omega-\tilde\vartheta\big)+|\chi|\,a^{\sigma-1}\sin\big(\omega+\psi+\tilde\vartheta\big)\Big], \end{aligned} \tag{62}$$

with $\tilde\vartheta=\tilde\vartheta(t)$, $\psi=\arg\chi$, and $|\chi|=|\chi(\sigma+it)|$ as before. The Iverson terms are now whole link lengths on each side separately, because the extra summand pair lies exactly along the two unit directions: $(m{+}1)^{-s}$ along $e^{-i\omega}$ and $\chi\,(m{+}1)^{s-1}$ along $e^{i(\omega+\psi)}$. Note the symmetry: swapping the two coefficients $a^{-\sigma}\leftrightarrow|\chi|\,a^{\sigma-1}$ swaps d_1 and d_2. At $\sigma=\frac12$, where $|\chi|=1$ and $\psi=-2\vartheta$, the two sine brackets coincide and each formula collapses to (59), since $\sin(\omega{-}\vartheta)/\sin\big(2(\omega{-}\vartheta)\big)=1/\,2\cos(\omega{-}\vartheta)$ and $a^{-1/2}=(2\pi/t)^{1/4}$. We verified that (61)–(62) reproduce the Cramer-solve route to machine precision; they are the same computation written out.

The accuracy across the strip (maximum $|d_1|$ error away from the pole windows described below):

σ	$[1,2)$	$[6,7)$	$[20,21)$
0.1	0.076	0.016	0.0046
0.3	0.035	0.0057	0.0013
0.5	0.006	0.0009	0.0002
0.7	0.025	0.0024	0.0004
0.9	0.045	0.0032	0.0004

The measured decay exponents match $O(T^{-\sigma-1})$ almost perfectly (1.07 at $\sigma=0.1$, 1.5 at $\sigma=0.5$, 1.8 at $\sigma=0.9$), and since d_1 itself scales like $T^{-\sigma}$, the *relative* error is $O(1/T)$ uniformly across the strip. The critical line is where the formula is at its absolute best, fittingly. And d_2 comes for free: the Cramer solve produces both components simultaneously from the same approximate R, and its errors are essentially identical to d_1's (e.g. 0.031 vs 0.035 at $\sigma=0.3$ on $[1,2)$), so no separate derivation is needed. Figure 12 shows both components at $\sigma=0.3$ and $\sigma=0.7$.

Two caveats. First, off the line the exact d_1 and d_2 have genuine narrow poles at the parallel-link heights $q\approx\frac14,\frac34$ (the bands of Figure 16); the first-term approximation smooths through those spikes, so within about ±0.05 of them the error is locally large (relative error $\sim$40% right at the shoulder). Everywhere else the table applies. Second, the formula is not limited to the strip: it works for any σ with relative error $O(1/T)$. For $\sigma>1$ it is excellent, while for $\sigma\le0$ the absolute error $T^{-\sigma-1}$ decays ever more slowly (at $\sigma=-\frac12$ it is still $\approx$0.06 at $T=20$), so "any σ, relative error $1/T$" is the honest statement, with the strip as the sweet spot.

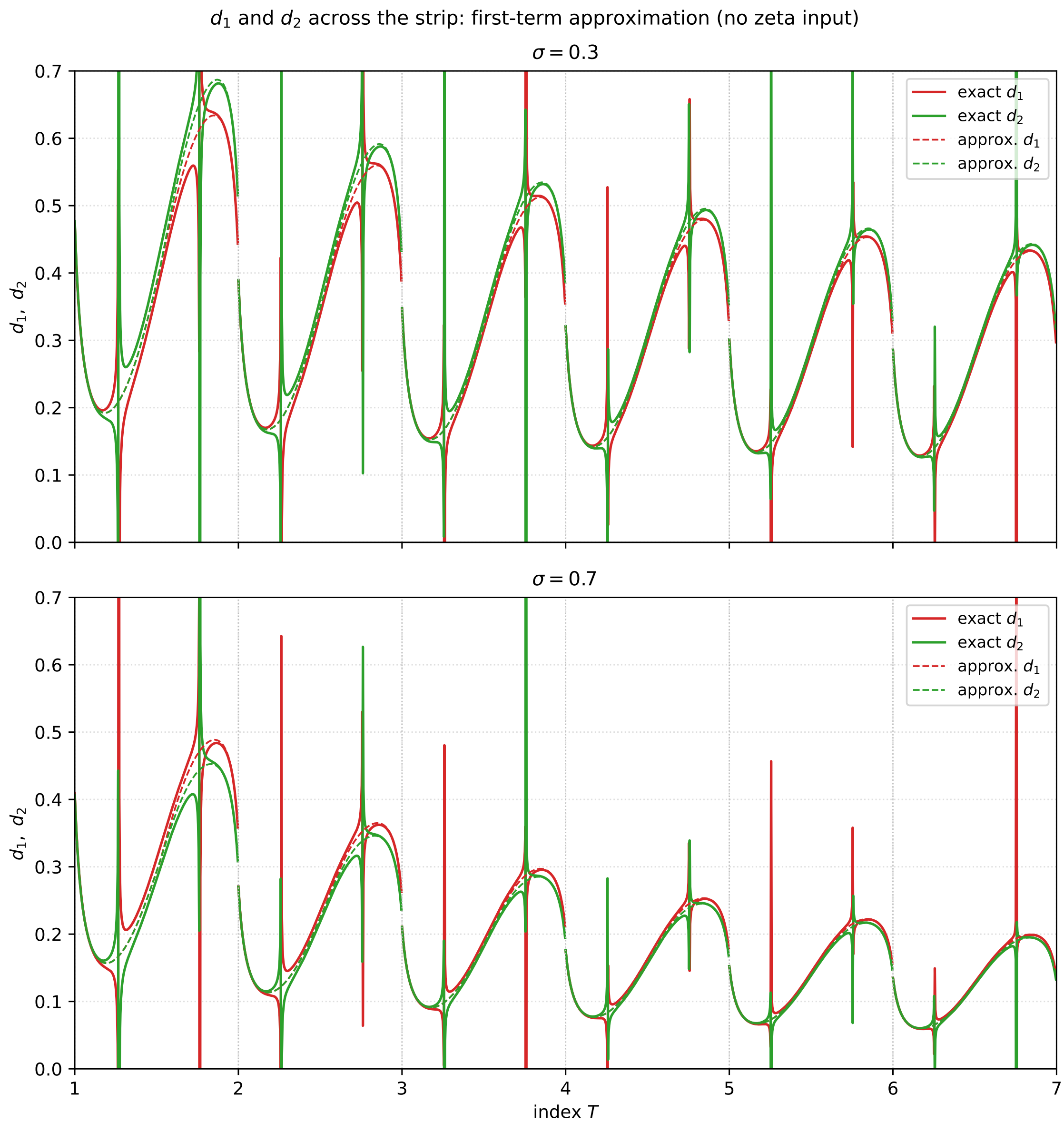

Figure 12: d_1 (red) and d_2 (green) across the strip at $\sigma = 0.3$ (top) and $\sigma = 0.7$ (bottom), $1 \leq T \leq 7$: exact values (solid) against the first-term approximation (60) fed through the Cramer solution (dashed, in the color of its exact counterpart). The narrow spikes leaving the frame near fractional heights $\frac{1}{4}$ and $\frac{3}{4}$ are the genuine off-the-critical-line poles of d_1 and d_2 (parallel links); the approximation smooths through them. Generated by `fig_d1_d2_general_sigma.py`.

8.4 Where are the d_1 and d_2 poles?

The poles of d_1 and d_2 (equivalently of R_{1ps} and R_{2ps}) appear to be at fractional heights $\frac{1}{4}$ and $\frac{3}{4}$, but they are not exactly there. Here we delve into where they are.

A pole occurs when the two bisector links are parallel; these are link $m = \lfloor T \rfloor$ of each chain, each carrying summand $\lceil T \rceil$, the summand just after the last one included in the partial sum. In other words, a pole occurs when the two next summands have the same argument, so that the cone spanned by the two unit directions $e^{-i\omega}$ and $e^{i(\omega+\arg\chi)}$ of (23) degenerates to a line and the common denominator $\sin(2\omega + \arg\chi)$ of (19)–(20) vanishes. Subtracting the two arguments from each other produces a formula that is zero at the poles:

$$2\,\omega(T) \;+\; \arg\chi\big(\sigma + i\,I(T)\big) \;=\; 2\pi k, \qquad k \in \mathbb{Z}, \tag{63}$$

with $\omega(T) = I(T)\log\lfloor T+1\rfloor$ as in §4. This formula was used to find the locations of the first 20 poles numerically (Table 2).

Odd Pole #	T **(odd)**	**Even Pole #**	T **(even)**
1	1.268870258145	2	1.763736013695
3	2.261710393365	4	2.759511841350
5	3.258504081544	6	3.757283144824
7	4.256679717535	8	4.755902912218
9	5.255501039250	10	5.754963311284
11	6.254676427788	12	6.754282121179
13	7.254067037740	14	7.753765522842
15	8.253598276826	16	8.753360246216
17	9.253226472440	18	9.753033785343
19	10.252924347790	20	10.752765174160

Table 2: First 20 poles for $\sigma = 0.6$, which is the same result as for $\sigma = 0.4$. The symmetry about $\sigma = \frac{1}{2}$ exists because $\arg\chi(\sigma+it) = -\arg\chi(1-\sigma+it)$ by the identity $\chi(s)\,\chi(1-s) = 1$, and the pole condition (63) uses $\arg\chi$ only as a phase offset; reflecting $\sigma \mapsto 1-\sigma$ therefore yields the same T-solutions. Note the poles occur at approximately $T = \lfloor T\rfloor + \frac{1}{4}$ and $T = \lfloor T\rfloor + \frac{3}{4}$ (equivalently, an integer $\pm\frac{1}{4}$), and approach those values as $T \to \infty$.

9 The geometry behind the result (experimental mathematics)

The decomposition $R = R_{1ps} + R_{2ps}$ was not first derived; it was first *seen*. Over roughly eight years we built and used an interactive visualization tool, Zest,[3] which draws the partial-sum chains of ζ in the complex plane and lets us sweep σ and T continuously while measuring what appears on the screen. This section reconstructs the series of observations made with that tool. It led, in order, to a distinguished link and point of the spiral (the *bisector link* and *bisector point*), and from their motion to the mapping $I(T)$ of §3. The final observation in that sequence, the *yin* and *yang* curves from which the weights d_1, d_2 of §4 were read off, is developed at length in §11.

Throughout we use the link and joint numbering of §6: joint k of a chain is the value of its partial sum after k summands, and link k runs from joint k to joint $k+1$, so that link k represents the $(k+1)$st summand and each remainder is the fractional link m. The *forward* chain (Σ_1) is anchored at joint 0, the origin. The *reverse* chain (Σ_2) is anchored at ζ and traversed backwards, so its joint m lands at $\zeta - \Sigma_2 = \Sigma_1 + R$. In the pictures the chains are drawn well past joint $m = \lfloor T \rfloor$: the sum, though ultimately divergent, first winds into an Euler-like spiral whose center lies near ζ; it seemingly converges before it diverges. Figure 13 shows the picture: the forward chain winds from the origin into a spiral centered near ζ, while the reverse chain winds from ζ into a mirror spiral centered near the origin, and the two link chains cross. If more links were drawn, the links would continue to spiral outward around ζ and cover the entire figure, because the zeta sum diverges on the critical line, unlike the Dirichlet L-functions of §10.2, whose characters have bounded partial sums and so those chains close in on their limit instead of covering the plane.[4] We stop drawing links at link number

$$L_N(T, S_n) = \left\lfloor \frac{I(T)}{\pi\,(2S_n + 1)} \right\rfloor, \tag{64}$$

where S_n is the spiral number, spiral number 0 being the last one (the biggest, at ζ). Note that ζ itself is not typically *on* this last link; the last link is the link closest to ζ. The same formula locates the other spirals and pseudo-spirals too: with $S_n = 1$, for example, it yields the link number of the link most in the center of the second-to-last spiral.

[3] Source code and standalone executables (Mac and PC) at `https://github.com/paul-stahura/zest`.

[4] At $t = 176.13$ on the critical line, the partial sums of $L(s, \chi_3)$ come within 0.021, 0.0021 and 0.00019 of the value by $N = 10^3$, 10^5 and 3×10^6, a drift inward like $N^{-1/2}$, while zeta's are off by 0.18, 1.80 and 9.83, a drift outward at $\sqrt{N}/t$. Convergence in $\sigma > 0$ is special to a non-principal character: the principal character returns ζ itself, and L-functions of higher degree have coefficients too large to converge anywhere in the critical strip.

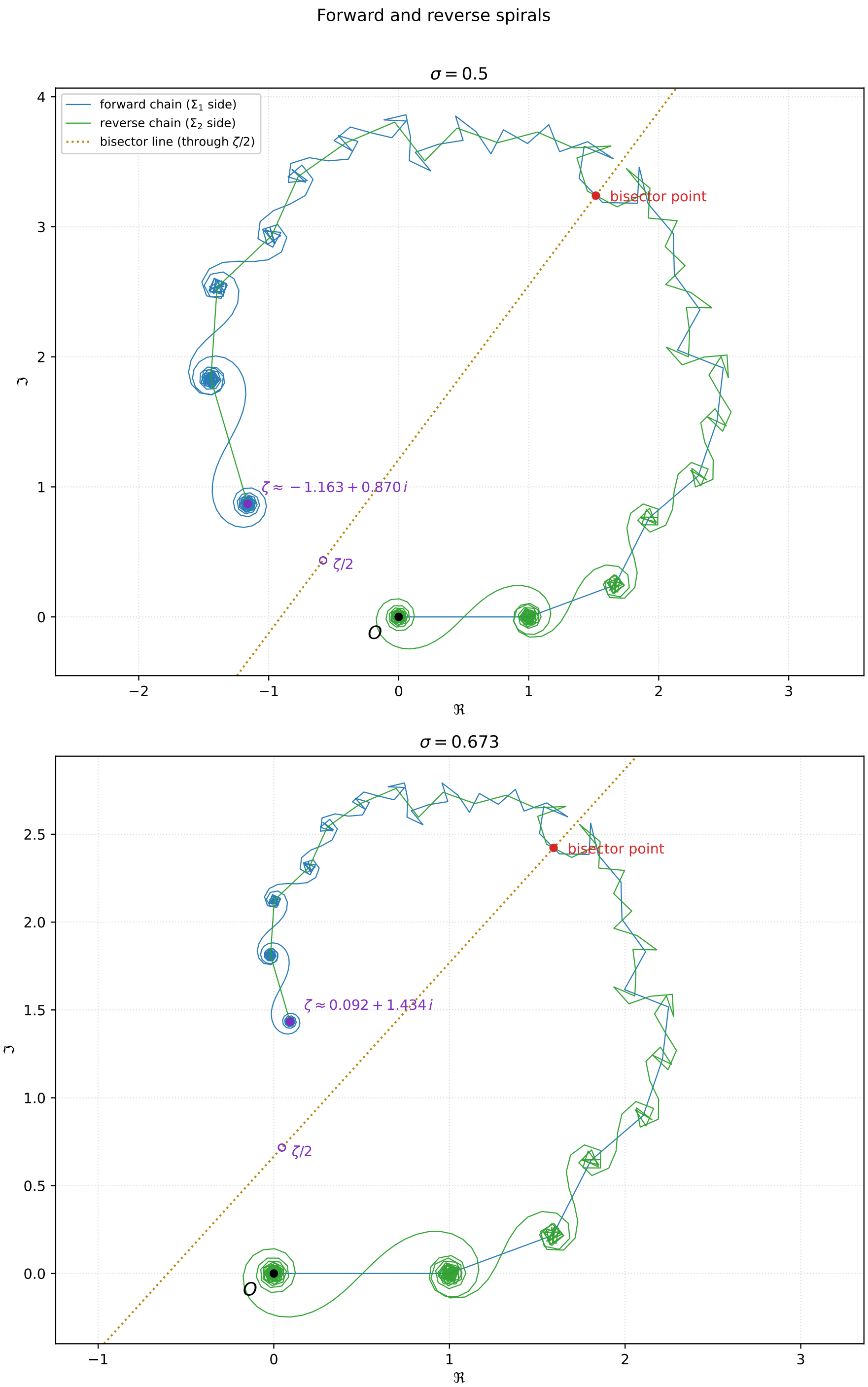

Figure 13: The forward spiral (blue, anchored at the origin O) and the reverse spiral (green, anchored at ζ) for $T = 14.3085$ (so $t = I(T) \approx 1377.33$ and $m = 14$), on the critical line $\sigma = \frac{1}{2}$ (top, $\zeta \approx -1.163 + 0.870i$) and off it at $\sigma = 0.673$ (bottom, $\zeta \approx 0.092 + 1.434i$). Each spiral is drawn out to link $\lfloor I(T)/\pi \rfloor = 438$, the link nearest its spiral center, for a total of 439 links per spiral (the count is independent of σ). The red dot is the bisector point $B_1 = \Sigma_1 + R_{1ps}$, where the two chains cross at their bisector links (§9.1); the dotted line is the bisector line through the bisector point and $\zeta/2$ (open circle). On the critical line (top) the whole configuration is symmetric about the bisector line; at $\sigma = 0.673$ (bottom) the symmetry is lost. Generated by `fig_full_spirals.py`.

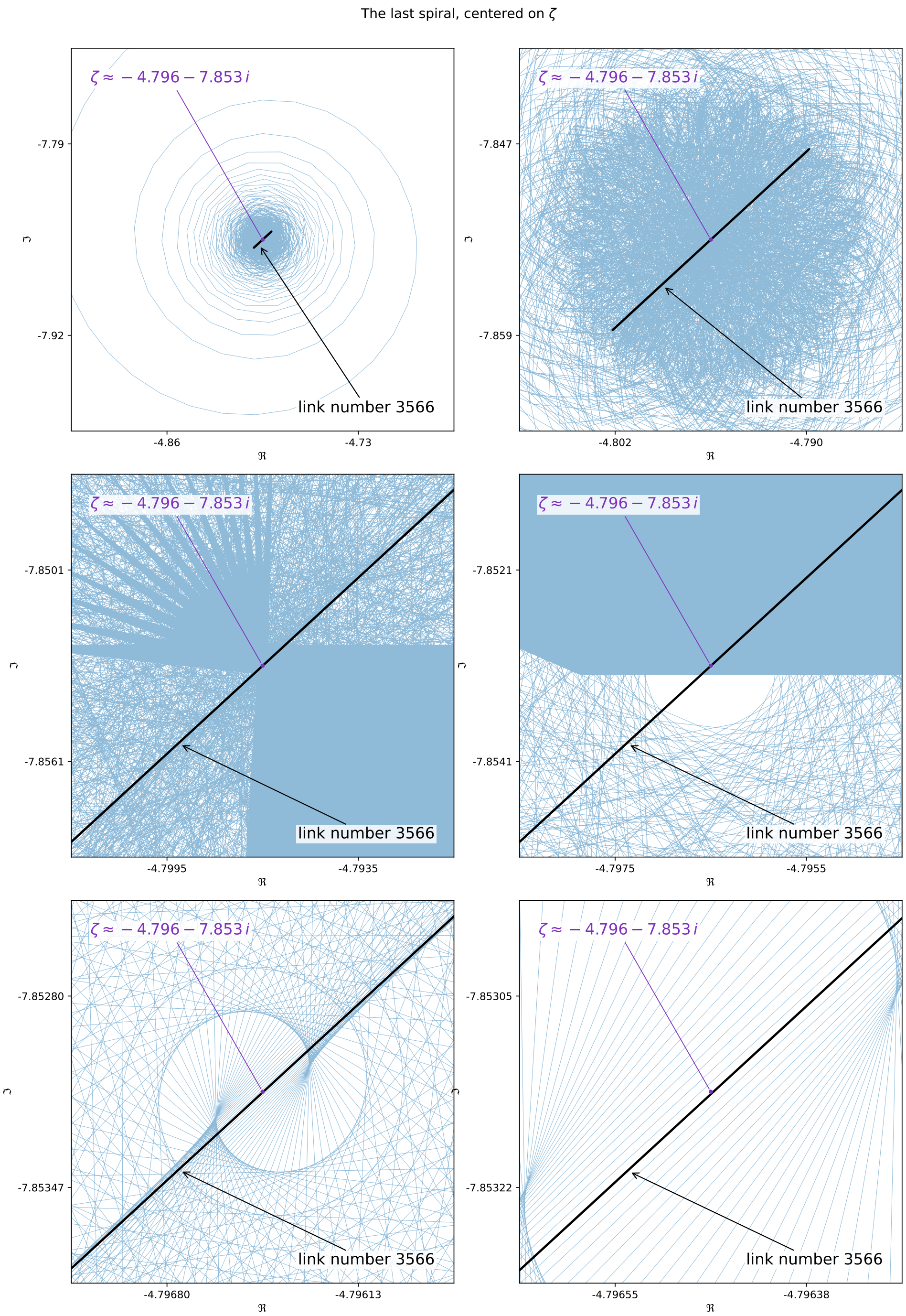

Figure 14: Zooming in on the last spiral, the spiral centered on ζ, of the forward chain at $T = 41.73$ (so $t = I(T) \approx 11204.74$, $\sigma = \frac{1}{2}$, $\zeta \approx -4.796 - 7.853i$). The chain carries $\lfloor I(T)/\pi \rfloor + 1 = 3567$ links; the last link is number 3566 by (64) with $S_n = 0$, and it is highlighted in black in every panel. Each panel is more zoomed in than the last (panels 1–6, left to right from the top). Near the spiral center each link turns by nearly π, so the links sweep back and forth across ζ and the joints settle onto a tight ring of radius about half a link length: at coarse scale the spiral renders as a dense annulus (panels 1–3), and the near-diameter links envelope a lens-shaped clearing around ζ (panels 4–5). Observe in panel 6 that ζ is ***not on*** the last link: the last link is only the link closest to ζ. Generated by `fig_last_spiral_zoom.py`.

9.1 The bisector link and the bisector point

Drawn at height T, each chain exhibits spirals and pseudo-spirals, and (as others have also observed [7, 11, 14, 8]) there is a one-to-one correspondence between the joints and the spiral centers, in a type of symmetry; Nickel puts it as the center-to-center distances and angles of the large linked spirals reproducing the initial steps exactly. The link at index $m = \lfloor T \rfloor$ sits at the boundary between the two regimes, and we call it the *bisector link*. Each chain has one: the *forward bisector link* (link m of the Σ_1 chain, running from Σ_1 along the $(m+1)$st summand direction) and the *reverse bisector link* (link m of the Σ_2 chain, running from $\Sigma_1 + R$ along its $(m+1)$st summand direction).

The two spirals intersect at their bisector links, and we call the point where the forward and reverse bisector links cross the *bisector point*. One qualification: the crossing can momentarily fail. Twice per unit of T, near fractional parts ≈ 0.25 and ≈ 0.75, the two bisector links become parallel and, off the critical line, do not intersect; these are the isolated instants at which $\sin(2\omega + \arg\chi)$, the denominator of (19)–(20), vanishes. Those are exactly the places where d_1 and d_2 have poles: the parallel-link condition and the pole condition are the same equation, which is why the precise locations were worked out in §8.4. On the critical line the two links are then collinear rather than merely parallel, so the bisector point survives there. In the notation of the earlier sections the bisector point is exactly

$$B_1 \;=\; \Sigma_1 + R_{1ps} \;=\; \Sigma_1 + R - R_{2ps}; \tag{65}$$

it is visible as the joint between the red and orange fractional links in Figures 2 and 3.

9.2 Legs

When $\sigma = \frac{1}{2}$ the spiral object has a type of symmetry about the line through the bisector point and $\frac{\zeta}{2}$, so the bisector point sits at the apex of an isosceles triangle whose base runs from the origin to ζ and whose equal sides are the two legs B_1 and $\zeta - B_1$ (this is Corollary 9.1 below, proved in Appendix B). Off the line we keep the same terminology even though the triangle's legs are then (usually) unequal.

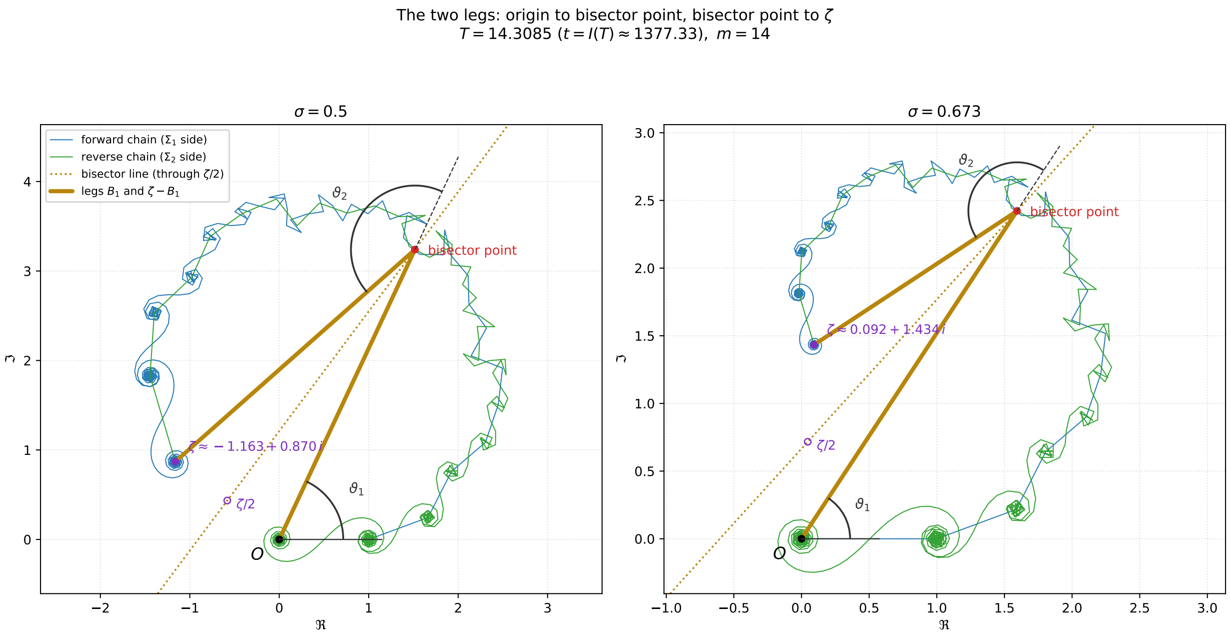

Figure 15: The two legs, drawn as thick dark-yellow segments over the spirals of Figure 13: one from the origin O to the bisector point (B_1), the other from the bisector point to ζ ($\zeta - B_1$). On the critical line (left, $\sigma = \frac{1}{2}$) the legs have equal length ($|B_1| = |\zeta - B_1| \approx 3.577$): the bisector point is the apex of an isosceles triangle whose base runs from O to ζ, and the dotted bisector line is its axis of symmetry. Off the critical line (right, $\sigma = 0.673$) the legs are unequal (≈ 2.899 and ≈ 1.797) at this T. Both panels also mark the two angles defined below: ϑ_1, at O from the positive real axis to Leg 1, and ϑ_2, at the bisector point from the extension of Leg 1 (dashed) to Leg 2. Generated by `fig_legs.py`.

On the critical line the two fractional summands are not independent: each is the other's complex conjugate, rotated by the functional-equation factor. Writing $R_{1ps} = d_1 e^{-i\omega}$ and $R_{2ps} = d_2 e^{i(\omega + \arg\chi)}$ as in §6 and using that $|\chi| = 1$ when $\sigma = \frac{1}{2}$, the equality $d_1 = d_2$ proved in §8.1 (Corollary 8.1) is exactly equivalent to the compact relation

$$R_{2ps} = \chi\,\overline{R_{1ps}} \qquad \left(\sigma = \tfrac{1}{2}\right). \tag{66}$$

Geometrically, (66) says the second fractional summand is the mirror image of the first (conjugation across the real axis) followed by the rotation χ. Because χ has unit modulus on the line this rotation is an isometry, so $|R_{1ps}| = |R_{2ps}|$ immediately. This is the functional-equation reflection $s \leftrightarrow 1 - s$ acting as a symmetry of the two remainders, and it is the source of the equal-leg phenomenon we now develop.

The two legs $B_1 = \Sigma_1 + R_{1ps}$ and $B_2 = \Sigma_2 + R_{2ps}$ of §6 meet at a "bisector point": Leg 1 runs from the origin to $\Sigma_1 + R_{1ps}$, and Leg 2 from there to ζ. Their lengths are

$$L_1(\sigma, T) = \left|\,\Sigma_1 + R_{1ps}\,\right|, \tag{67}$$

$$L_2(\sigma, T) = \left|\,\Sigma_2 + R_{2ps}\,\right|, \tag{68}$$

while ϑ_1 is the angle Leg 1 makes with the real axis and ϑ_2 is the angle from Leg 1 to Leg 2 (oriented so that a positive ϑ_2 rotates Leg 1 toward Leg 2):

$$\vartheta_1(\sigma, T) = \arg\bigl(\Sigma_1 + R_{1ps}\bigr), \tag{69}$$

$$\vartheta_2(\sigma, T) = \arg\left(\frac{\Sigma_2 + R_{2ps}}{\Sigma_1 + R_{1ps}}\right). \tag{70}$$

With these, ζ is simply the sum of the two legs,

$$\zeta(\sigma, T) = L_1\, e^{i\vartheta_1} \; + \; L_2\, e^{i(\vartheta_1+\vartheta_2)}, \tag{71}$$

or equivalently, in matrix form,

$$\zeta(\sigma, T) = \begin{pmatrix} 1 & i \end{pmatrix} \begin{pmatrix} \cos\vartheta_1 & \cos(\vartheta_1+\vartheta_2) \\ \sin\vartheta_1 & \sin(\vartheta_1+\vartheta_2) \end{pmatrix} \begin{pmatrix} L_1 \\ L_2 \end{pmatrix}, \tag{72}$$

where the row $\begin{pmatrix} 1 & i \end{pmatrix}$ turns the resulting real (x, y) components into the complex value $x + iy$.

Corollary 9.1. *When $\sigma = \frac{1}{2}$ the two legs are always equal in length: $L_1 = L_2$.*

This is immediate from Corollary 8.1 of §8.1: on the critical line every summand of Σ_2 is χ times the conjugate of the corresponding summand of Σ_1, and by (66), which is equivalent to $d_1 = d_2$, the same holds for the fractional summands. Hence $\Sigma_2 + R_{2ps} = \chi\,\overline{\Sigma_1 + R_{1ps}}$, and since $|\chi| = 1$ on the line, the two legs have the same length. A Lean proof is recorded in Appendix B.

The same reflection locates the apex exactly, which is what the dotted bisector line of Figures 13 and 15 draws.

Corollary 9.2. *When $\sigma = \frac{1}{2}$ the perpendicular projection of the bisector point B_1 onto the line through the origin carrying ζ is exactly $\frac{\zeta}{2}$: writing $\psi = \arg\chi$,*

$$\Re\bigl(e^{-i\psi/2} B_1\bigr)\, e^{i\psi/2} \;=\; \frac{\zeta}{2}. \tag{73}$$

Indeed $B_2 = \chi\overline{B_1}$ gives $e^{-i\psi/2}\zeta = e^{-i\psi/2}B_1 + \overline{e^{-i\psi/2}B_1} = 2\Re(e^{-i\psi/2}B_1)$, and $e^{i\psi/2}$ is the direction of ζ (this is the reality of the Riemann–Siegel Z function). So the isosceles triangle with base from O to ζ has its apex over the midpoint of that base, and the equal-leg statement $L_1 = L_2$ is the same fact read as a distance. This is also proved in Appendix B.

Consequences for zeta zeros. A zero of ζ in the critical strip requires *both* of the following:

$$\frac{L_1(\sigma, T)}{L_2(\sigma, T)} = 1 \qquad \text{and} \qquad \vartheta_2(\sigma, T) = \pi,$$

i.e. the two legs must be equal in length *and* Leg 2 must fold back onto Leg 1. By Corollary 9.1 the length condition holds automatically on the critical line; there, a zero occurs exactly when $\vartheta_2(\frac{1}{2}, T) = \pi$. Away from the critical line, the two legs are unequal except at exceptional points, so most of the strip is immediately ruled out for zeros. The leg-length ratio is symmetric across the line,

$$\frac{L_1(\sigma, T)}{L_2(\sigma, T)} = \frac{L_2(1-\sigma, T)}{L_1(1-\sigma, T)}. \tag{74}$$

Numerically we have never found a point with $\sigma \neq \frac{1}{2}$ where the ratio is 1 and $\vartheta_2 = \pi$ simultaneously; such a point would disprove the Riemann Hypothesis.

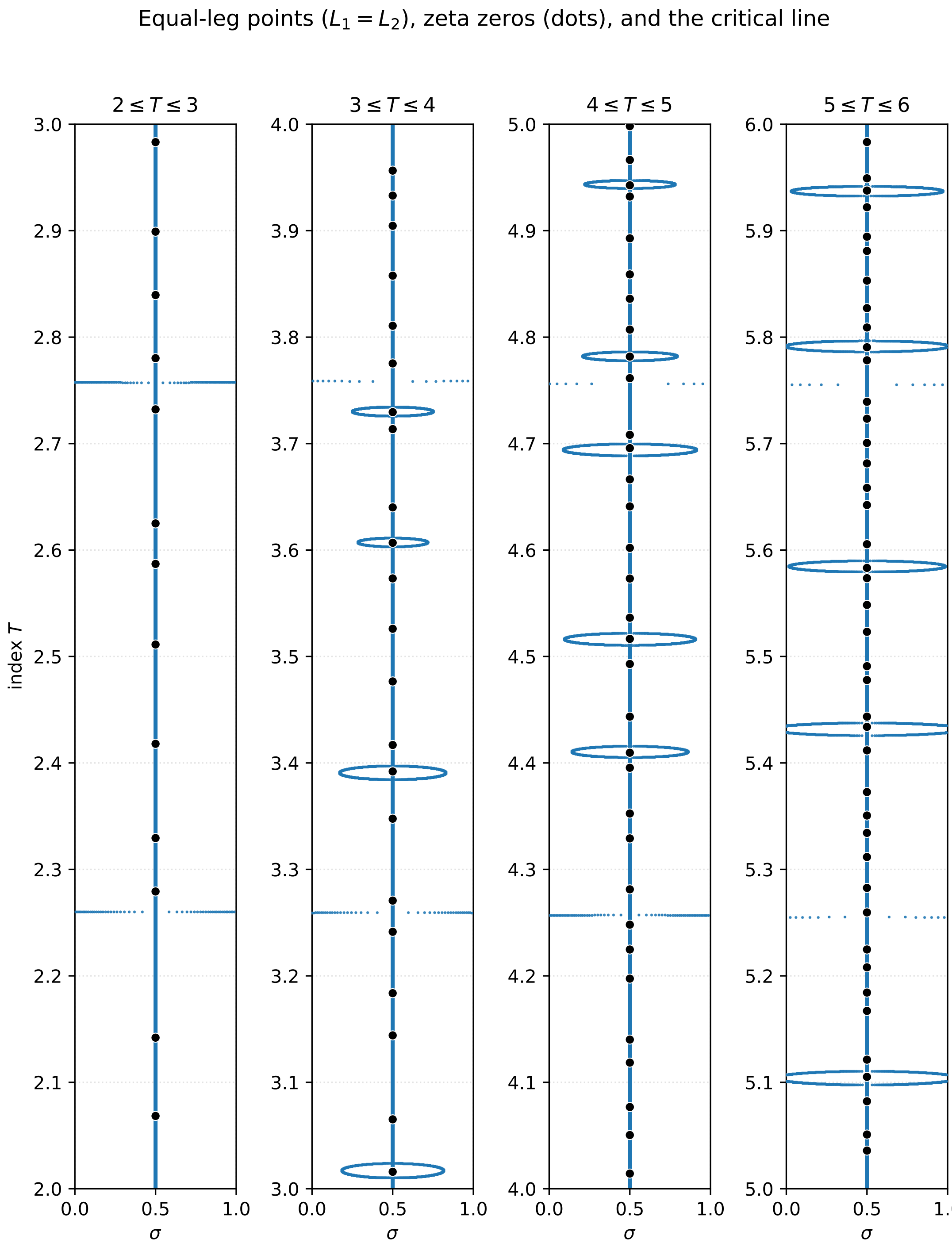

Figure 16: Four windows of the critical strip ($0 \leq \sigma \leq 1$, index T vertical): $2 \leq T \leq 3$, $3 \leq T \leq 4$, $4 \leq T \leq 5$, and $5 \leq T \leq 6$. The blue vertical line is the critical line $\sigma = \frac{1}{2}$, the black dots are the zeta zeros, and the small blue points are the equal-leg points: points (σ, T) where $L_1 = L_2$, i.e. where the bisector point $B_1 = \Sigma_1 + R_{1ps}$ is equidistant from the origin and from ζ. By Corollary 9.1 the entire critical line consists of equal-leg points; off the line the equal-leg locus forms isolated ovals and thin horizontal bands, and every zero sits on the critical line where the length condition is automatic. The horizontal bands, near fractional parts ≈ 0.25 and ≈ 0.75 of T, are the instants where the two bisector links become parallel and fail to intersect (§9.1): the bisector point has a pole there (d_1 and d_2 blow up), and as T passes through it the equal-leg point sweeps across the full width of the strip. The sweep gets faster as T grows, so the fixed-step sampling leaves fewer dots and the bands appear fainter. Data from the Zest tool point set *Zps Equal Leg Lengths*; generated by `fig_equal_legs_strips.py`.

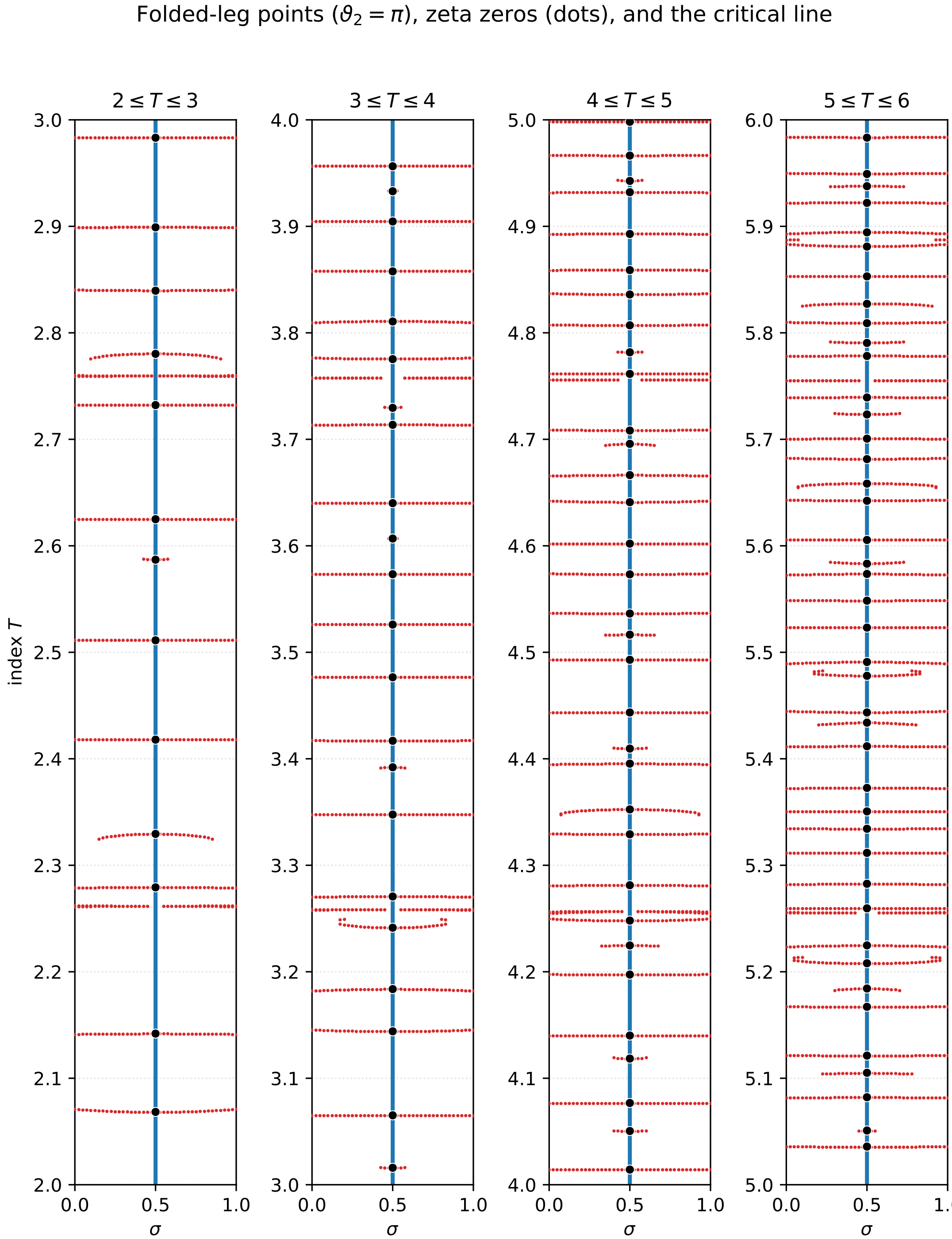

Figure 17: The same four windows of the critical strip as in Figure 16, now showing the *folded-leg* points (red): points (σ, T) where $\vartheta_2 = \pi$, i.e. where Leg 2 folds straight back onto Leg 1 and a zeta zero is therefore possible. The blue vertical line is the critical line and the black dots are the zeta zeros. The folded-leg locus forms nearly horizontal curves sweeping the strip; a zero requires *both* $\vartheta_2 = \pi$ and $L_1 = L_2$, and since the length condition holds automatically on the critical line (Corollary 9.1), every crossing of a red curve with the critical line is a zeta zero. Data from the Zest tool point set *Zps Leg Angle* $= PI$; generated by `fig_theta2_strips.py`.

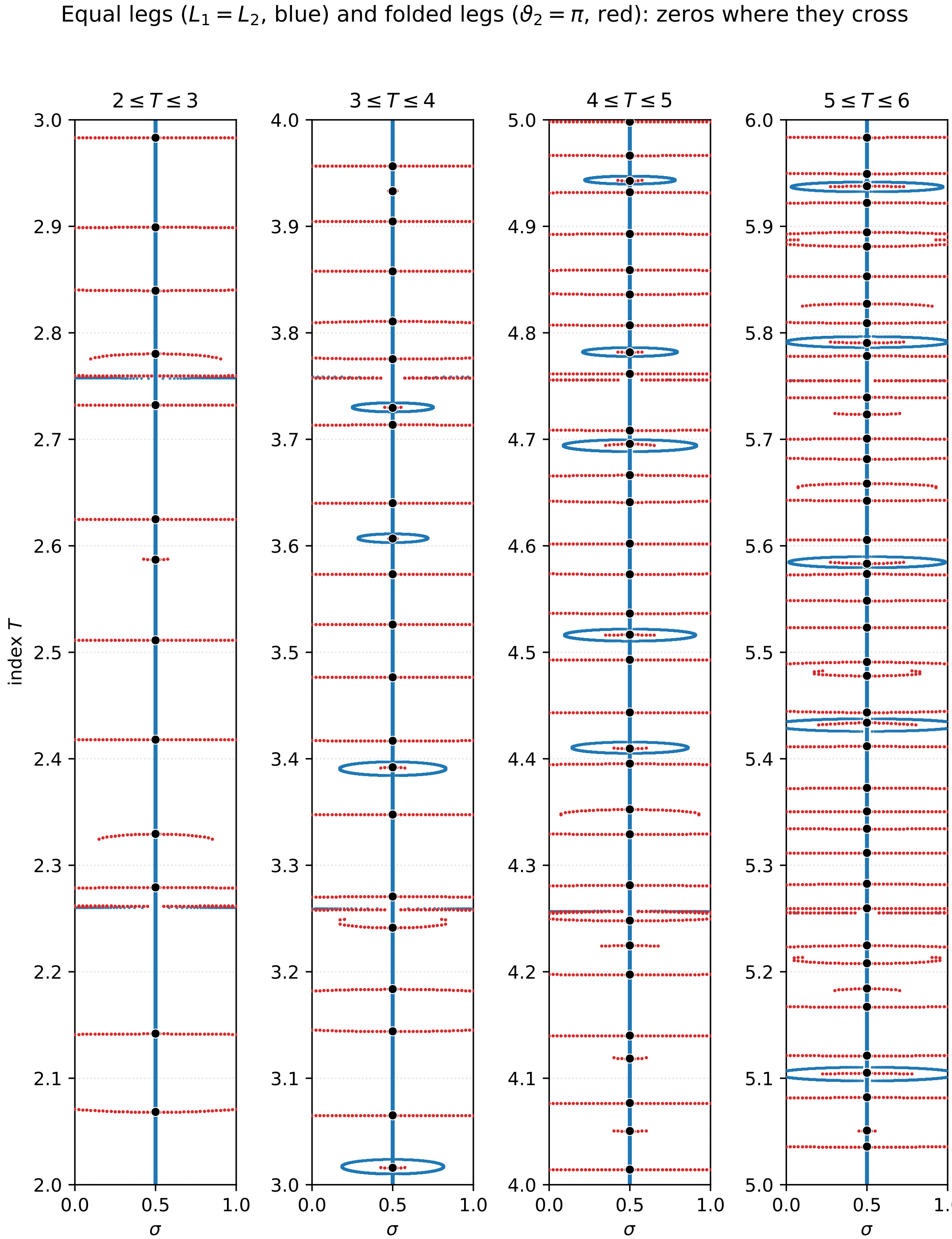

Figure 18: Figures 16 and 17 overlaid: the equal-leg locus $L_1 = L_2$ (blue), the folded-leg locus $\vartheta_2 = \pi$ (red), the critical line (blue, since it belongs to the equal-leg locus), and the zeta zeros (black dots). A zero requires both conditions at once, so zeros occur exactly where red meets blue. On the critical line the blue locus is the line itself (Corollary 9.1), and every red curve crossing it produces a zero; away from the line, the red curves and the blue ovals interleave without touching. A red/blue crossing off the critical line would be a counterexample to the Riemann Hypothesis. Generated by `fig_combined_strips.py`.

Figure 18 makes the stakes plain: wherever a region of the critical strip contains no red and no blue dots, a zero of ζ is impossible. On the critical line the two legs are always equal in length, so whenever $\vartheta_2 = \pi$ there, a zero occurs. Off the line the legs are seldom equal in length, but sometimes they are (the blue ovals and bands). If one could prove that, off the critical line, in the rare times when the legs are equal, that ϑ_2 is simultaneously never π, one would have proven the Riemann Hypothesis true.

9.3 The strip lines at ≈ 0.25 and ≈ 0.75 are not flat

The pole heights are not exactly flat. Viewed across the strip, the pole heights of §8.4 are nearly invariant ("flat") in σ, and symmetric about $\sigma = \frac{1}{2}$ (Table 2 and its caption). But they are *not exactly* flat: there is a slight, symmetric curvature in σ, visible in Figure 19.

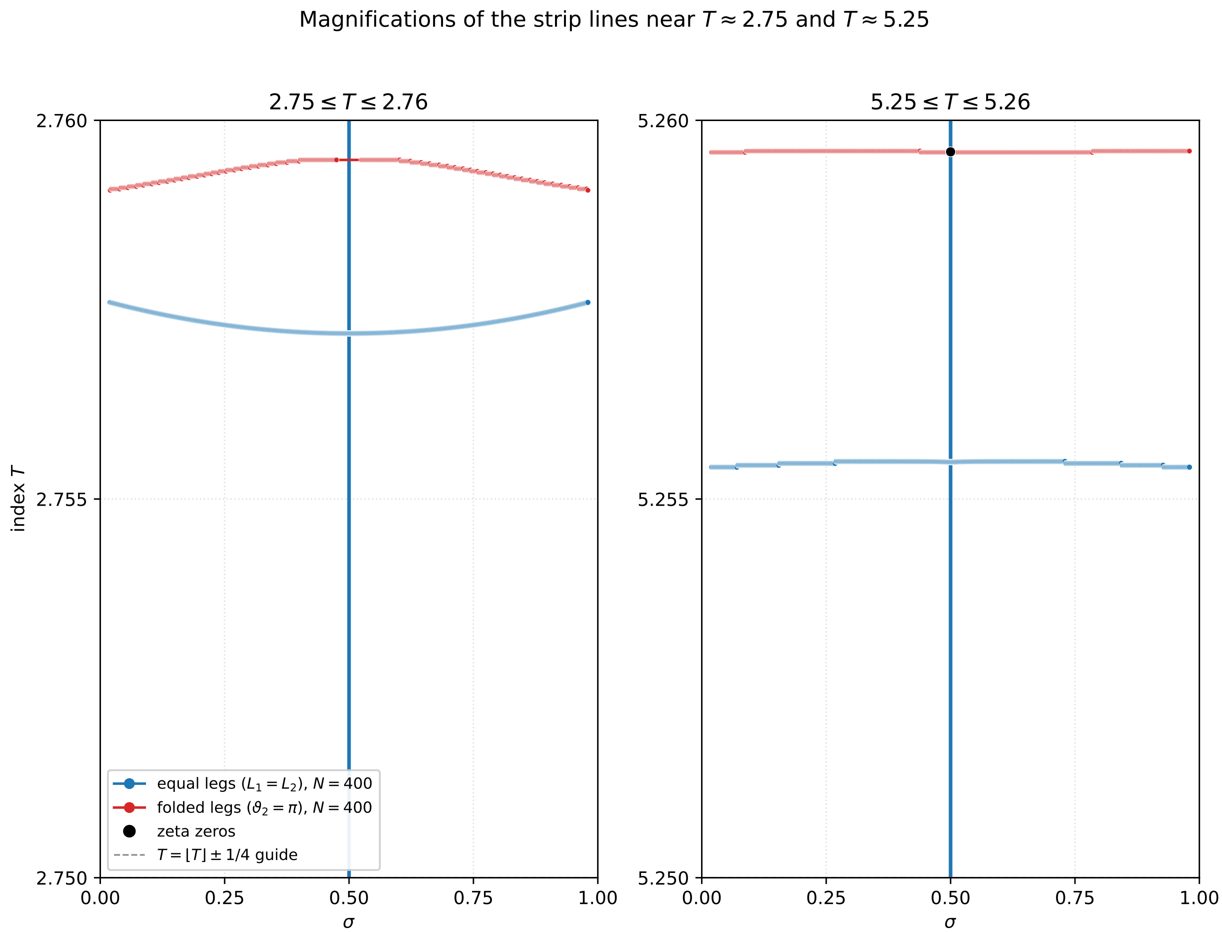

Figure 19: Two magnifications of the critical strip in the style of Figure 18: equal-leg locus $L_1 = L_2$ (blue) and folded-leg locus $\vartheta_2 = \pi$ (red), each recomputed at $N = 400$ values of σ across $(0, 1)$; critical line (blue vertical) and zeta zeros (black). Samples where the PS split is singular (tiny det) are omitted from the red locus, so the d_1 pole is not mislabeled as a fold. Left: $2.75 \le T \le 2.76$; the blue equal-leg band sits near $T \approx 2.757$ and the red folded band near $T \approx 2.759$, with a gap at $\sigma = \frac{1}{2}$ (no zero in this window). Right: $5.25 \le T \le 5.26$; blue near $T \approx 5.255$ and red near $T \approx 5.2596$, meeting the critical line at the zeta zero in the window (not at the pole band). The dashed horizontal guide marks $\lfloor T \rceil \pm \frac{1}{4}$; the blue strip lines bow slightly across σ, symmetric about $\sigma = \frac{1}{2}$. At this resolution the red and blue loci do not cross off the critical line. Generated by `fig_pole_heights_zoom.py`.

9.4 PS, AK and RS Legs and Angles

The equal-leg and folded-leg loci above were drawn with the exact bisector point $B_1 = \Sigma_1 + R_{1ps}$. The same two conditions $L_1 = L_2$ and $\vartheta_2 = \pi$ can be restated for any other choice of bisector point. Two natural alternatives come from Kuznetsov's split (44): take $B_1 = \Sigma_1 + R_{1ak}$, or take the midpoint remainder $B_1 = \Sigma_1 + (R_{1ak} + R_{2ak})/2$, which is the geometric analogue of Siegel's half-remainder $\frac{1}{2}R$ once $R_{1ak} + R_{2ak} \approx R$.

Figure 20 magnifies a short window of the strip, $4.65 \le T \le 4.80$, and overlays the three choices side by side. On the left, equal-leg loci: PS in blue, AK in green, and $R/2$ in purple. The three families form nested ovals about the critical line, but the nesting order is not fixed: already in this short window the lower oval has PS outermost and AK innermost, while the upper oval has AK outermost and $R/2$ innermost, and across $3 \le T \le 20$ all six orderings of PS, AK and $R/2$ occur with no simple alternating pattern. On the right, folded-leg loci $\vartheta_2 = \pi$: PS in red, AK in orange, and $R/2$ in purple (hollow). The zeros (black) still sit where a folded curve meets the critical line; the three remainder choices shift the off-line curves but agree on those critical-line crossings in this window.

Figure 21 is the same three-way overlay for a pair of neighboring equal-leg ovals near $T \approx 9.44$, drawn in the window $9.415 \le T \le 9.465$. The lower PS oval sits around the zero at $T \approx 9.431$ and the upper around $T \approx 9.451$; a third zero at $T \approx 9.440$ lies in the gap between them (not inside either oval). The AK and $R/2$ ovals nest with the PS locus as in Figure 20.

Intersecting equal-leg ovals and the Riemann Hypothesis. The nested ovals in Figures 20 and 21 suggest a comparison that uses only exact remainders. Write

$$B_{1ps} = \Sigma_1 + R_{1ps}, \qquad B_{1rs} = \Sigma_1 + R_{1rs}, \qquad R_{1rs} := \tfrac{1}{2}R,$$

where $R = R_{1ps} + R_{2ps} = \zeta - \Sigma_1 - \Sigma_2$ is Siegel's remainder, so R_{1rs} is the exact half-remainder ("rs" for Riemann–Siegel, parallel to the subscripts "ps" and "ak") and B_{1rs} is the midpoint of the remainder segment from Σ_1 to $\Sigma_1 + R$. Let $\mathcal{E}_{\mathrm{ps}}$ and $\mathcal{E}_{\mathrm{rs}}$ be the corresponding equal-leg loci in the strip: the sets of (σ, T) at which $|B_1| = |\zeta - B_1|$ for $B_1 = B_{1ps}$ and $B_1 = B_{1rs}$ respectively.

If $\zeta(\sigma + iT) = 0$, then for either choice of bisector point one has $L_2 = |\zeta - B_1| = |B_1| = L_1$, so

$$\frac{L_1}{L_2} = 1$$

holds simultaneously for both B_{1ps} and B_{1rs}. Every zero therefore lies in $\mathcal{E}_{\mathrm{ps}} \cap \mathcal{E}_{\mathrm{rs}}$. On the critical line this is automatic: Corollary 9.1 puts the whole line in $\mathcal{E}_{\mathrm{ps}}$, and the rs split is reflection-symmetric there as well: $\Sigma_2 = \chi\,\overline{\Sigma_1}$ and $R = \chi\,\overline{R}$ (the latter because $e^{-i\psi/2}R$ is real, Appendix B), so

$$\chi\,\overline{B_{1rs}} = \Sigma_2 + \tfrac{1}{2}R = \zeta - B_{1rs},$$

and since $|\chi| = 1$ this gives $\left|B_{1rs}\right| = \left|\zeta - B_{1rs}\right|$: the whole line lies in $\mathcal{E}_{\mathrm{rs}}$ as well. (Note that $B_{1rs} \neq \zeta/2$; only its perpendicular projection onto the base is $\zeta/2$, by Corollary 9.2, which is how Figure 37 draws it, at offset $h_{R/2}$ from $\zeta/2$.) The two loci therefore intersect everywhere along $\sigma = \frac{1}{2}$; that common set is expected and does not constrain zeros beyond the folded-leg condition already discussed.

Off the critical line the situation is different. An off-line zero would be an off-line point of $\mathcal{E}_{\mathrm{ps}} \cap \mathcal{E}_{\mathrm{rs}}$: a point where the PS and rs equal-leg ovals meet. Consequently, an off-line intersection of $\mathcal{E}_{\mathrm{ps}}$ and $\mathcal{E}_{\mathrm{rs}}$ is *necessary* for an off-line zero (it is not sufficient: the folded-leg condition $\vartheta_2 = \pi$, or equivalently $\zeta = 0$, is still required). The contrapositive is the statement of interest:

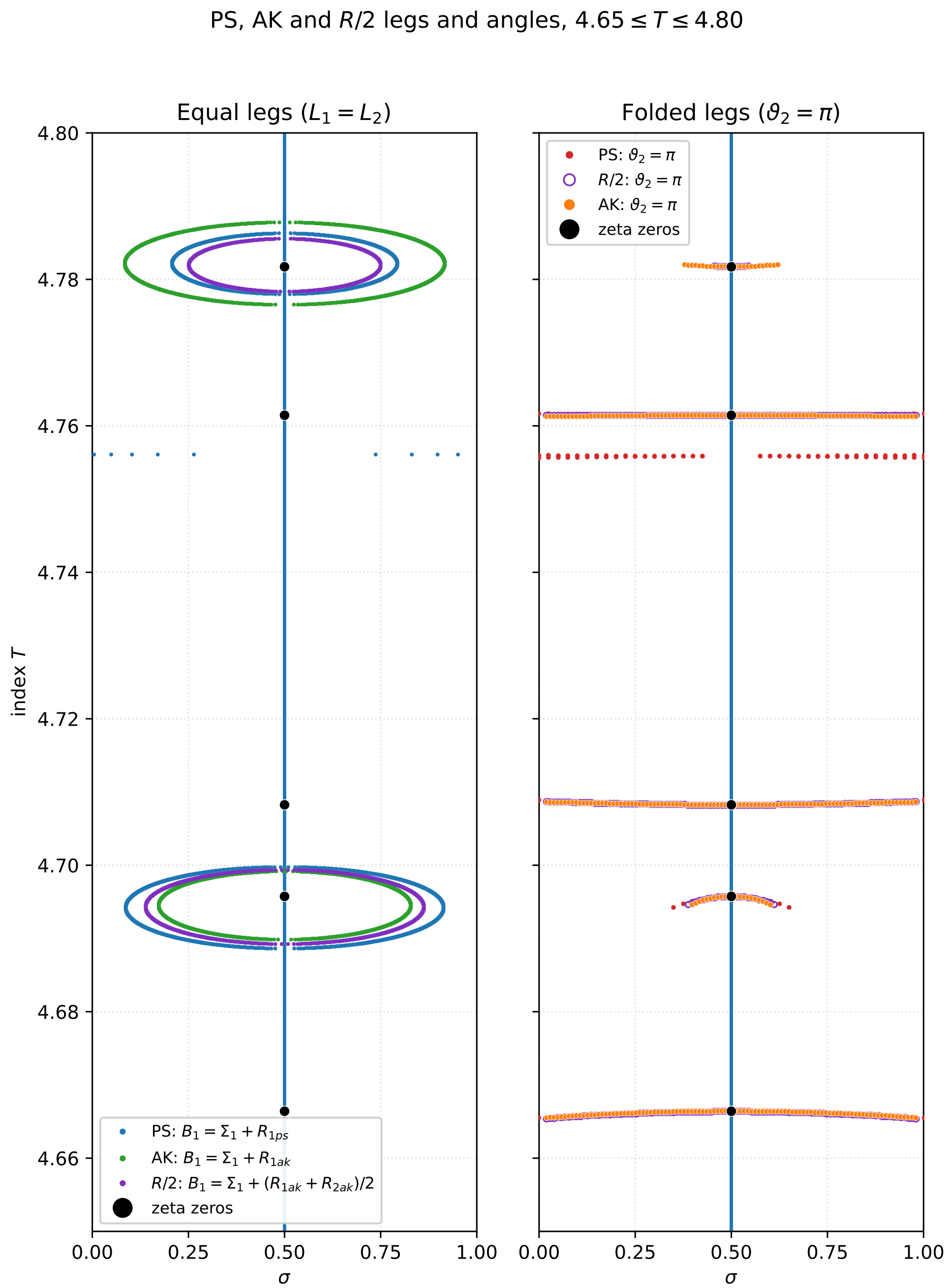

Figure 20: PS, AK and $R/2$ legs and angles in the window $4.65 \leq T \leq 4.80$. Left: equal-leg loci $L_1 = L_2$ for three choices of bisector point: PS $B_1 = \Sigma_1 + R_{1ps}$ (blue), AK $B_1 = \Sigma_1 + R_{1ak}$ (green), and $R/2$ $B_1 = \Sigma_1 + (R_{1ak} + R_{2ak})/2$ (purple). Right: folded-leg loci $\vartheta_2 = \pi$ for the same three choices (red, orange, and hollow purple). The blue vertical is the critical line; black dots are zeta zeros. Equal-leg and folded-leg loci computed from the definitions $L_1 = L_2$ and $\vartheta_2 = \arg((\zeta - B_1)/B_1)$ for each choice of B_1. Generated by `fig_ps_ak_r2_legs_angles.py`.

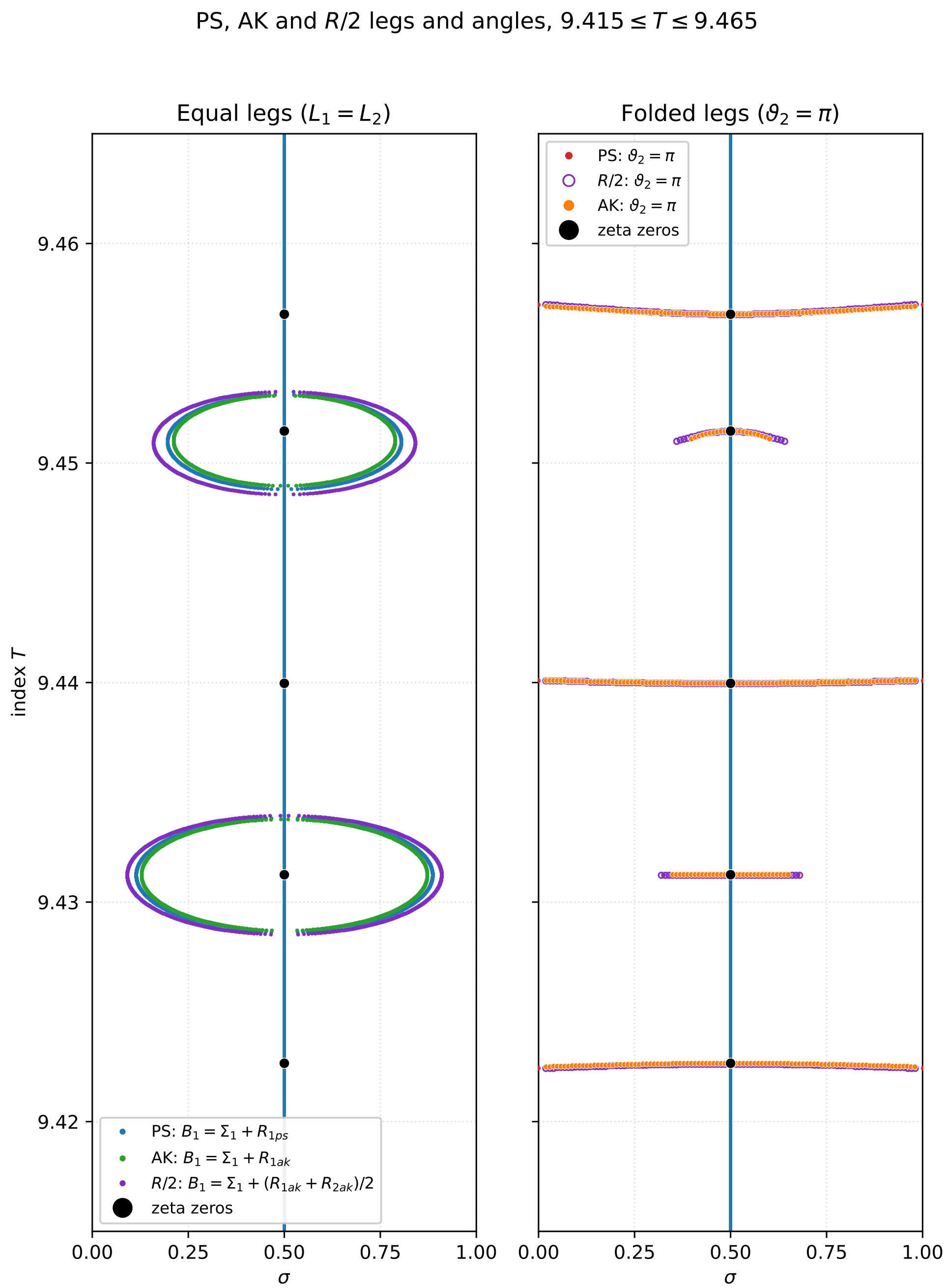

Figure 21: PS, AK and $R/2$ legs and angles for two neighboring equal-leg ovals near $T \approx 9.44$, in the window $9.415 \le T \le 9.465$. Left: equal-leg loci $L_1 = L_2$ for PS (blue), AK (green), and $R/2$ (purple). Right: folded-leg loci $\vartheta_2 = \pi$ for the same three choices (red, orange, and hollow purple). The blue vertical is the critical line; black dots are zeta zeros. The lower oval surrounds the zero at $T \approx 9.431$ and the upper the zero at $T \approx 9.451$; the zero at $T \approx 9.440$ sits in the gap between them. Loci computed as in Figure 20. Generated by `fig_ps_ak_r2_legs_angles.py` (`--out fig_ps_ak_r2_oval_9441 --t-lo 9.415 --t-hi 9.465`).

If $\mathcal{E}_{\rm ps}$ and $\mathcal{E}_{\rm rs}$ have no point in common with $\sigma \neq \frac{1}{2}$, then ζ has no zero off the critical line.

In particular, if the two loci are disjoint off the line throughout the strip, the Riemann Hypothesis follows; if they are disjoint off the line for all $T \leq T_{\max}$, then there is no off-line zero with index at most $T_{\max}$. The figures above are consistent with a strict nesting (the ovals approach one another but do not cross off the line). A proof that this nesting persists for all T, with due care at the pole heights of d_1 in §8.4 where B_{1ps} ceases to be defined in the usual way, would therefore be a proof of the Riemann Hypothesis.

In short: on the critical line the leg-length ratio is always one (Corollary 9.1, and likewise for B_{1rs}), so the zeros there are determined by the legs folding back, $\vartheta_2 = \pi$, for either the ps or the rs choice of bisector point. Off the line the length condition is no longer free: a zero can occur only where $L_1/L_2 = 1$ for the legs, and since that must hold for both choices it is a necessary condition that

$$\frac{L_1}{L_2} = 1 \tag{75}$$

simultaneously for the ps legs and for the rs legs.

The mean leg-imbalance metric. The equal-leg loci above are the zero sets of a continuous imbalance. For each of the three named splits of §7.2, write the two leg lengths

$$L_1 \;=\; |\Sigma_1 + R_1|, \qquad L_2 \;=\; |\Sigma_2 + R_2|, \tag{76}$$

where (R_1, R_2) is (R_{1ps}, R_{2ps}), $(R_{1rs}, R_{2rs}) = (\frac{1}{2}R, \frac{1}{2}R)$, or (R_{1ak}, R_{2ak}) respectively. The normalized length imbalance of a split is the scale-free quantity

$$\delta \;=\; \frac{|L_1 - L_2|}{L_1 + L_2 + \varepsilon}, \qquad \varepsilon = 10^{-18}, \tag{77}$$

so $\delta = 0$ if and only if $L_1 = L_2$ (the bisector condition for that choice of head). Averaging the three splits gives the plotted field

$$\bar{\delta} \;=\; \frac{\delta_{\rm ps} + \delta_{\rm rs} + \delta_{\rm ak}}{3}. \tag{78}$$

On the critical line each split has $L_1 = L_2$ (Corollary 9.1 and the parallel identities for rs and ak), so $\bar{\delta} \approx 0$ and the strip is dark purple along $\sigma = \frac{1}{2}$. Off the line $\bar{\delta}$ rises, and the equal-leg ovals of Figures 20–21 appear as the dark cellular structures hugging the critical line in Figure 22.

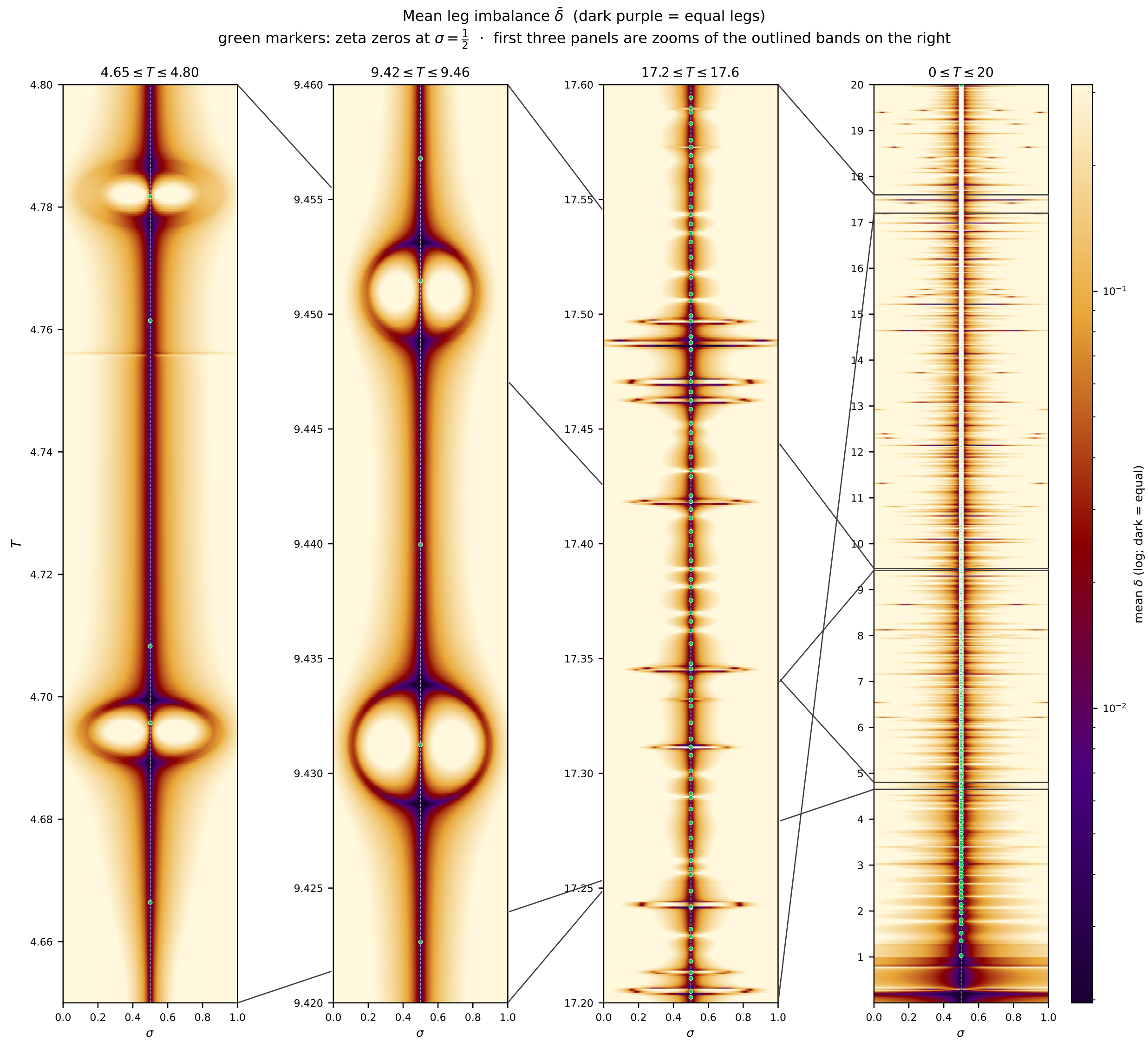

Figure 22: Mean leg imbalance $\bar{\delta}$ of (78) (dark purple = equal leg lengths). Horizontal axis $\sigma \in [0, 1]$; vertical axis the spiral index T. Color is $\bar{\delta}$ on a log scale. Green markers are zeta zeros at $\sigma = \frac{1}{2}$. The rightmost panel is the full strip $0 \leq T \leq 20$; the first three panels are zooms of the outlined bands ($4.65 \leq T \leq 4.80$, $9.42 \leq T \leq 9.46$, $17.2 \leq T \leq 17.6$). The dark vertical band on $\sigma = \frac{1}{2}$ is the simultaneous equal-leg condition for the ps, rs, and ak splits. Generated by `plot_leg_equality_four_panel.py`.

9.5 Toward a proof

The geometry assembled in this section suggests two routes to the Riemann Hypothesis, listed here in increasing order of how many remainder splits participate. (A third phenomenon, the exact collinearity of the three first-half remainders on the critical line, is a signature of the line itself rather than of the zeros; it is presented separately in §12.6.)

1. *Equal legs and fold-back together, for a single split.* A zero at (σ, T) forces, for any single choice of bisector point, both leg conditions of §9.2 at the same input: $L_1 = L_2$ *and* $\vartheta_2 = \pi$. On the critical line the length condition is automatic (Corollary 9.1), so zeros there are cut out by the fold-back alone. Off the line the two conditions trace independent families of curves (the equal-leg ovals and the nearly horizontal folded-leg curves of Figure 18), and a proof that the two families never pass through the same off-line point, for even one split, would prove the Riemann Hypothesis.

2. *Equal legs for two or more splits at once.* A zero makes the legs equal for *every* split simultaneously: when $\zeta = 0$, $L_2 = |\zeta - B_1| = |B_1| = L_1$ for any head B_1. An off-line zero would therefore be an off-line point of $\mathcal{E}_{\rm ps} \cap \mathcal{E}_{\rm rs}$ (and of $\mathcal{E}_{\rm ak}$ as well). This route drops the angle condition entirely and asks only that two equal-leg ovals meet off the line; the figures of §9.4 show the ovals nesting without crossing. This route is developed in detail below.

Route 2 in detail. An off-line zero at (σ, T) would require both equal-leg identities at the same input:

$$\big|\Sigma_1 + R_{1ps}\big| = \big|\Sigma_2 + R_{2ps}\big|, \tag{79a}$$

$$\big|\Sigma_1 + \tfrac{1}{2}R\big| = \big|\Sigma_2 + \tfrac{1}{2}R\big|. \tag{79b}$$

(Here (79a) is the ps condition and (79b) the rs condition with $R_{1rs} = \frac{1}{2}R$.) Using $R_{2ps} = R - R_{1ps}$ from Theorem 4.1, the same pair is

$$\big|\Sigma_1 + R_{1ps}\big| = \big|\Sigma_2 + R - R_{1ps}\big|, \tag{80a}$$

$$\big|\Sigma_1 + \tfrac{1}{2}R\big| = \big|\Sigma_2 + \tfrac{1}{2}R\big|. \tag{80b}$$

From the definitions of the partial sums, with $s = \sigma + iI(T)$,

$$\Sigma_2(\sigma, T) = \chi(s)\,\overline{\Sigma_1(1-\sigma, T)}, \tag{81}$$

so Σ_2 is determined by χ and the reflected first sum. From the cone decomposition (23) and the Cramer solution (19)–(20), with $\phi = \arg R$, $\psi = \arg\chi$, and $\omega = \omega(T)$,

$$R_{1ps} = |R|\,\frac{\sin(\omega - \phi + \psi)}{\sin(2\omega + \psi)}\,e^{-i\omega}. \tag{82}$$

Substituting (81) into (80b) and (81)–(82) into (80a) yields the pair

$$\big|\Sigma_1 + \tfrac{1}{2}R\big| = \big|\chi(s)\,\overline{\Sigma_1(1-\sigma, T)} + \tfrac{1}{2}R\big|, \tag{83a}$$

$$\left|\Sigma_1 + |R|\,\frac{\sin(\omega - \phi + \psi)}{\sin(2\omega + \psi)}\,e^{-i\omega}\right| = \left|\chi(s)\,\overline{\Sigma_1(1-\sigma, T)} + R - |R|\,\frac{\sin(\omega - \phi + \psi)}{\sin(2\omega + \psi)}\,e^{-i\omega}\right|. \tag{83b}$$

Equivalently, writing $u = e^{-i\omega}$ and $v = e^{i(\omega+\psi)}$ for the two unit directions in $R = d_1 u + d_2 v$ from (23), the same real coefficients are recovered by projecting R onto the (u, v)-frame:

$$R_{1ps} = \frac{\operatorname{Im}(\overline{v}\,R)}{\operatorname{Im}(\overline{v}\,u)}\,u, \tag{84}$$

and likewise $R_{2ps} = R - R_{1ps}$. Substituting (84) together with (81) into (80a) recovers (83b). A proof that (83) has no solution for $\sigma \neq \frac{1}{2}$ (away from the poles of d_1) would therefore prove that $\mathcal{E}_{\rm ps}$ and $\mathcal{E}_{\rm rs}$ never meet off the critical line, and with it the Riemann Hypothesis.

10 $I(T)$ functions

10.1 The origin of $I(T)$: the neighboring links in the bisector frame

Now change the point of view: hold the forward bisector link still. Translate joint m to the origin and rotate and scale so that the bisector link lies along the real axis from 0 to 1; everything else moves in this *bisector frame* as t increases. Watching the two links on either side of the bisector link, the link at $m-1$ behind it and the link at $m+1$ ahead of it, one sees a strikingly periodic picture: each neighboring link revolves around the stationary bisector link approximately *twice* (approaching exactly two revolutions as T grows), and then the whole pattern repeats. The instant the pattern repeats is precisely the handoff instant of §9.1: the neighboring link ***folds back*** onto the bisector link, the bisector point moves onto that next link and that next link becomes the new bisector link with its own two neighbors.

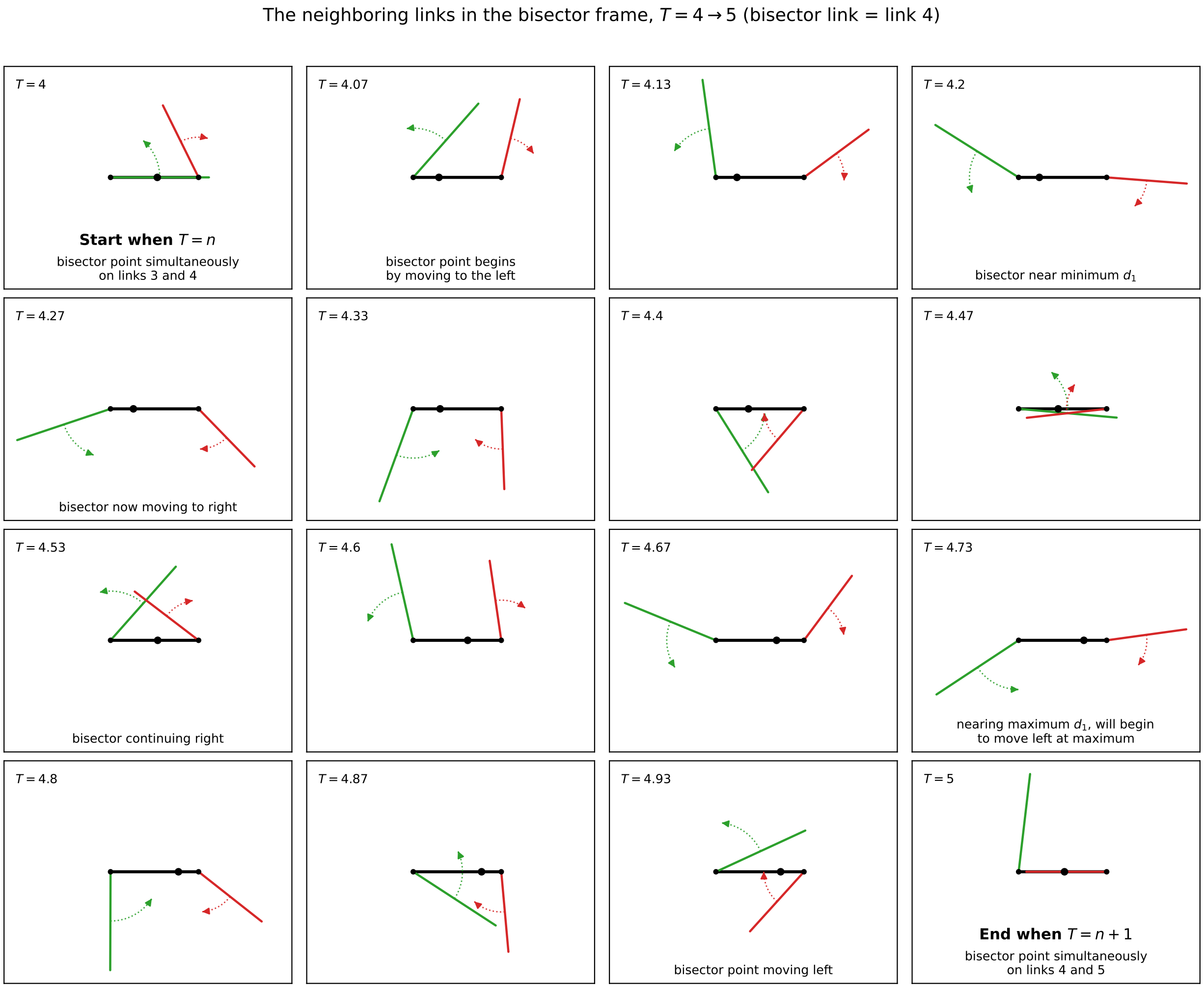

Figure 23: The two neighboring links in the bisector frame as T sweeps from 4 to 5 ($\sigma = \frac{1}{2}$), sixteen snapshots read left to right from the top left ($T = 4$) to the bottom right ($T = 5$). In every panel the bisector link, link $m = 4$ of the forward chain, is held fixed along the real axis from 0 to 1 (black); link 3 (green) hangs off joint 4 at the origin and link 5 (red) off joint 5 at the point 1, both scaled relative to the unit bisector link. The dotted curved arrow on each neighbor runs from the middle of the link to the middle of that link as the next panel depicts it, tracing the motion the link takes between panels. The black dot on the bisector link marks the bisector point, at the fraction $\lceil T \rceil^{\sigma} d_1$ along the unit link. As T advances, each neighbor revolves around the stationary bisector link approximately twice. At $T = 4$ (Start) the green link is folded back onto the bisector link (joint angle 9π, an odd multiple of π), so the bisector point lies on links 3 and 4 simultaneously: that is the handoff instant at which link 4 became the bisector link. At $T = 5$ (End) it is the red link that folds back (joint angle 11π), the bisector point lies on links 4 and 5 simultaneously, and link 5 takes over as the new bisector link. Generated by `fig_bisector_frame.py`.

This periodicity begs for a clock that ticks once per handoff, and that is where $I(T)$ came from: we sought a formula $t = I(T)$ whose input T passes through an integer at exactly the moments the bisector point moves from one link to the next. Two facts about the joint angle between the outgoing and incoming bisector links pin the formula down.

1. *From the zeta function.* Consecutive summands of Σ_1 differ in phase by $t \ln \frac{n+1}{n}$, so the turn at the joint where link $T-1$ meets link T (interpolating the index continuously) is

$$j_\angle(T) \;=\; t\,\log\!\Big(\tfrac{1}{T}+1\Big). \tag{85}$$

2. *From the observed motion.* At a handoff the two links fold back onto one another, so the joint angle there is an *odd multiple of* π (hence a summand π); and the neighboring link revolves around the stationary bisector link *twice* per unit of the index, so the odd multiple advances by two with each unit of T (hence the factor 2). Interpolating between handoffs, the joint angle at index T is

$$j_\angle(T) \;=\; \pi\,(2T+1). \tag{86}$$

Setting (85) equal to (86) and solving for t yields

$$\pi\,(2T+1) \;=\; t\,\log\!\Big(\tfrac{1}{T}+1\Big) \qquad\Longrightarrow\qquad t \;=\; \frac{\pi\,(2T+1)}{\log\!\big(\tfrac{1}{T}+1\big)} \;=\; I(T), \tag{87}$$

which is the mapping adopted in §3: under $t = I(T)$ the bisector point resides on link $\lfloor T \rfloor$, and T is an integer exactly when the bisector point moves from one link to the next.

Remark 10.1. Taylor-expanding the reciprocal logarithm in (87), $1/\log(1+\frac{1}{T}) = T + \frac{1}{2} - \frac{1}{12T} + \frac{1}{24T^2} - \cdots$, and collecting orders of T gives a family of invertible polynomial approximations to $I(T)$; keeping successively more terms:

terms	$I(T)\;\approx$	solved for T
2	$2\pi\big(T^2+T\big)$	$T = -\frac{1}{2} + \sqrt{\frac{t}{2\pi}+\frac{1}{4}}$
3	$2\pi\big(T^2+T+\frac{1}{6}\big)$	$T = -\frac{1}{2} + \sqrt{\frac{t}{2\pi}+\frac{1}{12}}$
4	$2\pi\big(T^2+T+\frac{1}{6}-\frac{1}{180T^2}\big)$	$T \approx -\frac{1}{2} + \sqrt{\frac{t}{2\pi}+\frac{1}{12}+\frac{\pi}{90t}}$

(The $1/T$ term of the expansion cancels, so the fourth term is already $O(T^{-2})$.) The 2-term form misses $I(T)$ by a constant $\approx \pi/3$; the 3-term form is accurate to $O(T^{-2})$ and the 4-term form to $O(T^{-3})$. At $T = 20$ their errors in T after inversion are 4×10^{-3}, 3×10^{-7} and 4×10^{-11} respectively. The first two rows invert exactly (quadratics); the 4-term row is inverted by substituting the 3-term inverse into the small T^{-2} correction. Comparing with Siegel's cutoff (9) shows $\sqrt{t/2\pi} \approx T + \frac{1}{2}$, so Siegel's m steps *near* the *half*-integers of T while ours steps *exactly at* the integers.

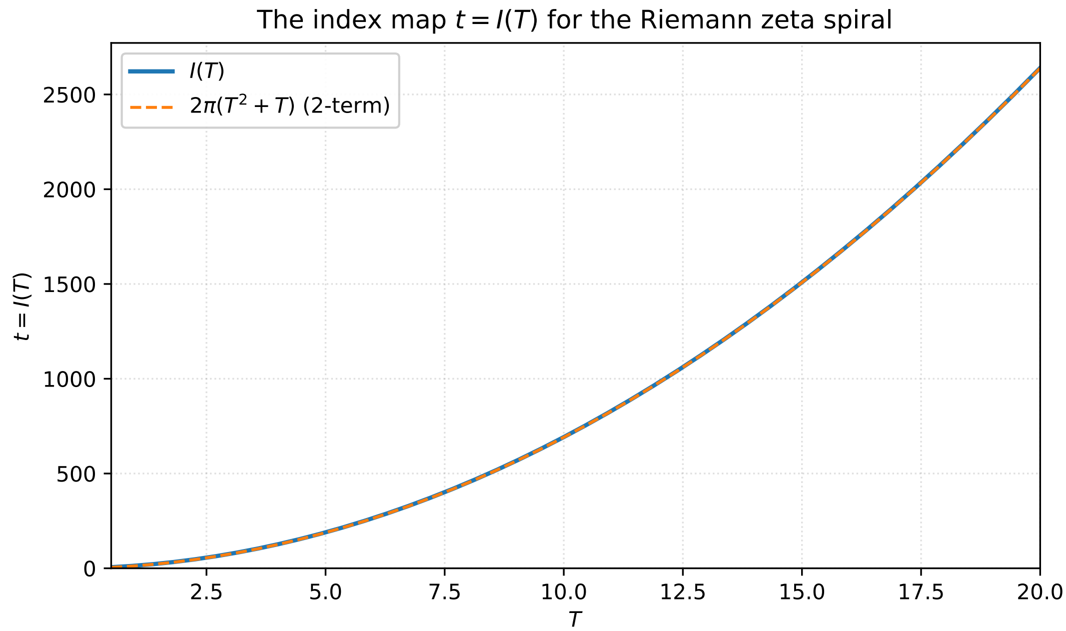

Figure 24: The index map $t = I(T)$ of (10) (solid), together with the two-term approximation $2\pi(T^2 + T)$ from the Taylor table above (dashed). Generated by `fig_IT.py`.

We stress that the polynomial forms of the table are conveniences only: everywhere in this paper $t = I(T)$ is used *exactly*. The defining property of the mapping, that the neighboring link folds back onto the bisector link *exactly* as T passes an integer and the bisector point transfers from one link to the next exactly at that instant, holds only for the exact $I(T)$. Under any approximation, including the one whose inverse is essentially Siegel's $\sqrt{t/2\pi}$, the fold instants drift off the integers by the corresponding error in the table: the clock still ticks once per handoff, but never precisely on the hour.

Remark 10.2 (comparing the classical t rescaling so that zeros have unit mean spacing, and $I(T)$)**.** In random-matrix and zeta-statistics work the zero-counting function, which maps each zero $\gamma \mapsto N(\gamma)$ (or uses $\tilde{\gamma} = \frac{\gamma}{2\pi} \log \frac{\gamma}{2\pi}$, the leading term of (131)), leaves the zeros with unit mean spacing, which lets one compare zero statistics at different heights. This mapping is called "unfolding the zeros". Our $I(T)$ is a change of variable in the same spirit, a reparameterization of the height t, but it runs in the opposite direction, compressing ever more zeros into each unit interval of the index T. For example, the interval $[5, 6]$ carries 42 zeros, $[10, 11]$ carries 106, $[20, 21]$ carries 256, and $[50, 51]$ carries 802. Unfolding the zeros flattens the count to unit spacing; the index does the reverse.

10.2 $I(T)$ functions for other L-functions besides the zeta function

The same change-of-variable idea extends from ζ to Dirichlet L-functions. On the critical line the partial sums of

$$L(s, \chi_p) = \sum_{n=1}^{\infty} \chi_p(n)\, n^{-s} \tag{88}$$

form a spiral just as for zeta, but now the character $\chi_p(n) = \left(\frac{n}{p}\right)$ (Legendre symbol, for a prime modulus p) inserts periodic sign flips and, when $p \mid n$, zero-length *phantom* links. Counting only the nonzero links as *links*, there are $p-1$ links and one phantom in every block of p summands, and the natural index T is the link index (phantoms receive no T).

Watching these chains as t varies, we noticed that the bisector point still moves from one link to the next, just as for zeta, but not always at folds: sometimes the bisector point moves from link to link at a *joint*, as the bisector line sweeps through it. So L-functions have other event types that zeta does not. An *event* is the instant the bisector point moves from one link to the next. There are two ways this can happen: either the two links fold back onto one another (the turning angle at their common joint is an odd multiple of π), so the bisector point slides across the joint from the old link onto the new one; or the bisector line sweeps through a joint, carrying the crossing from one link to its neighbor. For zeta only the first kind occurs: every handoff of §9.1 is a folded event, once per unit of T, and there are no bisector events. For $L(s, \chi_p)$ both kinds occur, recurring with period $\varphi(p) = p - 1$ in the integer index T:

- *folded* events, where the turning angle at the current joint is $\pm\pi$ (the chain reverses), at $T \equiv 0 \pmod{p-1}$;
- *bisector* events, where the joint lies on the perpendicular bisector of the segment from 0 to $L(s, \chi_p)$, at the remaining residues modulo $p-1$.

For $p = 3$ one has $\varphi(3) = 2$: even T are folds and odd T are bisectors. The map $t = I_p(T)$ is defined so that integer T lands at the t-value of its event.

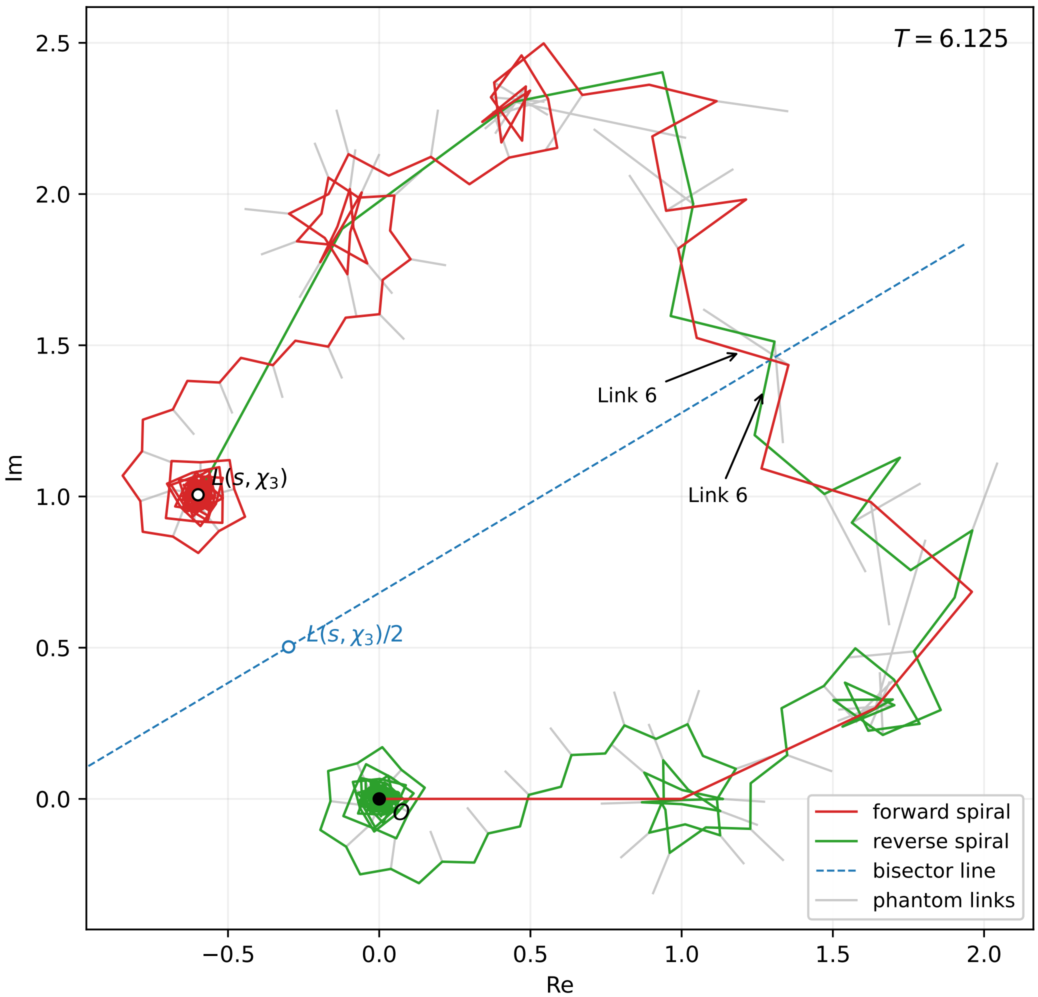

Figure 25: The two partial-sum spirals of $L(s, \chi_3)$ at $T = 6.125$ on the critical line ($t = I_3(T) \approx 176.13$), drawn with $\lfloor 2T(T+1)p \rfloor = 261$ summands. The forward chain (red) starts at the origin O and winds into its spiral around the analytic value $L(s, \chi_3)$ (open dot); the reverse chain (green) is its mirror image across the bisector line (dashed blue), the perpendicular bisector of the segment from O to $L(s, \chi_3)$ through the labeled midpoint $L(s, \chi_3)/2$, so its tail sits on $L(s, \chi_3)$ and its head lands at O. Every third summand has $\chi_3(n) = 0$: these zero-length phantom links are drawn light grey as the connectors the chain skips. The labeled link 6 of each chain is the current bisector link: the two chains cross on it, on the bisector line. Generated by `fig_L3_spirals.py`.

Because only a fraction $(p-1)/p$ of the summands are links, the smooth *base* formula generalizing (10) is

$$I_p^{\text{base}}(T) \; = \; \frac{4\pi T}{(p-1)\,\log \dfrac{pT + (p-1)}{pT - (p-1)}} \,. \tag{89}$$

Asymptotically $I_p(T) \sim \dfrac{2p}{(p-1)^2}\,\pi T^2$: the leading coefficient is $\frac{3}{2}\pi$ for $p = 3$ and $\frac{5}{8}\pi$ for $p = 5$, against 2π for zeta's own $I(T)$. (Zeta is not the $p = 1$ member of this family, since the factor $2p/(p-1)^2$ diverges there; for zeta every summand is a link and there are no phantoms.) The base does not run through the middle of the events: it touches I_p at the folds and lies below it in between, missing a one-signed bulge of period $p-1$ imposed by the character. For $p=3$ the contact is exact, since at even T the gate of (92) vanishes and $c=2$ turns (91) into the base; for $p=5$ the fitted correction returns to zero at $T \equiv 0 \pmod 4$ to within 2×10^{-3}. One restores the bulge by writing

$$I_p(T) \;=\; I_p^{\text{base}}(T) + C_p(T). \tag{90}$$

For $p=3$ the character is $[1,-1,0]$. The base specializes to $I_3^{\text{base}}(T) = 2\pi T\big/\log\frac{3T+2}{3T-2}$, and the correction is built from the character-modulated map

$$c = \frac{3+\cos(\pi T)}{2}, \qquad I_3^{\text{geom}}(T) = \frac{c\,\pi\,T}{\log\dfrac{3T+c}{3T-c}}, \tag{91}$$

together with a small odd-T offset $\alpha(T) = 0.390914 + 0.01712\,T^{-2}$ fitted to event scans:

$$I_3(T) = I_3^{\text{geom}}(T) + \tfrac{1}{2}\big(1-\cos\pi T\big)\,\alpha(T). \tag{92}$$

(Equivalently $I_3 = I_3^{\text{base}} + C_3$ with $C_3 = I_3^{\text{geom}} - I_3^{\text{base}} + \frac{1}{2}(1-\cos\pi T)\,\alpha$; the gate vanishes on even T, so $C_3 = 0$ on folds.) On odd $T = 1,\ldots,99$ the residual between scanned event times and (92) has RMSE about 8×10^{-6}.

For $p=5$ the same pattern gives $I_5^{\text{base}}(T) = \pi T\big/\log\frac{5T+4}{5T-4}$ and a fitted period-4 correction

$$C_5(T) = 0.6206 - 0.6576\cos\tfrac{\pi T}{2} + 0.0369\cos\pi T + \tfrac{0.169}{T^{3.12}}. \tag{93}$$

Figure 26 compares $I(T)$, $I_3(T)$, and $I_5(T)$ on a common range of T.

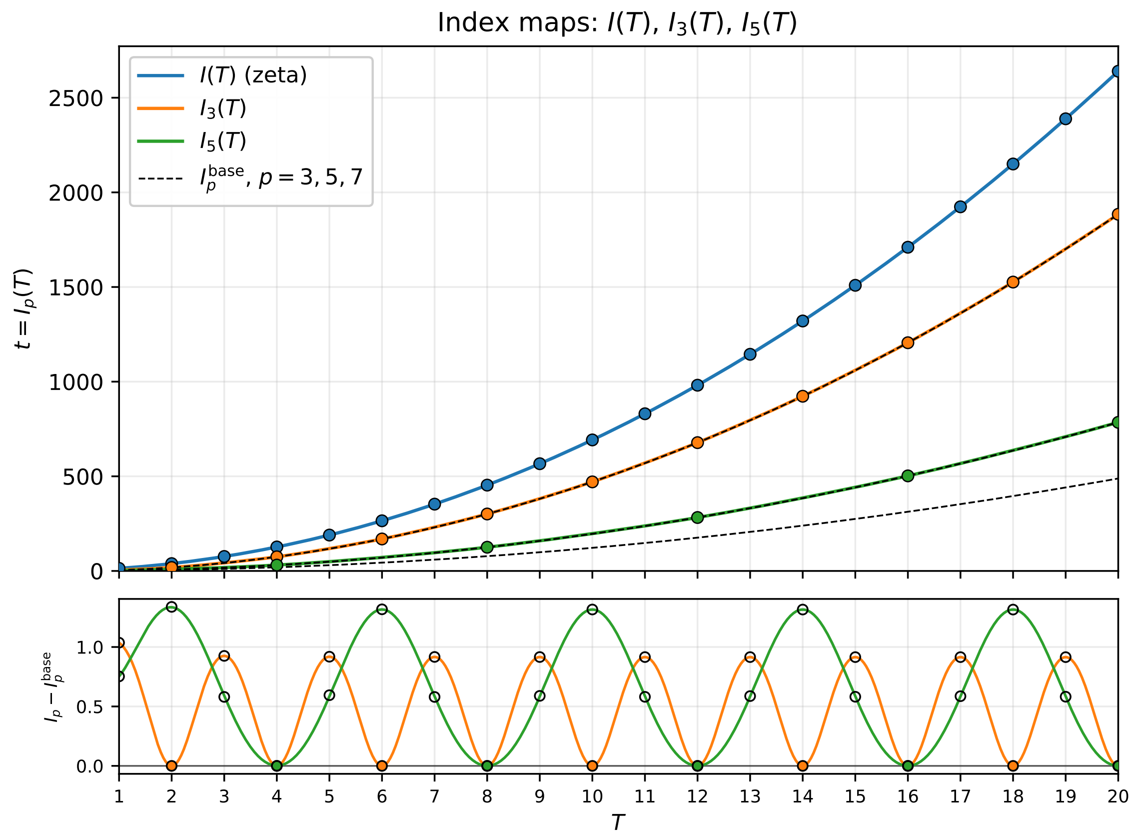

Figure 26: Top: index maps on $T \in [1, 20]$: zeta's $I(T)$ (blue), $I_3(T)$ for $p = 3$ (orange), and $I_5(T)$ for $p = 5$ (green). Dots mark the fold events, at $T \equiv 0 \pmod{p-1}$: every integer on the zeta curve, every other integer on I_3, and every fourth integer on I_5. Leading growth is quadratic with coefficients 2π, $\frac{3}{2}\pi$, and $\frac{5}{8}\pi$ respectively. Thin dashed black: the smooth base maps (89) for $p = 3, 5, 7$. The $p = 3$ and $p = 5$ bases are hidden under I_3 and I_5 at this scale; the $p = 7$ base is the lowest curve, drawn for comparison, no I_7 having been fitted here. Bottom: the character bulge on its own scale, $I_p - I_p^{\text{base}}$, which the top panel cannot resolve because it is of order 1 against values in the thousands. It is one-signed, returning to zero at the folds (dots) and rising to ≈ 0.92 between them for $p = 3$ and ≈ 1.33 for $p = 5$, and its amplitude barely decays: the $p = 3$ peaks are 0.9253, 0.9183, 0.9153 at $T = 3, 5, 11$. So the correction is a fixed additive offset, not a fixed relative one. Open black circles are the numerically scanned events at every integer T, measured against the same base: the folds sit on zero, the intermediate events ride the bulge, and all of them follow the modeled curves to within 1.7×10^{-5} for $p = 3$ and 1.0×10^{-2} for $p = 5$ on $T \leq 20$. Generated with the Dirichlet L-function companion note.

11 The yin and yang curves

As T increases the bisector point travels back and forth along the bisector link, and when T passes an integer it moves to the next link of the chain. It does so without any jump: the handoff happens exactly at the instant when the outgoing and incoming links fold back onto one another and momentarily occupy the same line in the plane (when $\sigma = \frac{1}{2}$ the two bisector links are then coincident; for $\sigma \neq \frac{1}{2}$ they are parallel). The path of the bisector point is therefore continuous both as a curve in the complex plane and as a position along the one-dimensional link structure.

The previous section watched the two links *neighboring* the bisector link in the bisector frame. Now watch the *reverse bisector link* in that same frame: holding the forward bisector link stationary while T increases, the reverse bisector link revolves around it, and each end of the revolving link follows a path reminiscent of a teardrop, the pair of paths together resembling a yin-yang (Figure 29). We call the paths of the two ends the *yin* and *yang* curves, and this section develops them: first their formulas, then the derivation of R_{1ps} and R_{2ps} from them, and finally their behavior as $T \to \infty$, where a familiar function makes an unexpected appearance.

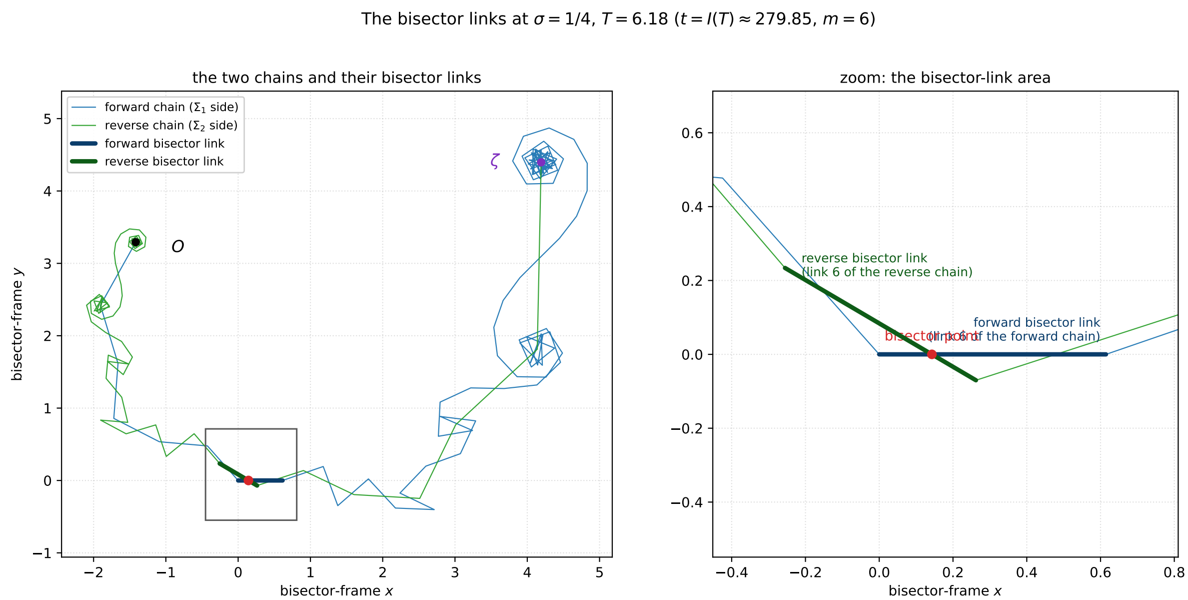

Figure 27: The two bisector links at $\sigma = \frac{1}{4}$, $T = 6.18$ ($t = I(T) \approx 279.85$, $m = 6$). Both panels are drawn in the coordinate frame of the *forward bisector link*: the whole picture is translated and rotated so that this link lies along the positive x-axis, with no rescaling. All lengths are true, so the link runs from 0 to $7^{-1/4} \approx 0.615$. *Left:* the forward chain (blue, starting at the world origin O) and the reverse chain (green, ending at ζ), each drawn out to the link nearest its spiral center; link 6 of each chain (the forward and reverse bisector links) is overdrawn with a thick stroke, and the red dot where they cross is the bisector point. *Right:* zoom into the gray window: the two bisector links crossing at the bisector point, which sits on the x-axis at distance d_1 from the origin. The bisector frame of this section additionally rescales this link to the unit interval $[0, 1]$ and watches the thick reverse link revolve around it. Generated by `fig_yinyang_spirals.py`.

The motivation is the question left open in §9.1: the two bisector links cross, but *where*? The answer matters because, by (65), the crossing (the bisector point) sits at distance d_1 from the joint Σ_1 along the forward bisector link and at distance d_2 from the joint $\Sigma_1 + R$ along the reverse

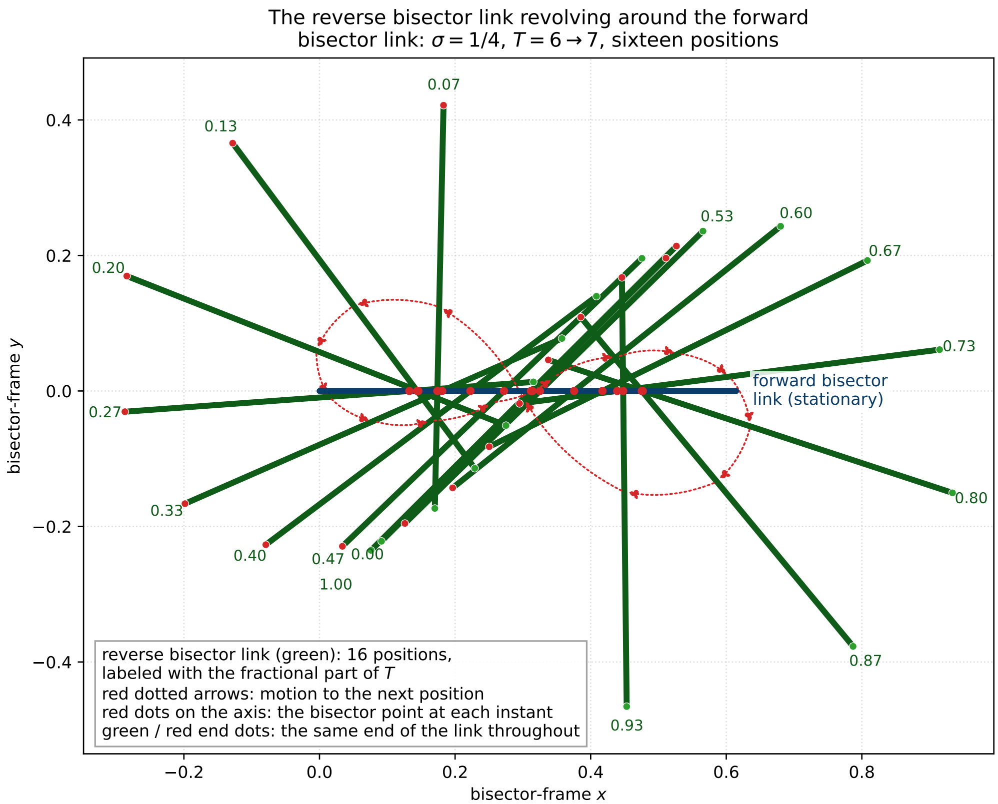

Figure 28: The two bisector links in the frame of the forward bisector link as T sweeps from 6 to 7 ($\sigma = \frac{1}{4}$, the frame of Figure 27, lengths unscaled). The forward bisector link is stationary (the single thick dark-blue segment on the x-axis) while the reverse bisector link (green, same thickness) is drawn at sixteen equally spaced values of T, each position labeled on its outer end with the fractional part of T; over the sweep it revolves fully around the stationary link, and at $T = 7$ (labeled 1.00) it returns nearly to its $T = 6$ position. The red dotted arrows run from the middle of each position to the middle of the next, tracing the motion. The small green and red dots mark the two ends of the revolving link consistently: green is always the joint-m end and red the joint-(m+1) end; they trace the yin and yang curves of Figure 29, in the same colors. Each red dot on the x-axis is the bisector point at one instant: the crossing of that position with the x-axis, at $(d_1, 0)$. Near the parallel instants (fractional part of T near $\frac{1}{4}$ and $\frac{3}{4}$) the revolving link is nearly parallel to the stationary one and the crossing races along the axis (the poles of d_1 of §8.4). Generated by `fig_bisector_sweep.py`.

bisector link. If we can *compute* the crossing, we obtain d_1 and d_2, and with them a new formula for ζ that uses the fractional summands in place of the usual remainders, all without evaluating ζ or either partial sum.

11.1 The formulas for the yin and yang curves

In the bisector frame the forward bisector link is pinned to $[0,1]$, so the only question is where the *reverse* bisector link is. In the world frame its chain is anchored at ζ, which is both moving and precisely the quantity we do not wish to assume known; worse, the reverse bisector link also moves relative to its own anchor. The resolution is that the reverse chain's joint m sits at $\Sigma_1 + R$: relative to the forward joint Σ_1 it is displaced by exactly R, and R has its own integral representation (16), independent of any evaluation of ζ. Applying the frame map

$$z \;\longmapsto\; (z - \Sigma_1)\,\lceil T\rceil^{\,\sigma+iI(T)} \tag{94}$$

(translate joint m to the origin, then divide by the link vector $\lceil T\rceil^{-\sigma-iI(T)}$), the two ends of the reverse bisector link land at

$$\begin{aligned} Y_{in1}(\sigma,T) \;&=\; R\,\lceil T\rceil^{\,\sigma+iI(T)}, &(95)\\ Y_{ang1}(\sigma,T) \;&=\; Y_{in1} \;-\; \chi\,\lceil T\rceil^{\,-1+2(\sigma+iI(T))}, &(96)\end{aligned}$$

the first being the image of $\Sigma_1 + R$ and the second the image of the far end $\Sigma_1 + R - \chi\lceil T\rceil^{\,\sigma+iI(T)-1}$. As T sweeps, these two endpoints orbit the stationary unit link along the yin and yang curves of Figure 29.

Because the frame map scales the bisector link to unit length, positions along it are *fractions*: the crossing we are about to compute is the fraction of the way along the link at which the bisector point sits, and multiplying that real fraction by the (unscaled) link vector recovers the fractional summand itself. The actual distance from the joint, the fraction times the true link length $\lceil T\rceil^{-\sigma}$, is the weight d_1.

The reverse side has its own frame and its own pair of curves. Pinning the *reverse* bisector link to $[0,1]$ instead (translate its joint $\Sigma_1 + R$ to the origin and divide by its link vector $-\chi\lceil T\rceil^{\,\sigma+iI(T)-1}$), the roles swap: now the forward bisector link revolves, its near end tracing

$$Y_{in2}(\sigma,T) = \frac{R\,\lceil T\rceil^{\,1-\sigma}}{\chi\,e^{\,i\omega}}, \qquad Y_{ang2}(\sigma,T) = Y_{in2} - \frac{\lceil T\rceil^{\,1-2\sigma}}{\chi\,e^{\,2i\omega}}, \tag{97}$$

with Y_{in2} the image of the joint Σ_1 and Y_{ang2} the image of the far end $\Sigma_1 + \lceil T\rceil^{-\sigma-iI(T)}$, and with $\omega = \omega(T) = I(T)\log\lceil T\rceil$ as always.

The next two subsections derive R_{1ps} from (Y_{in1}, Y_{ang1}) and R_{2ps} from (Y_{in2}, Y_{ang2}). Each derivation is in two parts: first the modulus, found from the axis crossing; then the direction, which is simply that of the next summand.

11.2 Derivation of R_{1ps} from the yin and yang functions

First, the modulus. From its joint, how far along the forward bisector link do the two links intersect? This distance is $|R_{1ps}|$, because R_{1ps} is the fractional part of summand $\lceil T\rceil$ of Σ_1, and that summand *is* the forward bisector link. In the frame the link lies along the real axis, so the intersection is the point where the segment from Y_{in1} to Y_{ang1} crosses the axis. By the standard segment–axis intersection formula that crossing is

$$Y_{in1} \;-\; \frac{\Im\,Y_{in1}}{\Im\,Y_{ang1} - \Im\,Y_{in1}}\,\bigl(Y_{ang1} - Y_{in1}\bigr), \tag{98}$$

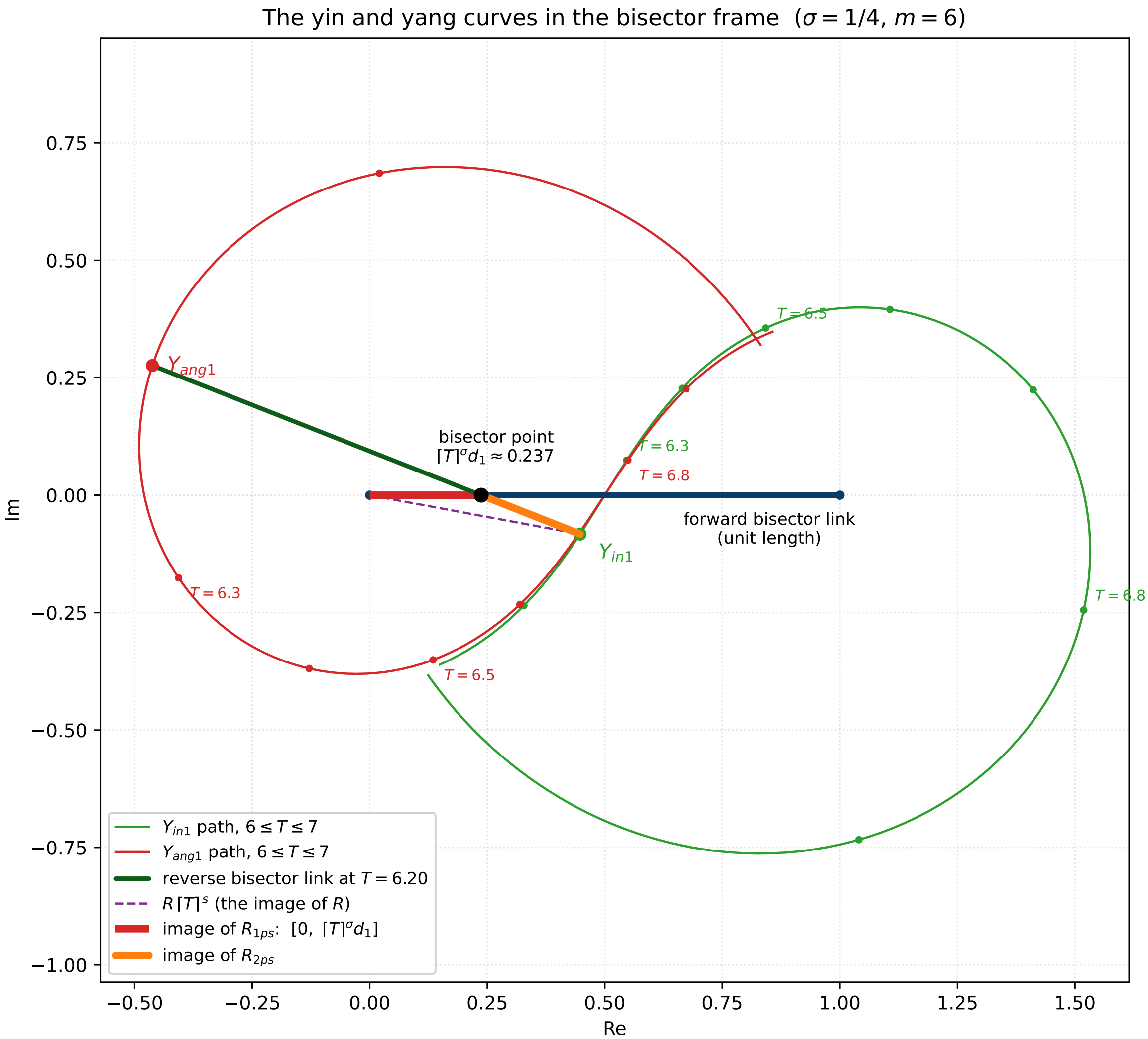

Figure 29: The yin and yang curves in the bisector frame, $\sigma = \frac{1}{4}$, $m = 6$. The forward bisector link is pinned to the unit interval $[0, 1]$ (dark blue). As T sweeps $6 \leq T \leq 7$, the near end of the reverse bisector link traces the yin curve Y_{in1} (green, one dot every 0.1 in T) and the far end traces the yang curve Y_{ang1} (red); the two teardrop paths together resemble a yin-yang. The reverse bisector link itself is drawn at $T = 6.20$ (dark green): it crosses the real axis at the bisector point (black dot), at the fraction $\lceil T\rceil^{\sigma} d_1 \approx 0.237$ of the way along the unit link. The dashed violet vector from 0 to Y_{in1} is the in-frame image of Siegel's remainder R; the crossing splits it into the images of R_{1ps} (red, the piece of the unit link from 0 to the bisector point) and R_{2ps} (orange, from the bisector point to Y_{in1}). Generated by `fig_yinyang.py`.

a real number. Substituting (95)–(96) and cancelling $Y_{ang1} - Y_{in1} = -\chi\lceil T\rceil^{\,-1+2(\sigma+iI(T))}$ from the ratio,

$$(98) \;=\; R\,\lceil T\rceil^{\,\sigma+iI(T)} \;-\; \frac{\Im\!\big(R\,\lceil T\rceil^{\,\sigma+iI(T)}\big)}{\Im\!\big(\chi\,\lceil T\rceil^{\,-1+2(\sigma+iI(T))}\big)}\,\chi\,\lceil T\rceil^{\,-1+2(\sigma+iI(T))}. \tag{99}$$

Writing $\lceil T\rceil^{\,\sigma+iI(T)} = \lceil T\rceil^{\sigma} e^{i\omega}$ and expanding the imaginary parts in components,

$$(98) \;=\; R\,\lceil T\rceil^{\,\sigma+iI(T)} \;-\; \lceil T\rceil^{\,1-\sigma}\,\frac{\Re R\,\sin\omega + \Im R\,\cos\omega}{\Re\chi\,\sin 2\omega + \Im\chi\,\cos 2\omega}\,\chi\,\lceil T\rceil^{\,-1+2(\sigma+iI(T))}, \tag{100}$$

and finally, writing $\varphi = \arg R$ and $\psi = \arg\chi$ and collapsing with the sine difference identity $\sin(2\omega+\psi)\cos(\omega+\varphi) - \cos(2\omega+\psi)\sin(\omega+\varphi) = \sin(\omega-\varphi+\psi)$, the whole expression reduces to

$$(98) \;=\; \lceil T\rceil^{\sigma}\,|R|\,\frac{\sin\!\big(\omega-\varphi+\psi\big)}{\sin\!\big(2\omega+\psi\big)} \;=\; \lceil T\rceil^{\sigma}\,d_1. \tag{101}$$

The reader will recognize the sine quotient: it is, symbol for symbol, the definition (19) of the weight d_1. *This is where that definition came from.* The crossing (101) is the "fractional amount" column of Table 1, the fraction of the way along the unit link at which the reverse link crosses, and dividing out the frame's scaling $\lceil T\rceil^{\sigma}$ leaves the actual distance from the joint, d_1.

Second, the direction. The forward bisector link points along the next summand of Σ_1: extending the summation cutoff by one,

$$\sum_{n=1}^{m+1} n^{-\sigma-iI(T)} \;-\; \sum_{n=1}^{m} n^{-\sigma-iI(T)} \;=\; \lceil T\rceil^{\,-\sigma-iI(T)} \tag{102}$$

(for non-integer T, so $m+1 = \lceil T\rceil$). Multiplying the fraction (101) by this link vector undoes the frame map and yields the fractional summand itself:

$$R_{1ps} \;=\; \Big(\lceil T\rceil^{\sigma} d_1\Big)\cdot\lceil T\rceil^{\,-\sigma-iI(T)} \;=\; d_1\,\lceil T\rceil^{\,-iI(T)}, \tag{103}$$

which is definition (17).

Other forms. Two equivalent ways to write R_{1ps} fall out of the same computation. Subtracting the crossing from Y_{in1} instead of building it up from 0 gives

$$R_{1ps} \;=\; R \;-\; |R|\,\frac{\sin(\omega+\varphi)}{\sin(2\omega+\psi)}\,e^{\,i(\omega+\psi)} \tag{104}$$

(the subtracted term is exactly R_{2ps}, anticipating Theorem 4.1), and keeping the imaginary parts unexpanded gives the compact

$$R_{1ps} \;=\; R \;-\; \chi\,e^{\,i\omega}\,\frac{\Im\!\big(R\,e^{\,i\omega}\big)}{\Im\!\big(\chi\,e^{\,2i\omega}\big)}. \tag{105}$$

11.3 Derivation of R_{2ps} from the yin and yang functions

The same procedure in the reverse frame, using (Y_{in2}, Y_{ang2}) of (97), locates the crossing of the (now revolving) forward bisector link with the real axis at the fraction

$$\frac{\lceil T\rceil^{\,1-\sigma}}{|\chi|}\,|R|\,\frac{\sin\!\big(\omega+\varphi\big)}{\sin\!\big(2\omega+\psi\big)} \;=\; \frac{\lceil T\rceil^{\,1-\sigma}}{|\chi|}\,d_2 \tag{106}$$

of the way along the unit reverse link, again the "fractional amount" column of Table 1, and again the sine quotient is, symbol for symbol, the definition (20) of d_2: this is where *it* came from. (The two normalized fractions are nearly symmetric: since $|\chi(\sigma+it)| = (t/2\pi)^{\frac{1}{2}-\sigma}\big(1+O(1/t)\big)$ and $\sqrt{t/2\pi} \approx T+\frac{1}{2}$, the prefactor $\lceil T\rceil^{\,1-\sigma}/|\chi|$ is asymptotically $\lceil T\rceil^{\sigma}$, matching the forward side.) Multiplying by the reverse link vector $\chi\,\lceil T\rceil^{\,\sigma+iI(T)-1}$ undoes the frame map:

$$R_{2ps} \;=\; \frac{\lceil T\rceil^{\,1-\sigma}}{|\chi|}\,d_2\cdot\chi\,\lceil T\rceil^{\,\sigma+iI(T)-1} \;=\; d_2\,\lceil T\rceil^{\,iI(T)}\,\frac{\chi}{|\chi|}, \tag{107}$$

which is definition (18). As with the forward side there are equivalent rewritings, for example

$$R_{2ps} \;=\; R \;-\; |R|\,e^{-i\omega}\,\frac{\sin\!\big(\omega-\varphi+\arg\chi(1-\sigma+iI(T))\big)}{\sin\!\big(2\omega+\arg\chi(1-\sigma+iI(T))\big)} \tag{108}$$

and

$$R_{2ps} \;=\; R \;-\; e^{-i\omega}\,\frac{\Im\!\Big(\dfrac{R}{\chi(1-\sigma+iI(T))\,e^{\,i\omega}}\Big)}{\Im\!\Big(\dfrac{1}{\chi(1-\sigma+iI(T))\,e^{\,2i\omega}}\Big)}, \tag{109}$$

where the reflected factor may be used interchangeably in the argument because $\arg\chi(1-\sigma+iI(T)) = \arg\chi(\sigma+iI(T))$: from $\chi(s)\chi(1-s)=1$ and the Schwarz reflection $\chi(\bar{s}) = \overline{\chi(s)}$ one gets $\chi(1-\sigma+it) = 1/\overline{\chi(\sigma+it)}$, which has the same argument as $\chi(\sigma+it)$ and reciprocal modulus.

The poles cancel in the sum. The two weights share the denominator $\sin(2\omega+\psi)$, and §8.4 located the isolated heights at which it produces genuine poles. In the *sum* d_1+d_2 those poles cancel: adding the numerators with the sum-to-product identity $\sin(\omega-\varphi+\psi)+\sin(\omega+\varphi) = 2\sin\!\big(\omega+\frac{\psi}{2}\big)\cos\!\big(\varphi-\frac{\psi}{2}\big)$ and writing $\sin(2\omega+\psi) = 2\sin\!\big(\omega+\frac{\psi}{2}\big)\cos\!\big(\omega+\frac{\psi}{2}\big)$, the offending factor divides out:

$$d_1+d_2 \;=\; |R|\,\frac{\cos\!\big(\varphi-\frac{\psi}{2}\big)}{\cos\!\big(\omega+\frac{\psi}{2}\big)}. \tag{110}$$

The zeros of the remaining denominator $\cos(\omega+\frac{\psi}{2})$ are matched by zeros of the numerator (at those instants R is parallel to the common link direction, forcing $\varphi \equiv -\omega \pmod{\pi}$), so every singularity of (110) is removable: the sum is smooth across all the poles of its two terms. Figure 30 shows this at $\sigma = 0.1$.

This is how the definitions of §4 were found: the sine expressions (19)–(20) are nothing other than the real-axis crossings of the yang segments, read off in the two bisector frames. Only afterwards was Theorem 4.1 proved, confirming that the two fractional summands so constructed add back to Siegel's R exactly. Note the economy of the construction: the bisector point, d_1, d_2, and both fractional summands are obtained from R and χ alone. Numerically we computed R to high accuracy by Kuznetsov's method (§7.1), which is how the pictures, and then the identity, were checked.

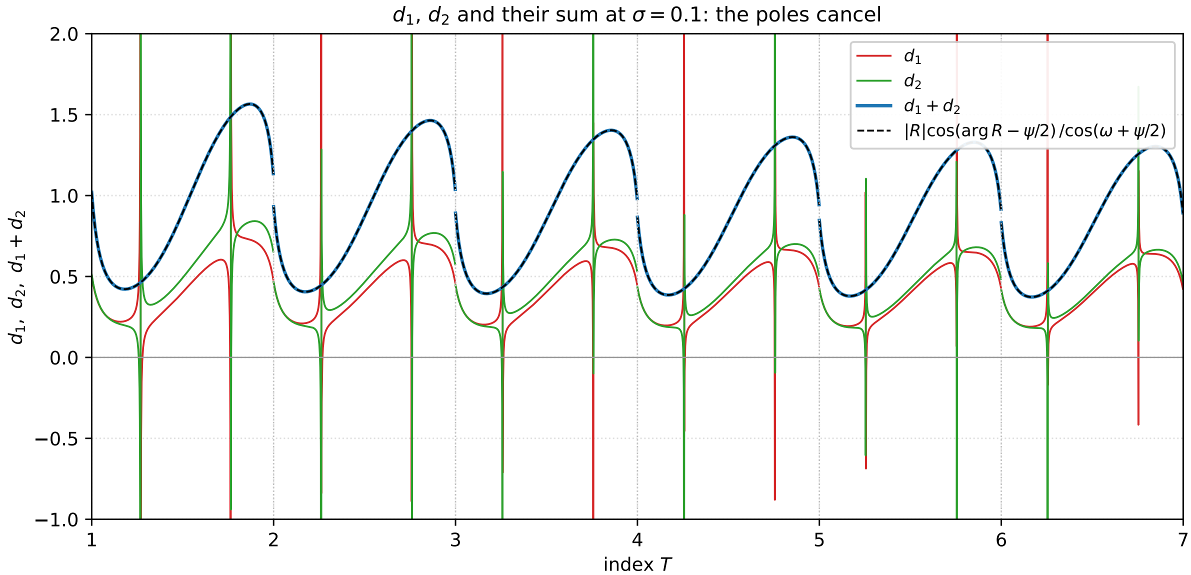

Figure 30: d_1 (red), d_2 (green) and their sum (blue) at $\sigma = 0.1$, $1 \le T \le 7$. Twice per unit of T the individual weights blow up at the parallel-link poles of §8.4, but the poles are equal and opposite and the sum passes smoothly through every one of them; the black dashed curve is the closed form (110), indistinguishable from the blue sum (they agree to machine precision on the plotted grid). Generated by `fig_d1_d2_sum.py`.

11.4 Comparison of yin and yang to Siegel's integral result

Siegel's 1932 paper [18] evaluates the integral in his equation 5 to what he calls the "desired result"

$$F(t) \;=\; \frac{1}{1-e^{-2\pi i t}} \;-\; \frac{e^{\pi i t^2}}{e^{\pi i t}-e^{-\pi i t}}, \tag{111}$$

which can also be written

$$F(t) = \frac{e^{2\pi i t} - e^{\pi i t(t+1)}}{e^{2\pi i t}-1} = \left(1 - e^{\pi i (t^2 - t)}\right) \sum_{n=0}^{\infty} e^{-2\pi i n t}, \tag{112}$$

the last form using the geometric series $1/(1-e^{-2\pi i t}) = \sum_{n\ge 0} e^{-2\pi i n t}$. Its complex conjugate has the compact form

$$\overline{F(t)} = \frac{e^{-\pi i t^2} - e^{-\pi i t}}{2\, i\, \sin(\pi t)}. \tag{113}$$

Remark 11.1 (regarding F as the source of Siegel's $f(s)$)**.** F is not merely adjacent to the function f of (46); it is what f is built from. In his §3 Siegel multiplies his equation 5, that is (111), by u^{-s} and integrates along the bisector of the first quadrant, and under that operation the two terms of F separate into the two halves of zeta. The pole term $1/(1-e^{-2\pi i u})$ yields $-(2\pi)^{s-1} e^{i\pi(1-s)/2}\, \Gamma(1-s)\, \zeta(1-s)$, while the Gaussian term yields f itself:

$$f(s) \;=\; \left(e^{\pi i s} - 1\right) \int_0^{e^{i\pi/4}\infty} \frac{u^{-s}\, e^{\pi i u^2}}{e^{\pi i u} - e^{-\pi i u}}\, \mathrm{d}u \qquad (\sigma < 0), \tag{114}$$

the restriction $\sigma < 0$ being needed only for convergence at the endpoint $u = 0$; f is continued analytically from there, and (46) itself converges for every s. The same separation can be read off (113): since $1/(1 - e^{2\pi it}) = -e^{-\pi it}/(2i \sin \pi t)$, the numerator term $e^{-\pi it^2}$ of $\overline{F}$ is the $f(s)$ side and $-e^{-\pi it}$ is the $\zeta(1 - s)$ side, so the two lobes carry the two halves in their numerator. The yin-yang function is therefore the common source of both $\Sigma_1 + R_{1ak}$ of (45) and its reflected counterpart, which is one route into the convergence proof left open in §11.6.

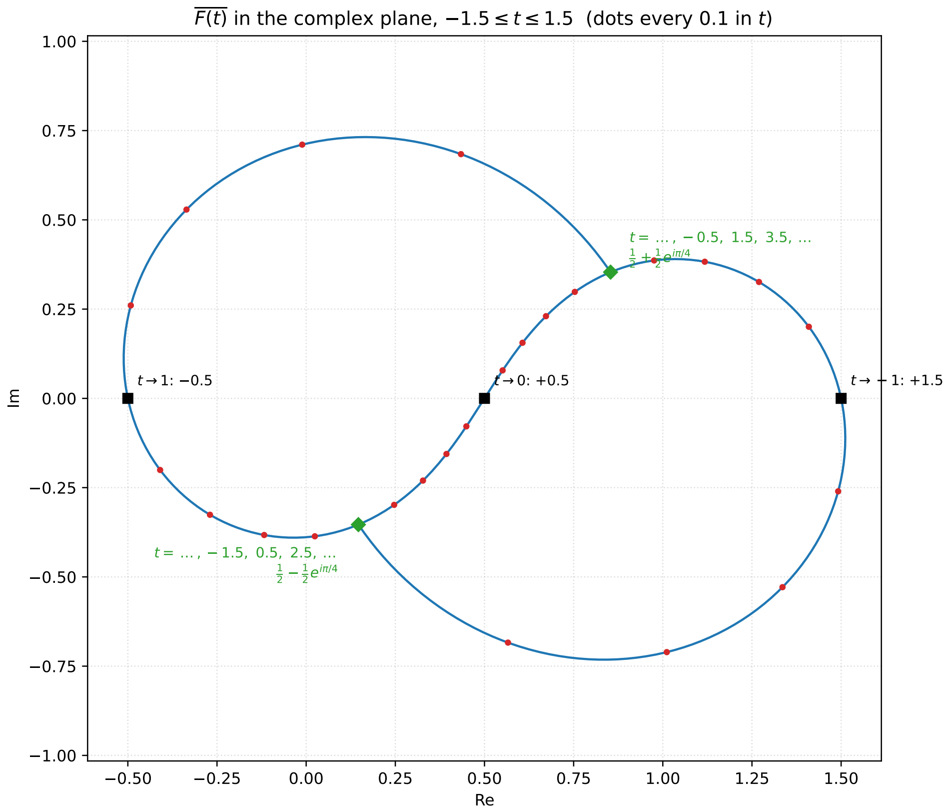

Figure 31: The trace of $\overline{F(t)}$ of (113) in the complex plane for $-1.5 \leq t \leq 1.5$ (red dots every 0.1 in t): the curve forms two congruent teardrop-like lobes, forming a shape reminiscent of the yin-yang symbol rooted in ancient texts like the *I Ching*. The curve is point-symmetric through $\frac{1}{2}$. The apparent poles at integer t are removable, with $\overline{F(t)} \to \frac{1}{2} - n$ as $t \to n$ (black squares). A curiosity: at *every* half-integer $t = k + \frac{1}{2}$ the exponent $t^2 = k(k+1) + \frac{1}{4}$ has $k(k+1)$ even, so $e^{-\pi it^2} = e^{-i\pi/4}$ always, and the whole expression collapses to just two values, $\overline{F}(k + \frac{1}{2}) = \frac{1}{2} - (-1)^k \frac{1}{2} e^{i\pi/4}$ (green diamonds): the curve returns to the point $\frac{1}{2} + \frac{1}{2}e^{i\pi/4} \approx 0.854 + 0.354i$ at $t = \ldots, -0.5, 1.5, 3.5, \ldots$ and to the point $\frac{1}{2} - \frac{1}{2}e^{i\pi/4} \approx 0.146 - 0.354i$ at $t = \ldots, -1.5, 0.5, 2.5, \ldots$, each revisited every $\Delta t = 2$, forever; they are the meeting points of the lobes. Generated by `fig_F_curve.py`.

We conjecture that as $T \to \infty$ the yin and yang curves converge to a universal limit curve built from this function, universal in that it does not depend on σ. The precise limit is most cleanly written through the Ψ function, which we do next.

11.5 The yin and yang curves are not symmetrical

While the name "yin-yang" invokes a pictorial similarity as intended, it also invites a symmetry that is not there. In the taijitu the two lobes are congruent, each carried onto the other by a half-turn about the center. In our case, the two curves are the two ends of one moving link, and (96) says exactly how they are related:

$$Y_{ang1} \;=\; Y_{in1} \;-\; \chi \lceil T \rceil^{\,2s-1}, \qquad s = \sigma + iI(T), \tag{115}$$

so the yang curve is the yin curve translated by $-v$, where $v = \chi\lceil T\rceil^{\,2s-1}$ is the revolving link's own vector in the frame, of length $|\chi|\,\lceil T\rceil^{\,2\sigma-1}$. Were v the same at every T the two curves would be congruent by construction. It is not. Over one unit of the index the direction of v swings through π and comes back, returning to where it began, with net turning zero. The length of v varies too, and it is exactly 1 only on the critical line, where $|\chi| = 1$ and the exponent $2\sigma - 1$ vanishes: the revolving link and the stationary link are then equal in length, which is the $d_1 = d_2$ symmetry of §9.2 seen in this frame.

What the taijitu has is a half-turn about the center of the picture, and here that center is the midpoint of the stationary link: the half-turn is $z \mapsto 1 - z$, which exchanges the link's two ends. So if the yin and yang curves were related as the taijitu's two lobes are, this map would carry one onto the other, and it is enough to draw $1 - Y_{ang1}$ on top of Y_{in1} and look. Figure 32 plots Y_{in1} against $1 - Y_{ang1}$ for $0 \le T \le 10$, one loop of each per unit of the index, on the critical line and at $\sigma = 0.05$ and $\sigma = 0.95$ above and below it. The reflection brings the two families almost on top of one another, and both converge to the single limit teardrop of §11.6, but at no finite T do they coincide. Each loop crosses its reflected partner twice in every unit of the index, running outside it along one arc and inside along the other, and the two enclose different areas. That difference in area narrows as T grows and widens away from the critical line, where the two lobes also trade which of them is the larger, near the line though not exactly on it. In the magnified windows of Figure 32 the yin curve (green) is the one running outside its reflected partner (red) in the top two rows, and the inside one in the bottom row, where the order has swapped.

What the critical line does give exactly is a mirror symmetry, though between the two frames rather than between the two curves of one frame. At $\sigma = \frac{1}{2}$ we have $\lceil T\rceil^{\,s} = \lceil T\rceil^{1/2} e^{i\omega}$ and, from $\zeta = \chi\overline{\zeta}$ and $\Sigma_2 = \chi\overline{\Sigma_1}$ with $|\chi| = 1$, also $R = \chi\overline{R}$. Comparing (95) with (97) then gives

$$\overline{Y_{in1}} = Y_{in2}, \qquad \overline{Y_{ang1}} = Y_{ang2} \qquad \big(\sigma = \tfrac{1}{2}\big), \tag{116}$$

which we confirm to 25 digits. Off the line both identities fail, the first by 6.4×10^{-2} and the second by 7.1×10^{-2} at $\sigma = \frac{1}{4}$, and by the same two amounts at $\sigma = \frac{3}{4}$.

An exact symmetry between the two curves themselves does arrive, but only in the limit, where the yang curve stops being a second curve at all: as $T \to \infty$ the yang settles onto the very teardrop the yin traces, half a unit of the index behind. So the asymmetry of this subsection is a finite-T effect, and the limit erases it. That limit curve, and the classical function it turns out to be built from, are the subject of the next subsection.

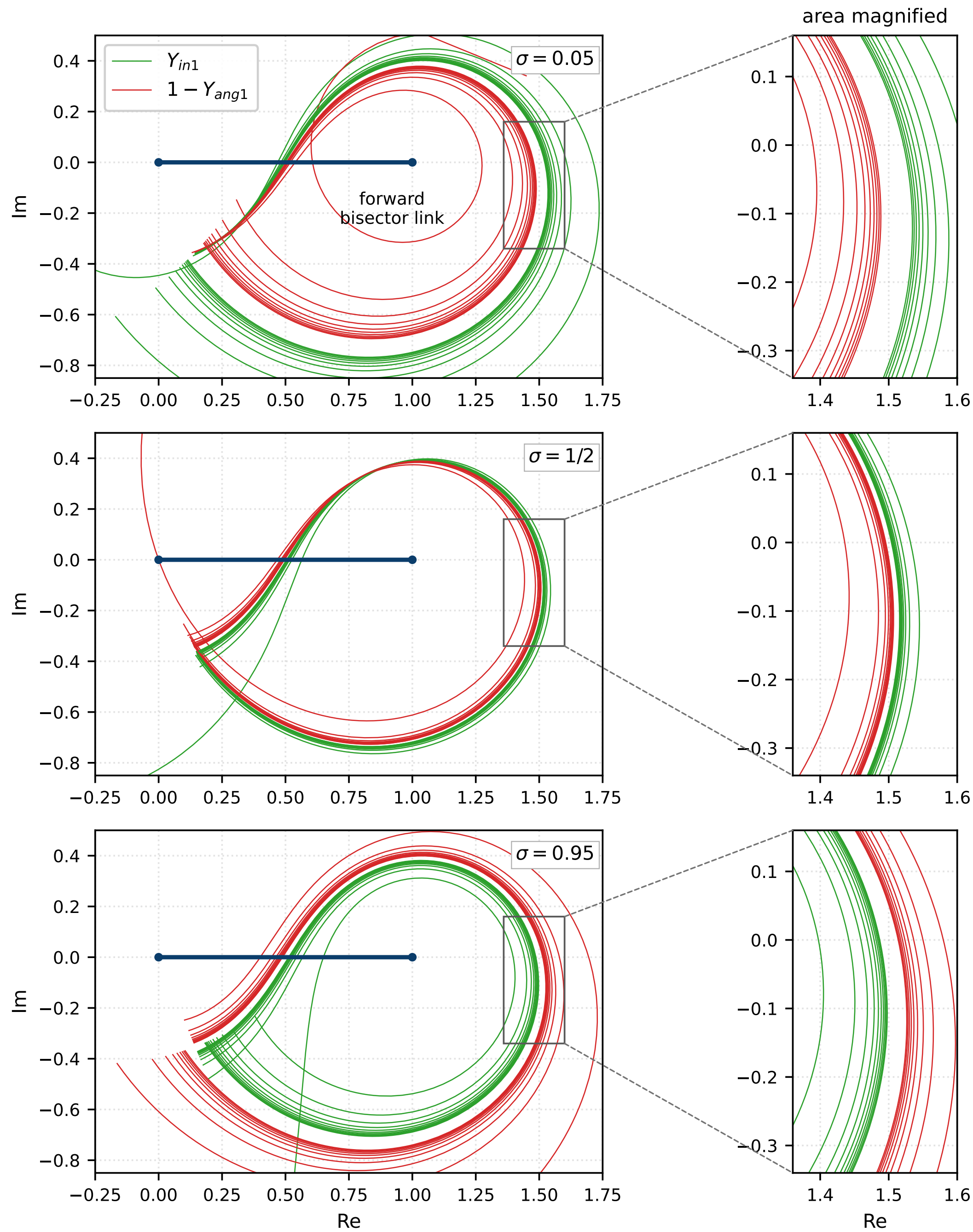

Figure 32: The yin curve (green) against the reflected yang curve $1 - Y_{ang1}$ (red), for $0 \le T \le 10$, which is one loop of each per unit of the index, out to $t = 692.2$. One row per σ, all on the same scale: $\sigma = 0.05$ on top, the critical line in the middle, $\sigma = 0.95$ below. The half-turn $z \mapsto 1 - z$ exchanges the ends of the stationary link (dark blue), so under the symmetry the picture suggests the two families would coincide. *Left:* all ten periods, the view cropped to the cluster. The stray arcs crossing it belong to the first period $0 \le T \le 1$, where $t < 13.6$ and the frame is still far from its limit; the later periods pile up on the limit teardrop, and they do so at every σ. *Right:* the boxed window magnified, chosen away from the two crossings. The families interleave and close on one another as T grows, and which of the two runs outside depends on σ: green outside red at $\sigma = 0.05$, red outside green at $\sigma = 0.95$. Generated by `fig_yinyang_asym.py`.

11.6 The limit curve and the Ψ function: the C_0 connection

Numerically, as $T \to \infty$ with the fractional part $q = T - \lfloor T \rfloor$ held fixed, the yin curve converges, regardless of σ, to

$$Y_{\inf}(q) \;=\; 1 - \Psi(q)\, e^{-2\pi i\,(q^2 - \frac{1}{16})}, \tag{117}$$

where

$$\Psi(x) \;=\; \frac{\cos\!\big(2\pi(x^2 - x - \frac{1}{16})\big)}{\cos(2\pi x)} \tag{118}$$

is the Riemann–Siegel correction function (it appears in §15.3 of Titchmarsh's book [20] and in §5.4.1 of Pugh's thesis [17]). The yang curve traces the *same* limit curve half a period behind: $Y_{ang1}(\sigma, T) \approx Y_{in1}(\sigma, T - \frac{1}{2}) \to Y_{\inf}(q - \frac{1}{2})$. The measured convergence is uniform in q away from the removable points below, at rate $O(1/T)$, and essentially independent of σ: the maximum of $|Y_{in1} - Y_{\inf}(q)|$ over a q-grid is 0.028, 0.015, 0.027 at $T \approx 10$ and 0.0076, 0.0042, 0.0075 at $T \approx 40$ for $\sigma = 0.2, 0.5, 0.9$ respectively. Figure 33 shows the convergence, including the pleasing fact that the very first yin path is the zeta function itself: for $0 < T < 1$ both partial sums are empty and $\lceil T \rceil = 1$, so $Y_{in1} = R = \zeta(\sigma + iI(T))$. The iconic zeta trajectory is yin orbit number zero. Proving the convergence, starting from Siegel's evaluation of his integral 5 (equations (111)–(113)), is future work.

Now the connection promised in the subsection title. The function (118) is, symbol for symbol, the universal amplitude

$$\Psi(x) \;=\; C_0(x) \;=\; \frac{\cos\!\big(2\pi(x^2 - x - \frac{1}{16})\big)}{\cos(2\pi x)} \tag{119}$$

of the Riemann–Siegel first term used in §8.2–8.3 (equations (59) and (60)). So the same universal function plays both roles: evaluated at $\hat{p} = \mathrm{frac}\sqrt{t/2\pi}$ it is the asymptotic *magnitude* of the remainder, and evaluated at $q = \mathrm{frac}(T)$ it is the asymptotic *shape* of the yin-yang orbit; by (117) the limiting yin curve keeps distance $|\Psi(q)| = |C_0(q)|$ from the point 1. Even the two parameters are the same clock read half a period apart: $\sqrt{t/2\pi} \approx T + \frac{1}{2}$ (the Taylor table of §10.1), so $\hat{p} \approx q \pm \frac{1}{2}$ (mod 1), precisely the lag between the yang and yin curves.

Two properties of the limit curve deserve note. First, it is closed: $Y_{\inf}(0) = Y_{\inf}(1) = 1 - \cos(\frac{\pi}{8})e^{i\pi/8}$. Second, the apparent singularities of Ψ at $q = \frac{1}{4}, \frac{3}{4}$ (the pole heights of §8.4) are *removable* on the curve: $\Psi \to \frac{1}{2}$ at both points, and the curve passes smoothly through the real values $Y_{\inf}(\frac{1}{4}) = \frac{1}{2}$ and $Y_{\inf}(\frac{3}{4}) = \frac{3}{2}$.

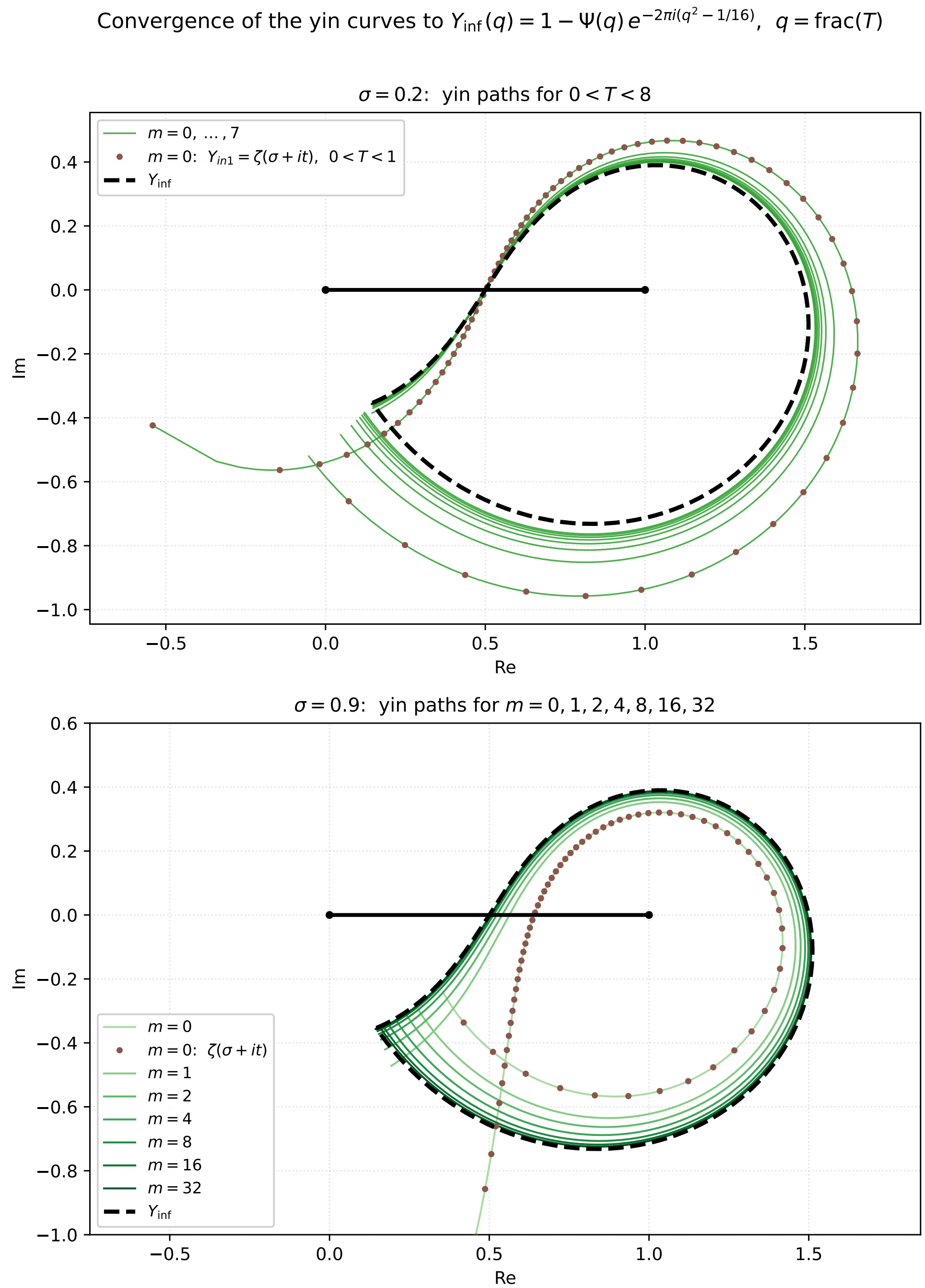

Figure 33: Convergence of the yin curves to the limit curve $Y_{\mathrm{inf}}(q) = 1 - \Psi(q)e^{-2\pi i(q^2-1/16)}$ (dashed black), with the unit bisector link in black; the two panels have identical window sizes. As everywhere in the bisector frame, the picture is scaled by the frame map (94) so that the forward bisector link always has length 1, running from 0 to 1 on the real axis. Top: $\sigma = 0.2$, the yin paths (green) of every unit interval of the index over $0 < T < 8$, $m = 0, \ldots, 7$; the $m = 0$ path, marked with spaced brown dots, is $\zeta(\sigma + it)$ itself, since for $0 < T < 1$ both partial sums are empty and $Y_{in1} = \zeta$. Bottom: $\sigma = 0.9$, the yin paths of the periods starting at $T = 0, 1, 2, 4, 8, 16, 32$ (light to dark green); by $T \approx 32$ the path is visually indistinguishable from Y_{inf} even this far off the critical line. The $m = 0$ path, again $\zeta(\sigma + it)$ and again marked with brown dots, leaves the frame at the bottom: $\sigma = 0.9$ sits close to the pole of ζ at $s = 1$, so this first orbit dives to about $-2.7i$ before the later orbits settle onto the teardrop; the window is clipped to the convergence region. The yang curves (omitted for clarity) trace the same limit half a period behind. Generated by `fig_yinyang_infinity.py`.

12 Further observations

12.1 The area swept out by the unit bisector link at infinity

In the language of Kakeya sets: what area does a unit-length needle sweep when one end rides $Y_{\mathrm{inf}}(t)$ and the other rides $Y_{\mathrm{inf}}(t-\frac{1}{2})$ (the limiting positions of the two ends of the reverse bisector link) as t runs from 0 to 1? The "needle" here is nothing abstract: it is the reverse bisector link itself, sweeping once around the stationary forward bisector link over one unit of the index, exactly the motion depicted in Figure 28. By the half-period lag, this swept area is twice the area enclosed by Y_{inf}, which we now compute.

Let $z(t) = Y_{\mathrm{inf}}(t)$ on $t \in [0,1]$, a closed curve passing smoothly through its two removable points. Writing $z(t) = x(t) + iy(t)$, the signed enclosed area is

$$\mathcal{A}_{\mathrm{signed}} = \frac{1}{2}\int_0^1 \big(x\,y' - y\,x'\big)\,dt = \frac{1}{2}\int_0^1 \Im\big(\overline{z}\,z'\big)\,dt. \tag{120}$$

Set $A(t) = 2\pi\big(t^2 - \frac{1}{16}\big)$, so that $z - 1 = -\Psi(t)\,e^{-iA(t)}$. A short computation gives the key simplification

$$\Im\Big(\overline{(z-1)}\,(z-1)'\Big) \;=\; -A'(t)\,\Psi(t)^2, \qquad A'(t) = 4\pi t \tag{121}$$

(the $\Psi'\,\Psi$ cross term is real and drops out). Translation by 1 does not change enclosed area, so

$$\mathcal{A}_{\mathrm{signed}} = -\frac{1}{2}\int_0^1 A'(t)\,\Psi(t)^2\,dt = -2\pi\int_0^1 t\,\Psi(t)^2\,dt, \tag{122}$$

and the geometric (positive) area is

$$\boxed{\mathrm{Area} = 2\pi\int_0^1 t\,\frac{\cos^2\!\big(2\pi(t^2 - t - \frac{1}{16})\big)}{\cos^2(2\pi t)}\,dt \;\approx\; 1.0341672002955850005.} \tag{123}$$

(The integral is split at the removable points $t = \frac{1}{4}, \frac{3}{4}$ for the numerical evaluation; with the given parameter direction the signed area comes out negative, and the magnitude is the enclosed area above.) The needle of the opening question therefore sweeps about 2.0683.

12.2 Connection between the yin and yang functions and the link numbers

In our numerical experiments with the tool we noticed that the bisector link, viewed in its local frame, is always parallel to one particular link of the "conveyor belt", the stretch of links between the last two spirals (spirals 0 and 1 in the numbering of (64)). That particular link follows the bisector link as the index varies between integers, and its link number is

$$\lfloor T\rfloor\,\lfloor T+2\rfloor \;=\; (m+1)^2 - 1, \tag{124}$$

so the follower's summand is number $(m+1)^2$, the square of the bisector link's summand $m+1$, whose phase $-t\log\big((m+1)^2\big)$ advances at exactly twice the bisector link's rate. The first few link numbers in the sequence are $3, 8, 15, 24, 35, 48, 63, 80, 99, 120, 143, \ldots$: it is sequence A005563 in the OEIS [16], though no connection of that sequence to the zeta function is listed there. The follower transitions from one of these links to the next exactly when the index T is an integer (the same instants at which the bisector point hands off), and the two links are momentarily coincident near $T + \frac{1}{2}$.

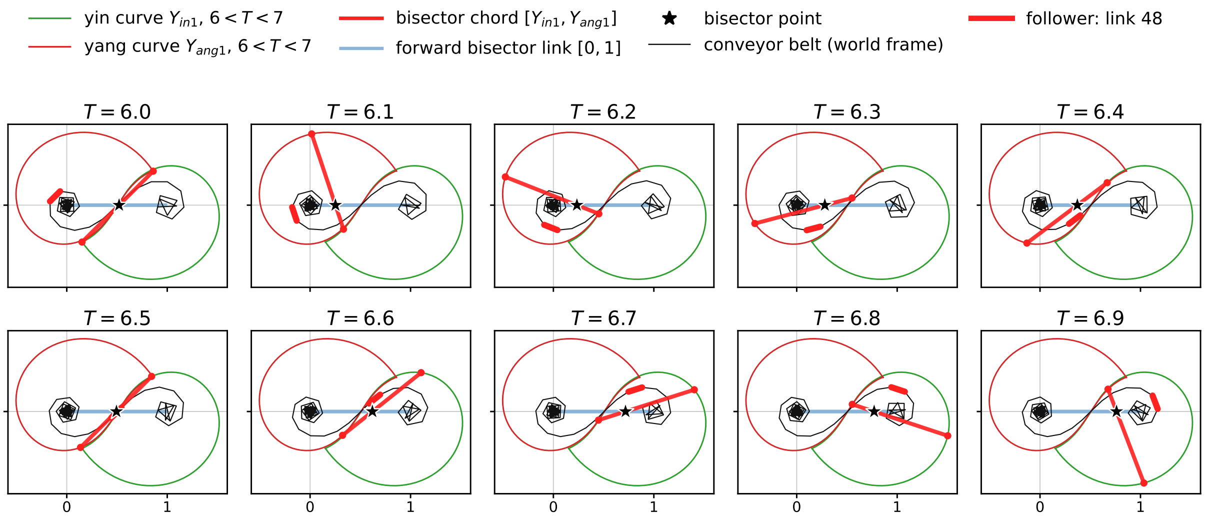

Figure 34: The follower link and the bisector chord over one unit of the index, $\sigma = \frac{1}{2}$ and $T = 6.0, 6.1, \ldots, 6.9$ (so $m = 6$ and the follower is link 48, the 49th summand). Each panel superposes two frames. In the world frame: the conveyor belt (thin black), the reverse chain's joints $\zeta - \Sigma_2(n)$ for n running from $L_N(T,1)$ to $L_N(T,0)$ of (64), which is the stretch between spirals 1 and 0; inside it the follower, link 48, is drawn thick in neon red. In the bisector frame (94): the yin curve Y_{in1} (thin green) and the yang curve Y_{ang1} (thin red) over $6 < T < 7$, the same in all ten panels; the chord $[Y_{in1}, Y_{ang1}]$ at this panel's T (neon red, endpoints dotted); the forward bisector link $[0, 1]$ (light blue); and the bisector point (star), where the chord's line crosses that unit link. The chord turns through a half revolution across the ten panels and the follower link turns with it, always parallel by (125) and always $\lceil T \rceil$ times shorter. Near $T = 6.5$ the two very nearly coincide. Generated by `fig_conveyor_follower.py`.

The observation is an exact identity, and the doubled[5] phase rate explains it completely. The chord of the yin and yang curves, that is the reverse bisector link seen in the frame (94), is $Y_{ang1} - Y_{in1} = -\chi \lceil T \rceil^{2s-1}$ by (96), with $s = \sigma + iI(T)$ as usual, while link $(m+1)^2 - 1$ of the reverse chain, in the world frame, is the summand vector $-\chi\big(\lceil T \rceil^2\big)^{s-1} = -\chi \lceil T \rceil^{2s-2}$. The two differ by the factor $\lceil T \rceil^{-1}$, which is real and positive, so

$$\text{link } (m+1)^2 - 1 \;=\; \frac{Y_{ang1} - Y_{in1}}{\lceil T \rceil} \qquad \text{for every } \sigma \text{ and every } T: \tag{125}$$

the follower is parallel to the chord, points the same way, and is shorter by exactly the factor $\lceil T \rceil$. No other link of the chain has this property, since the phases of neighboring links differ by $t \log \frac{n+1}{n}$, which is not a multiple of π in general. Figure 34 shows the pair over one unit of the index. Numerically the angle between the two vectors is zero to machine precision and the length ratio is $\lceil T \rceil^{-1}$ to twelve digits at every (σ, T) tested, on and off the critical line. What is *not* exact is the position: the follower lies on the chord only momentarily. Measuring the perpendicular distance from the follower's midpoint to the chord line at $\sigma = \frac{1}{2}$, $m = 6$, it falls from 0.512 at

[5] Every summand of the chain is a vector whose phase is $-t \log n$, so as t grows that vector rotates at rate $\log n$ radians per unit t. Writing $\omega = t \log \lceil T \rceil$ as elsewhere in the paper, the bisector link carries summand $\lceil T \rceil$ and so sits at phase ω, while the follower carries summand $\lceil T \rceil^2$ and sits at phase $t \log\big(\lceil T \rceil^2\big) = 2\omega$, double because squaring the summand doubles its logarithm.

$T = 6.0$ to 0.008 at $T = 6.5$ and rises again to 0.512 at $T = 7.0$, and at the minimum the follower's midpoint sits at fraction 0.503 of the chord. This is the "momentarily coincident near $T + \frac{1}{2}$" of the first paragraph.

12.3 Length ratios: Σ_1 to Σ_2, and R_{1ps} to R_{2ps}

The four terms of (24) come in two natural pairs: the two partial sums, and the two fractional links. Within each pair the two lengths are equal on the critical line, and the two remarks below record what their ratio does off it. The pairs behave quite differently, and in both cases the ratio obeys the same reflection law about $\sigma = \frac{1}{2}$.

Remark 12.1 (regarding the ratio of the lengths of Σ_1 and Σ_2)**.** On the critical line $\sigma = \frac{1}{2}$ we have $|\chi| = 1$, so the Σ_2 links have the same lengths $(n+1)^{-1/2}$ as the Σ_1 links; each link is then a rotation composed with a uniform link-length shrink $\sqrt{n/(n+1)}$, and $\arg\chi = -2\vartheta(t)$, with ϑ the Riemann–Siegel theta function. The two chains are thus congruent link for link there, so their end-to-end displacements agree. Away from the line they do not, and the discrepancy has a closed form. Write $\ell_{\Sigma_1} = |\Sigma_1|$ and $\ell_{\Sigma_2} = |\Sigma_2|$ for the lengths of the net displacements of P_{Σ_1} and P_{Σ_2}, and put $\Sigma_m(z) = \sum_{n \le m} n^{-z}$. Since $\sum_{n\le m} n^{\,s-1} = \overline{\Sigma_m\big((1-\sigma)+it\big)}$, the second leg is the first one read at the reflected abscissa $1-\sigma$, scaled by $|\chi|$:

$$\frac{\ell_{\Sigma_1}}{\ell_{\Sigma_2}} = \frac{\big|\Sigma_m(\sigma+it)\big|}{\big|\chi(\sigma+it)\big|\,\big|\Sigma_m\big((1-\sigma)+it\big)\big|}. \tag{126}$$

Since $\chi(s)\chi(1-s) = 1$ and $|\chi(\bar{s})| = |\chi(s)|$, the formula gives, at fixed T (hence fixed $t = I(T)$ and $m = \lfloor T \rfloor$),

$$\frac{\ell_{\Sigma_1}(\sigma)}{\ell_{\Sigma_2}(\sigma)} \cdot \frac{\ell_{\Sigma_1}(1-\sigma)}{\ell_{\Sigma_2}(1-\sigma)} = 1, \tag{127}$$

which we have verified numerically to twelve digits at several T. So the ratio is reciprocal-symmetric about $\sigma = \frac{1}{2}$: whatever it does on the right half of the strip, it does the reciprocal of on the left, with the value pinned to 1 at the center by the link-for-link congruence just described.

Remark 12.2 (regarding the ratio of the lengths of R_{1ps} and R_{2ps})**.** The same question for the two fractional links has a shorter answer, because in the quotient of (19) and (20) both $|R|$ and the common denominator $\sin(2\omega+\psi)$ cancel:

$$\frac{|R_{1ps}|}{|R_{2ps}|} \;=\; \frac{d_1}{d_2} \;=\; \frac{\sin(\omega+\psi-\arg R)}{\sin(\omega+\arg R)}. \tag{128}$$

This ratio is pure angle data, carrying no length scale at all: it is the law of sines in the triangle of Figure 3, recording how the direction of R divides the cone spanned by the two link directions $e^{-i\omega}$ and $e^{\,i(\omega+\psi)}$. Four consequences are worth recording. First, on the critical line $\arg R = \frac{\psi}{2}$, the two sines coincide and the ratio is 1, which is Corollary 8.1 again. Second, the reflection law (127) holds verbatim for d_1/d_2: at fixed T we have verified $(d_1/d_2)(\sigma)\cdot(d_1/d_2)(1-\sigma) = 1$ numerically to 10^{-17} and better. Third, because the scale cancels, this ratio is far tamer than (126): over $1 < T < 20$ and the whole strip its extremes are 0.387 and 2.583 (a reciprocal pair, as the second point requires), whereas $\ell_{\Sigma_1}/\ell_{\Sigma_2}$ ranges over several orders of magnitude there. Fourth, the parallel-link poles of §8.4 cancel out of it: where $2\omega+\psi = k\pi$, (128) gives $d_1/d_2 = (-1)^{k+1}$, exactly ± 1, even though d_1 and d_2 separately diverge (at $\sigma = \frac{1}{10}$, $T = 4.256594$ we compute $d_1 = -d_2 \approx -4.07 \times 10^{36}$). A negative value is off-line news of a different kind: it says R has left the cone, which happens in a narrow window around each pole. The ratio does reach 0 and ∞, but only on the isolated loci where R is exactly parallel to one of the two link directions; for instance $d_1 = 0$ at $\sigma = \frac{1}{10}$, $T = 4.2607679$, whose mirror at $\sigma = \frac{9}{10}$ has $d_2 = 0$. Off the critical line the locus $d_1 = d_2$ is the family of ovals plotted in Figure 22.

12.4 ϑ_1, ϑ_2 and the zero-counting function

The well known Riemann–von Mangoldt formula for the number of zeros of ζ with imaginary part in $(0, t]$, in its exact form, reads

$$N(t) \;=\; \frac{\vartheta(t)}{\pi} \;+\; 1 \;+\; S(t), \qquad S(t) \;=\; \frac{1}{\pi}\arg\zeta\left(\tfrac{1}{2}+it\right), \tag{129}$$

with ϑ the Riemann–Siegel theta function of §8.2. This is an identity ([20, Theorem 9.3]), obtained by applying the argument principle to ξ around a rectangle: $\vartheta(t)/\pi + 1$ is the contribution of the Γ and $\pi^{-s/2}$ factors, and $S(t)$ is the contribution of ζ itself along the critical line. Stated in terms of ϑ the formula is exact: (129) carries no error term at all; approximation enters only when ϑ is replaced by its expansion, and inexactness only when S is dropped. Figure 35 shows the two pieces over one unit of the index.

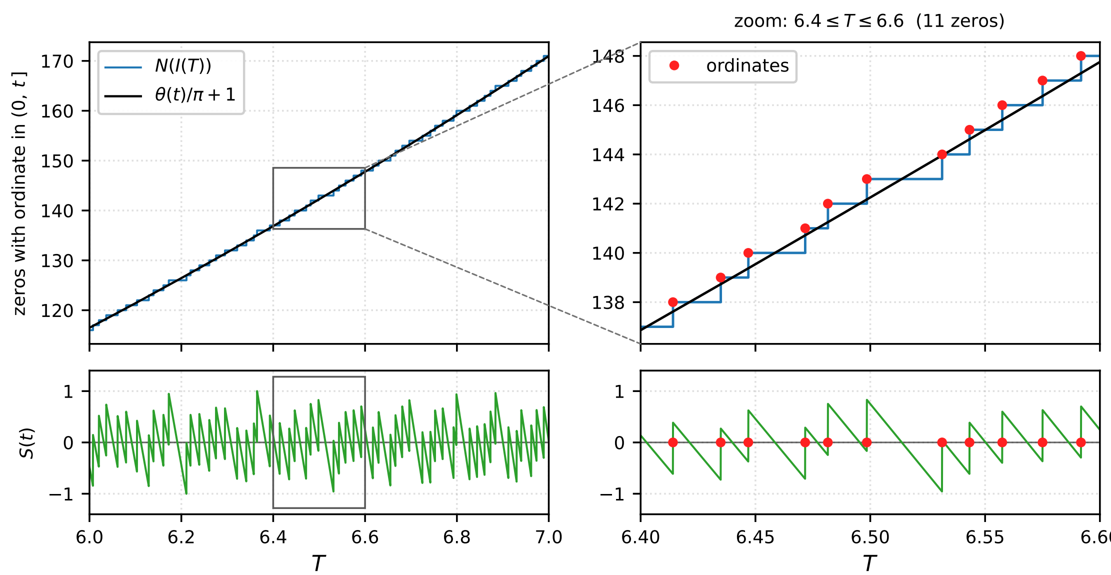

Figure 35: *Left:* the staircase $N(I(T))$ (blue) over $6 \le T \le 7$, which is $264.9 \le t \le 352.9$ and carries the 55 ordinates γ_{117} through γ_{171}, against the smooth part $\vartheta(t)/\pi+1$ of (129) (black). The vertical gap between them is S, which here stays within $[-1.002, +1.001]$. *Right:* the boxed window $6.4 \le T \le 6.6$ magnified, with the eleven ordinates it holds, γ_{138} through γ_{148}, marked in red at the top of each riser and placed at $T = I^{-1}(\gamma)$. The risers are unevenly spaced while the smooth part climbs steadily. Below each panel is the gap itself, S (green), which at this magnification shows plainly as a sawtooth: a unit jump at every ordinate and a steady slide between them. Generated by `fig_staircase_67.py`.

All of the discontinuity is in S. N is a step function and $\vartheta(t)/\pi + 1$ is analytic, so S carries every jump: it climbs by one (by the multiplicity, in principle) at each ordinate (the imaginary part

γ of a zero) and falls smoothly in between, at the rate

$$S'(t) \;=\; -\frac{\vartheta'(t)}{\pi} \;=\; -\frac{1}{2\pi}\log\frac{t}{2\pi} \;+\; O(t^{-2}), \tag{130}$$

which is the mean zero density. Across an ordinate S jumps up by one; between ordinates it slides back down along that line. The sawtooth is why S has mean zero without being small: each unit jump is repaid by a decline whose steepness is the density itself. At an ordinate the definition of S fails, since ζ vanishes there and its argument is undefined; one sets $S(\gamma) = \frac{1}{2}[S(\gamma^-) + S(\gamma^+)]$, and (129) then holds for every $t > 0$.

Three counting curves against the staircase. Expanding ϑ by Stirling and dropping S turns (129) into the Riemann–von Mangoldt formula in its approximate form,

$$N_{\rm rvm}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8} + \frac{1}{48\pi t} + \frac{7}{5760\pi t^3} \tag{131}$$

(the $\frac{7}{8}$ collects the $-\frac{\pi}{8}$ of the expansion, divided by π, with the standalone 1 of (129)). Nothing of consequence is discarded in truncating there: at $t = 1000$ even the first correction retained, $1/(48\pi t)$, is only 6.6×10^{-6}, while $S(1000) = +0.384$. The whole of the error is S. Set also

$$N_{\rm ps}(T) = \frac{1}{\pi}\vartheta\big(I(T)\big) + \frac{1}{\pi}\vartheta_1\big(\tfrac{1}{2}, T\big) + \frac{3}{2}, \tag{132}$$

which replaces S by the angle ϑ_1 that leg 1 makes with the real axis (§9.2), the leg running from the origin to the bisector point $B_1 = \Sigma_1 + R_{1ps}$. These two behave quite differently against the true count $N(I(T))$, and a third curve N_*, built below in (139) from a different splitting of ζ, repairs the one defect of $N_{\rm ps}$.

$N_{\rm rvm}$ is analytic and increasing, so it can only approximate the staircase in the mean, never landing on an integer at an ordinate, and its error is exactly the sawtooth S. $N_{\rm ps}$ instead meets an integer *exactly* at every ordinate, which we verify to 2.8×10^{-14}. The reason is Corollary 9.2: on the critical line

$$B_1\, e^{i\vartheta(t)} = \tfrac{1}{2}Z(t) + ih, \qquad h \in \mathbb{R}. \tag{133}$$

Multiplying by $e^{i\vartheta}$ rotates the whole configuration into the frame in which ζ is real, the frame that defines $Z = e^{i\vartheta}\zeta$; it is the rotation $e^{-i\psi/2}$ of Corollary 9.2 written with $\vartheta = -\frac{\psi}{2}$, since $\chi = e^{-2i\vartheta}$ on the line. We call it the *ϑ-frame*. There the base $O \to \zeta$ lies along the real axis, and (133) reads a split point off as a pair of coordinates: half the base along it, and the offset h across it. So at a zero Z vanishes, $B_1 e^{i\vartheta}$ is purely imaginary, $\vartheta_1 = -\vartheta \pm \frac{\pi}{2}$, and (132) is an integer. The argument runs both ways: since $\Re\big(e^{i\vartheta}B_1\big) = \frac{1}{2}Z$ at every t, and not merely at the zeros, (132) is an integer only where Z vanishes, so the integer points of $N_{\rm ps}$ are the ordinates and nothing else. The counting is therefore done by the chain's own geometry rather than by $\arg\zeta$: leg 1 stands perpendicular to the direction of ζ precisely when ζ vanishes. $N_{\rm rvm}$ carries no information about individual zeros, while $N_{\rm ps}$ locates every one of them.

Where $N_{\rm ps}$ misses, and why. Two things keep (132) from being a counting function. First, ϑ_1 is an angle, so (132) is determined only up to an even integer, and each passage of ϑ_1 through $\pm\pi$ breaks the curve; there are nine such wraps over $3 < T < 5$, and $\frac{3}{2}$ is the constant that puts the curve on the staircase. Second, and this is the real defect, the curve sometimes meets the wrong integer. Differentiating (133) at an ordinate, where Z vanishes, gives

$$\frac{d}{dt}\arg\big(B_1 e^{i\vartheta}\big)\bigg|_{Z=0} = -\frac{Z'(t)}{2h}, \tag{134}$$

so the crossing carries the angle forward or backward with the sign of $-hZ'$. Consecutive simple zeros of Z alternate in the sign of Z', so the crossings all turn the same way exactly when h alternates in sign too. Where h nearly vanishes it keeps its sign across two ordinates instead; the angle turns back on itself, the count fails to advance, and the second of the pair reads one too high. Over the first 400 ordinates $|h|$ has median 1.081 at the 350 that N_{ps} places correctly and 0.319 at the 50 it misses, and those 50 are exactly the crossings that run backwards, which we call *retrograde*. They are also exactly the wraps: the nine wraps of ϑ_1 over $3 < T < 5$ counted above sit one apiece on the nine retrograde ordinates of that range. Figure 36 shows the cost over a longer stretch: five retrograde ordinates in $5.9 < T < 6.7$, and after each one N_{ps} runs a unit above the staircase until the following ordinate restores it. The two ways of describing the failure, a broken branch and a backward crossing, are one event. Over $3 < T < 5$, which holds 55 zeros, both curves stay within one of the staircase all the same: the largest deviation is 0.946 for N_{rvm} and 0.999 for N_{ps}.

Reading ϑ_2 in place of ϑ_1 cannot help. On the critical line $B_2 = \chi\overline{B_1}$ and $\arg\chi = -2\vartheta$, so

$$\vartheta_2 = -2\big(\vartheta + \vartheta_1\big) \pmod{2\pi}, \tag{135}$$

whichever remainder defines the split. Every sawtooth built from ϑ_2 is therefore the same function as the one built from ϑ_1 and retrogrades in the same places: among the 63 ordinates of $3.0 \le T \le 5.2$ the splits at R_{1ps}, at $R/2$ and at R_{1ak} run backwards at 10, 9 and 12 of them. The retrograde belongs to the split, not to the branch.

The velocity split, and a curve that lands on every ordinate. The cure is to split ζ so that the offset h of (133) alternates by construction. With ϑ' the derivative of (130), exactly $\frac{1}{2}\operatorname{Re}\frac{\Gamma'}{\Gamma}\big(\frac{1}{4}+\frac{it}{2}\big) - \frac{1}{2}\log\pi$, put

$$B_1^* = \zeta(s) + \frac{\zeta'(s)}{2\vartheta'(t)}, \qquad B_2^* = \zeta(s) - B_1^* = -\frac{\zeta'(s)}{2\vartheta'(t)} = \frac{i}{2\vartheta'(t)}\frac{d\zeta}{dt}, \tag{136}$$

so leg 2 is the velocity of the chain's endpoint along the line, quarter-turned and scaled by $1/2\vartheta'$. On $\sigma = \frac{1}{2}$ this split is again reflection-symmetric, $B_2^* = \chi\overline{B_1^*}$, so its legs are exactly equal at every height,

$$\left|\zeta + \frac{\zeta'}{2\vartheta'}\right| = \frac{|\zeta'|}{2\vartheta'}, \tag{137}$$

which is the identity $\operatorname{Re}\big(\zeta\overline{\zeta'}\big) = -\vartheta'|\zeta|^2$, itself no more than the statement that $Z = e^{i\vartheta}\zeta$ is real. Differentiating that statement gives $e^{i\vartheta}\zeta' = -iZ' - \vartheta' Z$, and hence (133) holds for the new split with

$$e^{i\vartheta}B_1^* = \frac{Z}{2} - \frac{i\,Z'}{2\vartheta'}, \qquad h^* = -\frac{Z'}{2\vartheta'}. \tag{138}$$

Now h^* alternates because Z' does, and (134) returns $-Z'/2h^* = \vartheta'$: every crossing turns forward, and at exactly the mean density. Setting

$$N_*(T) = \frac{1}{\pi}\vartheta\big(I(T)\big) + \frac{1}{\pi}\vartheta_1^*(T) + \frac{3}{2}, \qquad \vartheta_1^* = \arg B_1^*, \tag{139}$$

we have a curve that is integer-valued at the ordinates and nowhere else, since $\operatorname{Re}\big(e^{i\vartheta}B_1^*\big) = \frac{1}{2}Z$ vanishes only there, and that advances by exactly one from each ordinate to the next. It lands on every one of the first 400 ordinates, $N_*(I^{-1}(\gamma_n)) = n$ to 2.8×10^{-13}, against 50 misses for N_{ps}; over $0.6 \le T \le 8$ it is strictly increasing on a grid of 3000 points, no wrap occurs ($|\vartheta_1^*|$ stays below

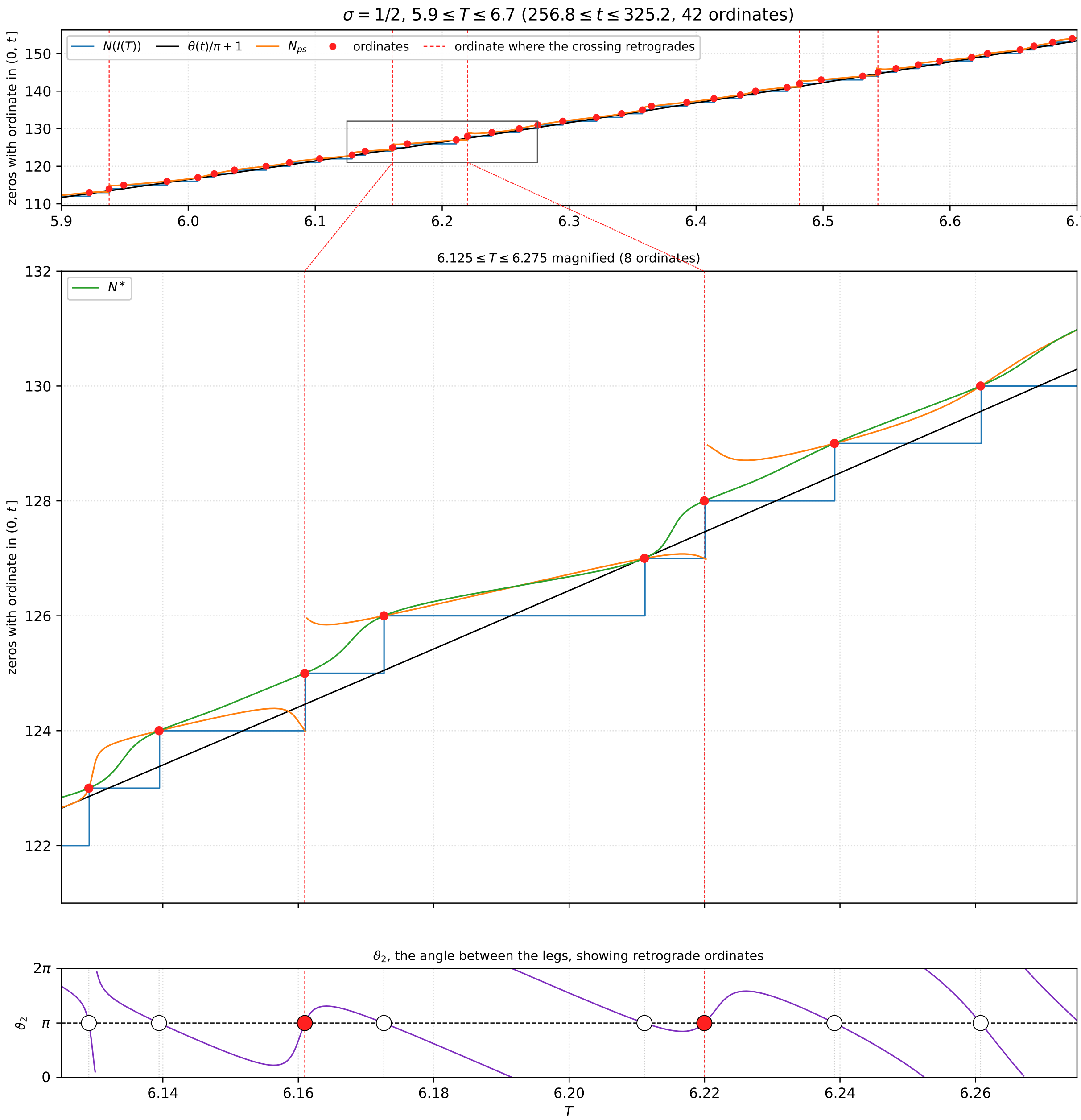

Figure 36: Staircase, in blue, is the exact Riemann–von Mangoldt formula (129), and the approximate Riemann–von Mangoldt formula (131) is the black curve, that formula with S dropped. *Top:* $5.9 \leq T \leq 6.7$, which is $256.8 \leq t \leq 325.2$ and holds the 42 ordinates γ_{113} through γ_{154}, five of them retrograde, with N_{ps} of (132) in orange, broken at its wraps. *Middle:* the boxed window $6.125 \leq T \leq 6.275$ magnified, holding 8 ordinates of which γ_{125} and γ_{128} are retrograde. The velocity split N^* of (139) is drawn in green. *Bottom:* the folding angle ϑ_2 of §9.2 on the same T range, reduced to $[0, 2\pi)$. Every passage through π (dashed) is at an ordinate; the two filled dots are the two retrograde ordinates of the panel above, where ϑ_2 rises through π, turns around, and comes back down through it. Generated by `fig_nps_staircase.py`.

0.60π), and it rises by 237.64 across the 237 zeros of that range. Height costs it nothing and no branch correction is needed there either: at $\gamma_{10000} = 9877.78$ and at $\gamma_{50000} = 40433.69$ the same formula with the same constant $\frac{3}{2}$ still returns exactly 10000 and 50000. Where N_{rvm} threads the staircase without touching it, and N_{ps} locates zeros without counting them, N_* does both. Like any count read off Z it sees the zeros on the line, which in the range plotted are all of them.

Remark 12.3 (the two offsets $h_{R/2}$ and h^*)**.** The offsets $h_{R/2}$ and h^* of the $R/2$ split and the velocity split are both drawn in Figure 37, and they differ only by a weighting of the links. For the $R/2$ split $e^{i\vartheta}R$ is real, so h is the sine companion of the main sum in the ϑ-frame, $h_{R/2} = \mathrm{Im}\big(e^{i\vartheta}\Sigma_1\big) = \sum_{n\le m} n^{-1/2}\sin(\vartheta - t\log n)$, while differentiating the real part of that rotated main sum and dividing by $-\vartheta'$ gives

$$h^* = \sum_{n\le m} n^{-1/2}\Big(1 - \frac{\log n}{\vartheta'}\Big)\sin\big(\vartheta - t\log n\big) + \cdots\,,$$

the tail being the remainder's derivative; the displayed sum by itself reproduces h^* to 2.3×10^{-2} over the first 400 ordinates. The weight is the phase rate $\vartheta' - \log n$ of link n measured in units of ϑ', so it runs from 1 on the first link down to nearly 0 on the last, where $\log m \approx \vartheta'$. The counting angle is carried by the same links; each one contributes in proportion to how much of the rotation rate it retains.

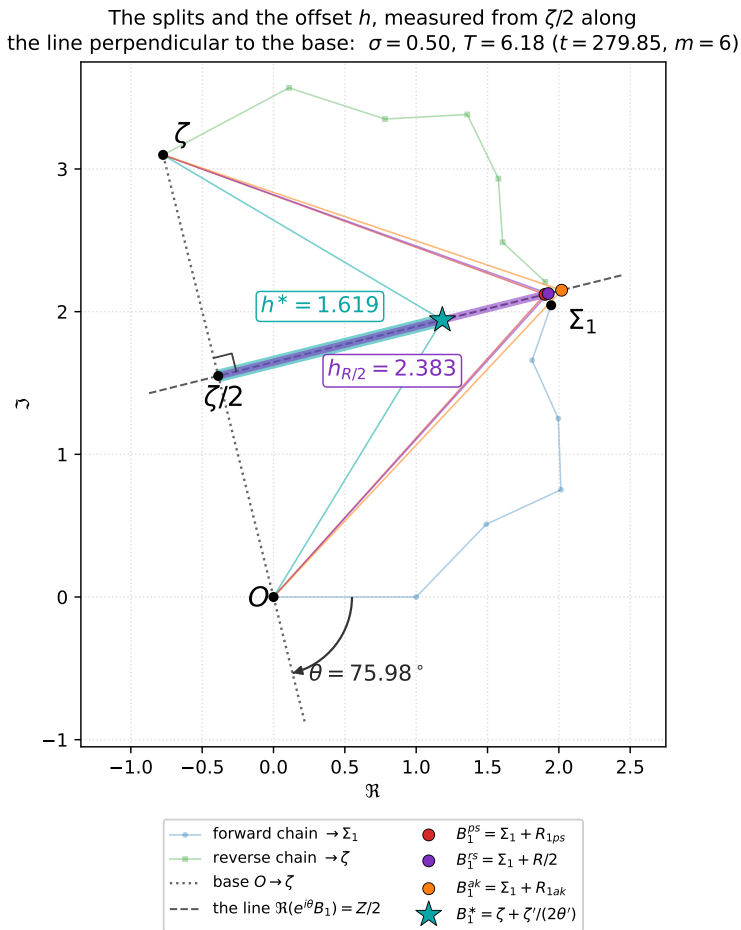

Figure 37: The offset h of (133), drawn at $T = 6.18$. Because $\Re\big(e^{i\vartheta}B_1\big) = Z/2$ for every split, all of the split points lie on one line when $\sigma = \frac{1}{2}$, the dashed line through $\zeta/2$ perpendicular to the base $O \to \zeta$ (the right angle is marked). What distinguishes one split from another is h, the signed distance from $\zeta/2$ along that line, drawn as a thick stretch of the line for the $R/2$ split (purple, $h_{R/2} = 2.383$) and for the velocity split (teal, $h^* = 1.619$); the two legs of each split are drawn thin in its own colour and the two chains fainter still. The arc at O is ϑ, swept clockwise off the positive real axis onto the base line, since $\arg\zeta \equiv -\vartheta \pmod{\pi}$: with $\vartheta = 390.884$ reducing to 75.98° modulo π, the base line leaves the axis at -75.98°, drawn dotted below O, and ζ itself lies on the far side of O at 104.02° because $Z = -3.194$ is negative at this height. Generated by `fig_splits_h.py`.

Remark 12.4 (N_* is monotonically increasing if RH is true)**.** Integer at the ordinates and nowhere else, and up by one from each to the next: what (139) leaves open is whether the curve ever runs backwards in between. It does not, and the reason is a known consequence of the Riemann hypothesis. With $W = e^{i\vartheta}B_1^* = \frac{1}{2}Z - \frac{iZ'}{2\vartheta'}$ from (138) and $\pi N_* = \arg W + \frac{3\pi}{2}$, differentiating and measuring against the local mean density ϑ' gives the slope in closed form,

$$\rho = \frac{1}{\vartheta'}\frac{d}{dt}\arg W = \frac{\pi}{\vartheta'\,I'(T)}\frac{dN_*}{dT} = \frac{Z'^2 - ZZ'' + \dfrac{\vartheta''}{\vartheta'}ZZ'}{\vartheta'^2Z^2 + Z'^2}, \tag{140}$$

which returns 1 at $Z = 0$ and $-Z''/\vartheta'^2 Z$ at $Z' = 0$, as it must. Since ϑ' and I' are positive, ρ

and dN_*/dT carry the same sign; the denominator of (140) is positive and $\vartheta''/\vartheta' = O(1/t\log t)$, so up to that negligible term the curve increases exactly where the Laguerre expression $Z'^2 - ZZ''$ is positive.

That expression belongs to Ξ. Writing $\Xi(t) = \xi(\frac{1}{2} + it) = -c(t)Z(t)$ with $c(t) = \frac{1}{2}\big(\frac{1}{4} + t^2\big)\pi^{-1/4}\big|\Gamma(\frac{1}{4} + \frac{it}{2})\big| > 0$, algebra gives $\Xi'^2 - \Xi\Xi'' = c^2\big[(Z'^2 - ZZ'') - (\log c)''Z^2\big]$. If the hypothesis holds then every zero of Ξ is real, its Hadamard product gives $-(\log\Xi)'' = \sum_k (t-\gamma_k)^{-2}$, and therefore

$$Z'^2 - ZZ'' = Z^2\Big[\sum_k \frac{1}{(t-\gamma_k)^2} + (\log c)''\Big], \tag{141}$$

the sum running over the zeros of Ξ of both signs. The bracket is positive with room to spare: it is of the order of the squared zero density, carried mostly by the two ordinates flanking t, while $(\log c)'' = O(t^{-2})$ takes away almost nothing. At $t = 100.5$ it reads 2.2003, of which the 600 zeros with $|\gamma| \le 542$ supply 2.1972 and $(\log c)''$ removes 1.7×10^{-4}.

Monotonicity is what makes the curve a counting function outright, rather than a curve that merely meets the integers in the right places. N_* is continuous, it equals n exactly at γ_n, and it is an integer nowhere else; if it also increases then between consecutive ordinates it is trapped strictly between n and $n+1$, and taking the floor recovers (129) on the nose,

$$\lfloor N_*(T)\rfloor = N\big(I(T)\big), \tag{142}$$

with no error term and no sawtooth. We find no exception at 5721 grid points across $14 \le t \le 300$ nor at 301 across $9877 \le t \le 9880$. Three things are wanted for it, and each is worth naming. The count is read off Z, so what it counts is zeros on the critical line, and it is the hypothesis again that makes those all of them. The zeros must be simple: at a double zero Z and Z' vanish together, $W = 0$, the argument is undefined, and the curve would climb by one where the count climbs by two; RH does not supply simplicity. And ϑ_1^* must be read on a continuous branch. Subtracting (131) from (139) gives $\vartheta_1^* = \pi\big(N_* - N_{\mathrm{rvm}} - \frac{1}{2}\big)$, so the principal branch serves precisely while the exact count and the smooth one stay within one of each other; since S is unbounded that must eventually fail, and past it the wrap count has to be carried explicitly, which is the bookkeeping N_{ps} already needs at every height. There is room to spare at these heights: over $14 \le t \le 300$ the angle $|\vartheta_1^*|$ reaches only 0.506π.

Read the other way, (140) can only fail where $ZZ'' > Z'^2$, which at a critical point of Z means Z and Z'' of one sign: a positive local minimum or a negative local maximum, a pair of zeros that comes to the axis and fails to reach it. The Laguerre inequality $\Xi'^2 \ge \Xi\Xi''$, which Laguerre proved for every real entire function with only real zeros, is precisely what forbids that. Being a consequence of the hypothesis and not a theorem about ζ, it leaves the monotonicity conditional; and being only the first of a family of inequalities of every order, which Csordas and Varga [5] show to be equivalent to RH when taken all together, it is weaker than the hypothesis. A failure of monotonicity would therefore disprove RH, while a proof of monotonicity would not prove it. Nothing is close to the boundary. On grids of spacing 0.1 the slope never turns negative and dips no lower than 0.431 over $10 \le t \le 200$, 0.403 over $200 \le t \le 300$, 0.375 over $1000 \le t \le 1030$ and 0.521 over $9877 \le t \le 9880$ beside γ_{10000}. The near misses do not threaten it either, and they push ρ the opposite way from what one might guess. At Lehmer's pair [12] $\gamma_{6709} = 7005.06287$ and $\gamma_{6710} = 7005.10056$ the hump between the two sits at $t = 7005.08186$ with $Z = 3.97 \times 10^{-3}$, and there $\rho = 458$, because $-Z''/\vartheta'^2 Z$ grows without bound as a hump flattens onto the axis. What such a pair costs is margin, not sign: $Z'^2 - ZZ''$ reads 0.089 at that hump against 1.53 and 1.38 just outside the pair. Verified by `check_nstar_mono.py`.

Remark 12.5 (regarding the slope of N_* and the size of $|Z|$)**.** Since $\pi N_* = \arg W + \frac{3\pi}{2}$ and the positive factor $\frac{1}{2}$ in W leaves an argument alone, $\pi N_* - \frac{3\pi}{2} = \arg\bigl(Z - iZ'/\vartheta'\bigr)$ is the polar angle of the plane point $(Z, -Z'/\vartheta')$, the Prüfer angle familiar from oscillation counting with ϑ' in the role of the frequency. The slope of the counting curve is therefore fixed by Z, Z' and Z'' at that height alone, and measured against the local mean density it is the ρ of (140). Two values of it are exact. At an ordinate Z vanishes and (134) with $h^* = -Z'/2\vartheta'$ gives $\rho = 1$, which we confirm to twelve digits at γ_1, γ_{100}, γ_{1000} and γ_{10000}; the whole of the variation therefore lives strictly between zeros. At an extremum of Z, where $Z' = 0$ leaves $W = \frac{1}{2}Z$ real,

$$\rho = -\frac{Z''}{\vartheta'^2 Z} \qquad (Z' = 0), \tag{143}$$

so flatness at a hump is exactly a small curvature-to-height ratio, in units of ϑ'^2: a broad tall hump gives ρ well below 1 and a sharp small one gives ρ well above it. The consequence is that the flatness of N_* ranks the size of $|Z|$ between the zeros. Over the 100 gaps of $14 \le t \le 238$, ρ at the hump against the height of that hump has rank correlation -0.987, with $|Z|_{\max} \propto \rho^{-1.17}$; two decades higher the same measurement reads -0.943 and $\rho^{-1.30}$ at γ_{1000}, and -0.944 and $\rho^{-1.20}$ at γ_{10000}. Since $\rho = 1$ at every ordinate, the first local datum there is $\rho'(\gamma_n)$, and that single number, read at one zero, ranks the height of the hump ahead of it at -0.80, -0.66 and -0.73 in the three blocks: the more steeply the slope falls away as one leaves a zero, the taller the hump being entered. Figure 38 plots all of this, the slope against $|Z|$ in two windows and then the two rankings. What the slope is really measuring is the local phase rate of Z rather than its amplitude, since for $Z = A\cos\varphi$ with $\varphi' = \vartheta'$ one gets $\rho = 1$ whatever A may be; $\rho < 1$ says the phase is turning more slowly than ϑ', which is the same statement as a wide gap, and it is the sparseness of the zeros that lets $|Z|$ grow. Consistently with that, ρ at the hump and the gap width are collinear to -0.99 in rank, so the slope should be read as a pointwise version of the spacing rather than as independent information about $|Z|$. Verified in `check_nstar_slope.py`.

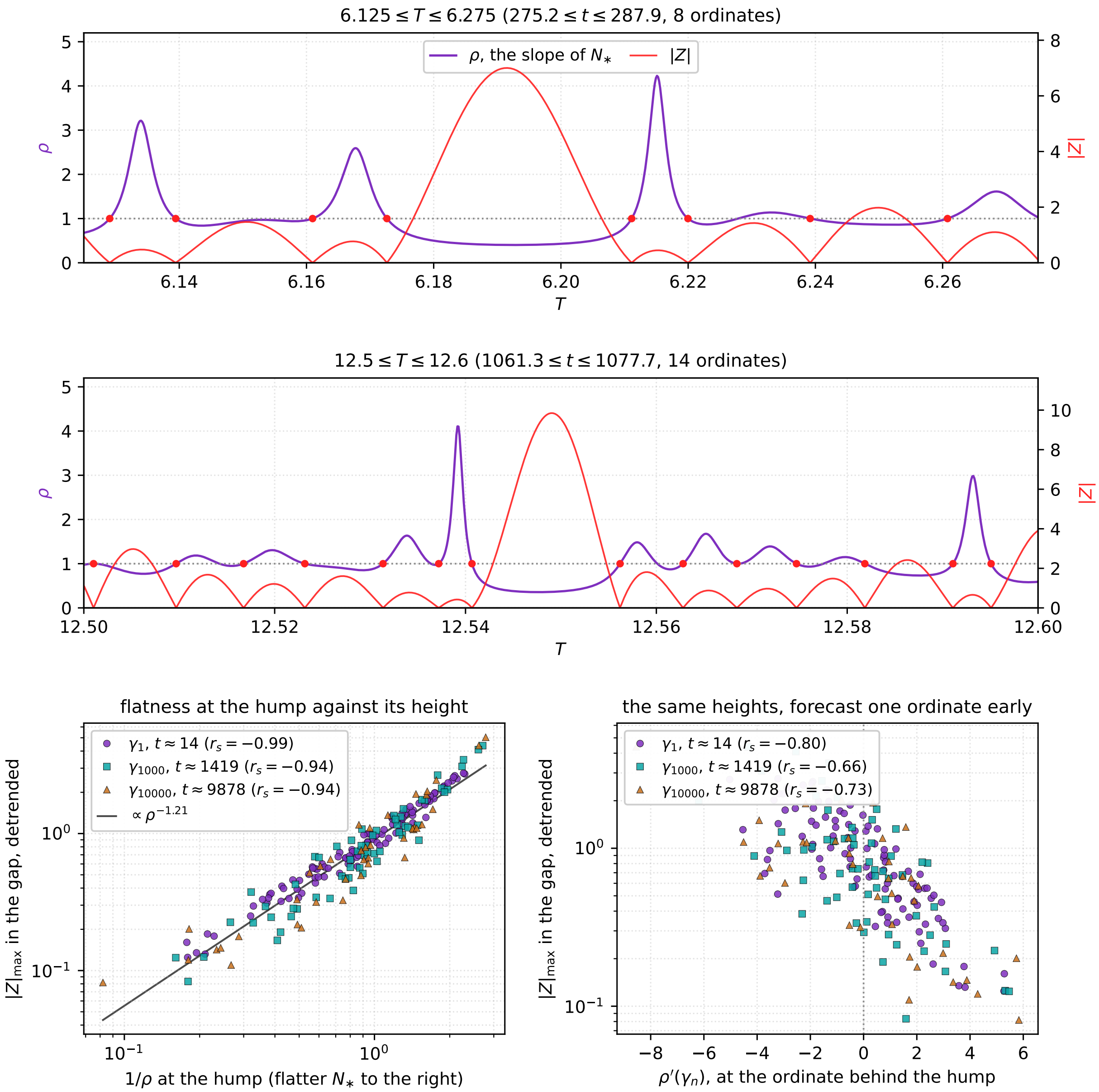

Figure 38: The slope ρ of (140) against the size of $|Z|$ between ordinates, everything at $\sigma = \frac{1}{2}$. *Top:* ρ in purple on the magnified window of Figure 36, with $|Z|$ in red on the right-hand scale. The red dots mark the ordinates, where $\rho = 1$ exactly; between them ρ dips, and it dips deepest where $|Z|$ rises highest. *Second:* the same reading twice as high in the index, over $12.5 \le T \le 12.6$. The hump that reaches $|Z| = 9.85$ near $T = 12.549$ sits on the flattest stretch of the window, where ρ bottoms at 0.356, while each ordinary hump carries only a shallow arch of ρ just above 1 and the two narrowest gaps throw up spikes of 4.11 and 2.99. *Bottom left:* one point per gap for three blocks of ordinates spanning two decades of t, the height of the hump against $1/\rho$ at that hump, both on log scales, with the heights divided by a 31-gap moving average so the slow growth of $|Z|$ with height does not enter. The line is the fit over all three blocks; the legend quotes the rank correlation against ρ for each block separately. *Bottom right:* the same hump heights against $\rho'(\gamma_n)$, read at the ordinate behind the hump rather than at the hump itself, scaled by the mean spacing π/ϑ'. Generated by `fig_nstar_slope.py`.

12.5 The envelope of $\left|\zeta\left(\frac{1}{2},t\right)\right|$

On the critical line the chain is isosceles for every reflection-symmetric split, by Corollary 9.1, so its two legs share one length $L_1 = |B_1| = |B_2|$ and the base is the chord they subtend across the folding angle ϑ_2 of §9.2. From $\zeta = B_1(1+e^{i\vartheta_2})$,

$$\left|\zeta\left(\tfrac{1}{2}+it\right)\right| = 2L_1\left|\cos\tfrac{\vartheta_2}{2}\right|, \tag{144}$$

so the modulus of ζ splits into a length carried by the links and an angle read at the joint between the legs. Bounding the angular factor bounds ζ. Write $u = \vartheta_2/\pi$ reduced to $[0,2)$. Then $\left|\cos\frac{\pi u}{2}\right| \le 1$ at once, and since $\cos\frac{\pi u}{2}$ is concave on $0 \le u \le 1$ it stays above the chord joining its endpoints there, giving $\left|\cos\frac{\pi u}{2}\right| \ge |1-u|$ on the whole of $[0,2)$ by the reflection $u \mapsto 2-u$, with equality only at $u = 0,1,2$. Hence

$$2L_1\left|1-\frac{\vartheta_2}{\pi}\right| \;\le\; \left|\zeta\left(\tfrac{1}{2}+it\right)\right| \;\le\; 2L_1. \tag{145}$$

The upper envelope is attained where the two legs point the same way and the chain is folded straight; the lower one is attained at every ordinate, where $u=1$ and both sides vanish, and again where u reaches 0 or 2 and the two envelopes meet. The triangular factor $|1-\vartheta_2/\pi|$ is the same sawtooth whose passages through zero do the counting in §12.4, appearing here with an amplitude on it. Over $6 \le T \le 13$, which is $264.9 \le t \le 1144.6$, we confirm (144) to 1.7×10^{-20} and both inequalities of (145) at every one of 2001 sample points, the two legs agreeing to 2.0×10^{-20}.

Which split to use is settled by how often the upper envelope is reached. Equation (133) gives $2L_1 = \sqrt{Z^2+4h^2}$, so $|\zeta| = 2L_1$ exactly where the offset h vanishes. For the split at the partial summand h vanishes only inside a gap whose two ends carry opposite h, which is the alternation that the count of §12.4 needs and that fails at a retrograde ordinate, so the envelope is reached in some gaps and missed in others: 3 of the 7 gaps of $6.125 \le T \le 6.275$, and 7 of the 13 gaps of $12.5 \le T \le 12.6$. The velocity split has no such gaps. Its offset $h^* = -Z'/2\vartheta'$ vanishes at every extremum of Z, so

$$2L_1^* = \sqrt{Z^2+\left(\frac{Z'}{\vartheta'}\right)^2} \;\ge\; |Z| = \left|\zeta\left(\tfrac{1}{2}+it\right)\right|, \tag{146}$$

with equality on every hump: 7 touches in 7 gaps and 13 in 13 in those two windows, matched to 10^{-25}. This is the familiar envelope of an oscillation of instantaneous frequency ϑ', with Z'/ϑ' as the quadrature companion of Z, and it is tighter on average as well, the mean of $|\zeta|/2L_1$ over $6 \le T \le 13$ being 0.635 against 0.586 for the split at the partial summand.

Both readings are polar coordinates of one function. The rotated bisector point $W = e^{i\vartheta}B_1^* = \frac{1}{2}Z - \frac{iZ'}{2\vartheta'}$ of (138) has modulus L_1^* and argument $\pi\left(N_* - \frac{3}{2}\right)$ by (139), and $Z = 2\,\mathrm{Re}\,W$, so

$$Z(t) = -2\,|W|\,\sin\bigl(\pi N_*(T)\bigr), \tag{147}$$

which we confirm to 6×10^{-13}, the precision of the counting curve itself. This is a restatement rather than a new fact, but it puts the two subsections together: the amplitude of the velocity split is the envelope of ζ and its phase is the counting curve, so the zeros are the moments when the phase passes an integer and the extrema are the moments when the amplitude is attained. Remark 12.5 reads the rate of that phase; the envelope (146) reads its amplitude. Verified in `check_envelope.py`.

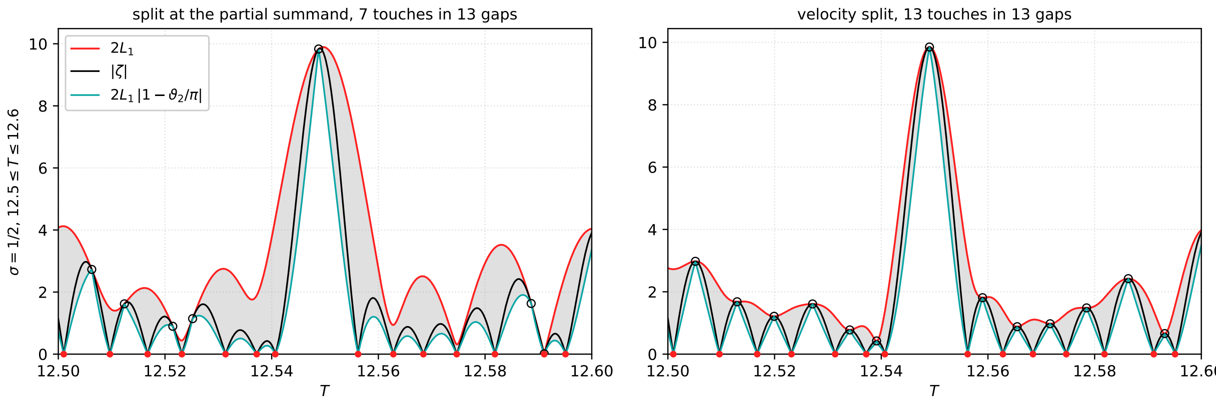

Figure 39: The two envelopes of (145), at $\sigma = \frac{1}{2}$, over $12.5 \le T \le 12.6$, the second window of Figure 38, which is $1061.3 \le t \le 1077.7$ and holds 14 ordinates. In each panel $|\zeta|$ is black, the upper envelope $2L_1$ red, the lower envelope $2L_1|1-\vartheta_2/\pi|$ a teal sawtooth, and the grey band between them is where $|\zeta|$ is confined. Red dots on the axis are the ordinates, where the lower envelope touches zero along with $|\zeta|$, and the open circles are the touches of the upper envelope, one for each vanishing of the offset h. *Left:* the split at the partial summand, $B_1 = \Sigma_1 + R_{1ps}$, whose h fails to alternate at a retrograde ordinate, so it reaches its bound in only 7 of the 13 gaps and several humps rise nowhere near it. *Right:* the velocity split, whose h^* vanishes at every extremum of Z, so the upper envelope is touched in every gap. Generated by `fig_envelope.py`.

Remark 12.6 (regarding hunting for large values with the first few links)**.** Neither the envelope nor the slope of Remark 12.5 can be used to scout, since $2L_1$ needs $R = \zeta - \Sigma_1 - \Sigma_2$ and $2L_1^*$ needs Z and Z', so both are functions of ζ at a point one has already evaluated. What can be computed without ζ is the reason the envelope grows. Leg 1 is mostly the link sum, so the size of $2L_1$ is a question of how well the link phases $t\log n$ align, and the ceiling is the aligned case,

$$\left|\zeta\left(\tfrac{1}{2}+it\right)\right| \;\le\; 2L_1 \;\le\; 2\sum_{n\le m} n^{-1/2} + 2|R_{1ps}| \;\approx\; 4\sqrt{T} \;\approx\; 4\left(\frac{t}{2\pi}\right)^{1/4} = \frac{4}{(2\pi)^{1/4}}\,t^{1/4} \;\approx\; 2.53\,t^{1/4}, \tag{148}$$

a bound that costs nothing to evaluate. Read in t it is nothing new: the exponent $\frac{1}{4}$ is the convexity, or trivial, bound $\mu(\frac{1}{2}) \le \frac{1}{4}$ of Lindelöf [13], where $\mu(\sigma)$ is the least exponent with $\zeta(\sigma+it) = O(t^{\mu})$, and (148) recovers it with an explicit constant. Nor can this path do better. The last step is the triangle inequality, which keeps only the aligned case and so discards every cancellation among the phases $t\log n$; in the geometry of this paper $4\sqrt{T}$ is the arc length of the chain laid out straight, while $|\zeta|$ is the distance between its endpoints once it has curled into its spirals. Every improvement past $\frac{1}{4}$ comes from that discarded cancellation, beginning with the $\frac{1}{6}$ that Hardy and Littlewood obtained by Weyl's method on this same sum and standing today at $\frac{13}{84}$ [2]; the Lindelöf hypothesis is $\mu(\frac{1}{2}) = 0$. Two things follow, both drawn in Figure 40. First, the alignment can be read from a truncation. Scoring an interval by $S_K(t) = \left|\sum_{n\le K} n^{-1/2}e^{-it\log n}\right|$ and then evaluating Z only at the strongest peaks of that score recovers most of the interval's largest $|Z|$: averaged over the 19 unit intervals $5 \le T \le 24$, which is $190 \le t \le 3771$, the three strongest peaks of the five-link score reach 0.967 of it and the single strongest peak of the ten-link score reaches 0.961. The point is not the saving at these heights but that K need not grow with t while m does. Second, the interval maxima themselves are nearly regular: $\max|Z|$ over $[T, T+1]$ fits $2.93\,T^{0.485}$, its ratio to $\sqrt{T}$ has mean 2.823 and standard deviation 0.123 across those intervals, and it holds a fraction of the ceiling

([148](#)) that ranges from 0.97 down to 0.76, the two extremes falling at $T = 9$ and $T = 22$. Most of that spread is the ceiling rather than the maxima: the exact sum $\sum_{n \le m} n^{-1/2} = 2\sqrt{m} + \zeta(\frac{1}{2}) + o(1)$ climbs toward its own asymptote $2\sqrt{m}$ across the range, from 0.72 of $4\sqrt{T}$ at $T = 5$ to 0.86 at $T = 23$, since $\zeta(\frac{1}{2}) = -1.4604$ still weighs heavily at $m = 5$. Against $\sqrt{T}$ itself the maxima are flat, 2.711 at $T = 5$ and 2.670 at $T = 23$, which is the same fact as the fitted exponent 0.485 being essentially the ceiling's $\frac{1}{2}$: at these heights the maxima simply ride the trivial bound. In these coordinates the Lindelöf hypothesis is precisely the statement that they eventually stop, that $\max|Z|/\sqrt{T}$ falls like $T^{-1/2+\varepsilon}$, the chain curling up so efficiently that its endpoints stay bounded while its arc length grows like $\sqrt{T}$. Nothing of that is visible over 19 intervals with $m \le 23$ links, and nothing here bears on it either way. So the answer to whether the next unit of the index will beat the last is yes by default, which it is in 13 of the 18 consecutive comparisons, and the exceptions are what needs predicting: on the maxima divided by $\sqrt{T}$ the peak of the ten-link score ranks the intervals at $+0.72$, while the three-link score, whose peak is nearly the same in every interval, carries nothing at -0.11. That last figure rests on 19 intervals at modest height and should be read as an indication only. Verified in `check_large_values.py`.

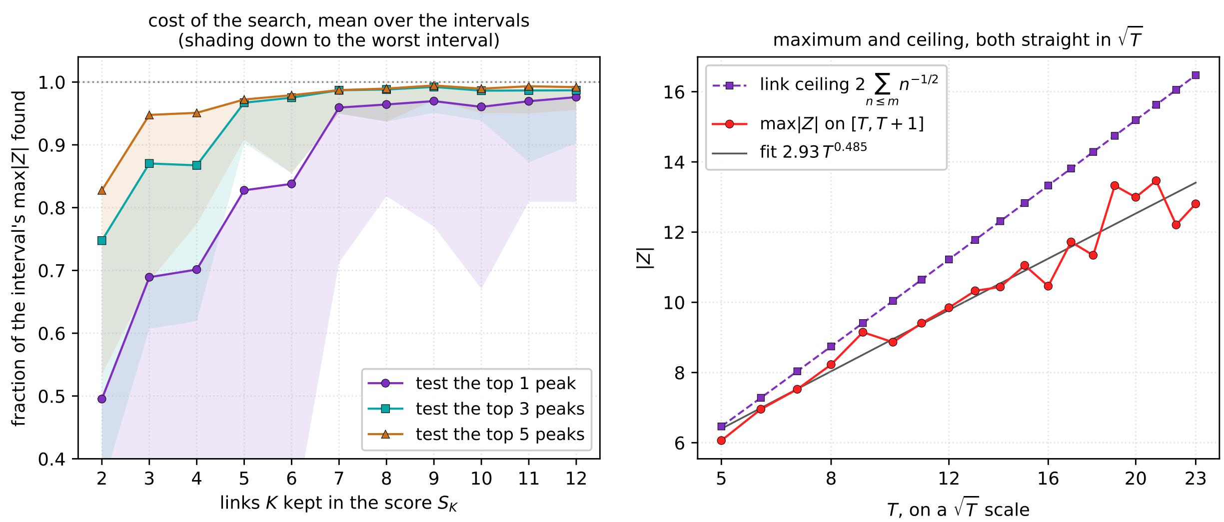

Figure 40: Reading large values off the links, over the 19 unit intervals $5 \le T \le 24$. *Left:* the fraction of an interval's largest $|Z|$ found by evaluating Z only at the strongest one, three or five peaks of the K-link score S_K, averaged over the intervals, with the shading reaching down to the worst single interval. Seven links suffice for a single candidate to land within 5%. *Right:* the interval maximum against $\sqrt{T}$, on which it is nearly straight, with the link ceiling of ([148](#)) above it and the fitted power law through it. The gap between the two curves is the alignment the links never quite achieve. Generated by `fig_large_values.py`.

12.6 Collinearity of the three first-half remainders

Collinearity of three points is unchanged by a common translation, so the shared tail Σ_1 of the three heads $B_{1ps}, B_{1rs}, B_{1ak}$ of §9.4 cancels, and the three remainder vectors R_{1ps}, $R_{1rs} = \frac{1}{2}R$, and R_{1ak} can be compared directly as points in the plane. Figure 41 plots two independent, scale-invariant measures of how far these three points are from lying on a common line. The first is the triangle flatness

$$F = \frac{2\,|\text{area}|}{\text{diam}^2}, \tag{149}$$

where the area is that of the triangle with the three points as vertices and the diameter is the largest of the three pairwise distances. The second is the aspect ratio $\lambda_{\min}/\lambda_{\max}$ of the covariance matrix of the three points (a principal-component measure). Both quantities are zero exactly when the points are collinear, and both are invariant under translation, rotation, and rescaling, so the shrinking of the remainders with growing T cannot masquerade as collinearity. Each panel is a grid of 1501 σ-samples by 1801 T-rows over $0 \le \sigma \le 1$, $3.9 \le T \le 5.1$, computed from the exact remainder formulas, with the σ-sample nearest $\frac{1}{2}$ snapped exactly onto the critical line. The color scale is logarithmic, and that is what separates *exact* collinearity (values at the double-precision floor, 10^{-16} to 10^{-12}) from the merely small values, 10^{-4} to 10^{-1}, that fill the rest of the strip.

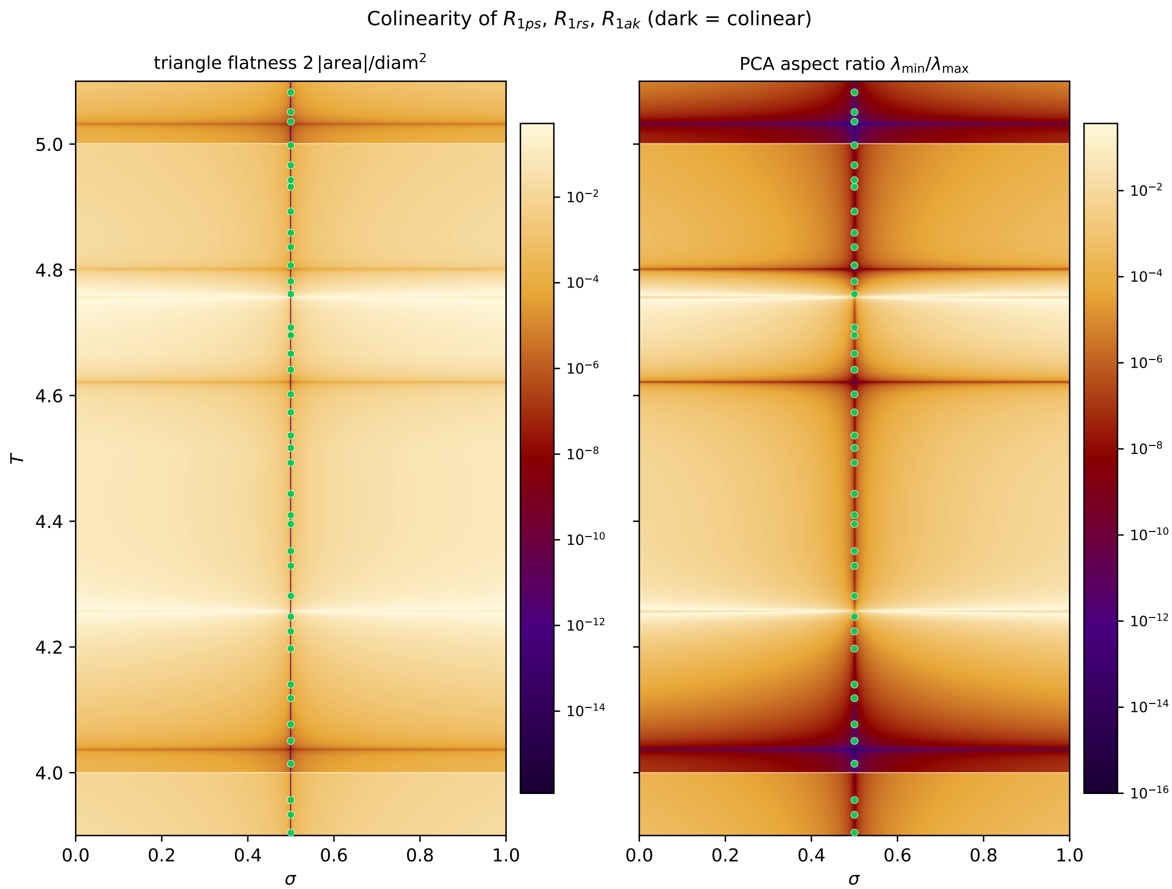

Figure 41: Collinearity of the three first-half remainders R_{1ps}, $R_{1rs} = \frac{1}{2}R$, R_{1ak} (dark = collinear). Horizontal axis $\sigma \in [0,1]$; vertical axis the spiral index $T \in [3.9, 5.1]$. Left: triangle flatness (149); right: PCA aspect ratio $\lambda_{\min}/\lambda_{\max}$. Both panels use a log color scale; green markers are zeta zeros at $\sigma = \frac{1}{2}$. The dark column pinned at the numerical floor along $\sigma = \frac{1}{2}$ shows the three points are exactly collinear everywhere on the critical line; off the line collinearity occurs only along the isolated dark horizontal bands. Generated by `fig_colinearity_zoom.py` from data produced by `remainder-colinearity-heatmap-zoom.mjs`.

Both panels show the same two facts. First, the column $\sigma = \frac{1}{2}$ sits at the numerical floor at every T: on the critical line the three first-half remainders are exactly collinear, not merely nearly so. Second, off the line the measures are many orders of magnitude larger everywhere except along isolated dark horizontal bands (near $T \approx 4.03$, 4.62, 4.80, and 5.03 in this window), where the triangle momentarily passes through a flat configuration as its signed area changes sign. The measured values are symmetric under $\sigma \mapsto 1 - \sigma$, a reflection of the functional equation.

The same pattern persists across the whole strip. Figure 42 repeats the four-panel layout of Figure 22 (the same three zoom bands and the same full strip $0 \leq T \leq 20$), with the PCA aspect ratio as the plotted field. As in the leg-imbalance panels, the dark column on $\sigma = \frac{1}{2}$ runs unbroken through every window, while off the line the only dark features are the isolated horizontal collinearity bands.

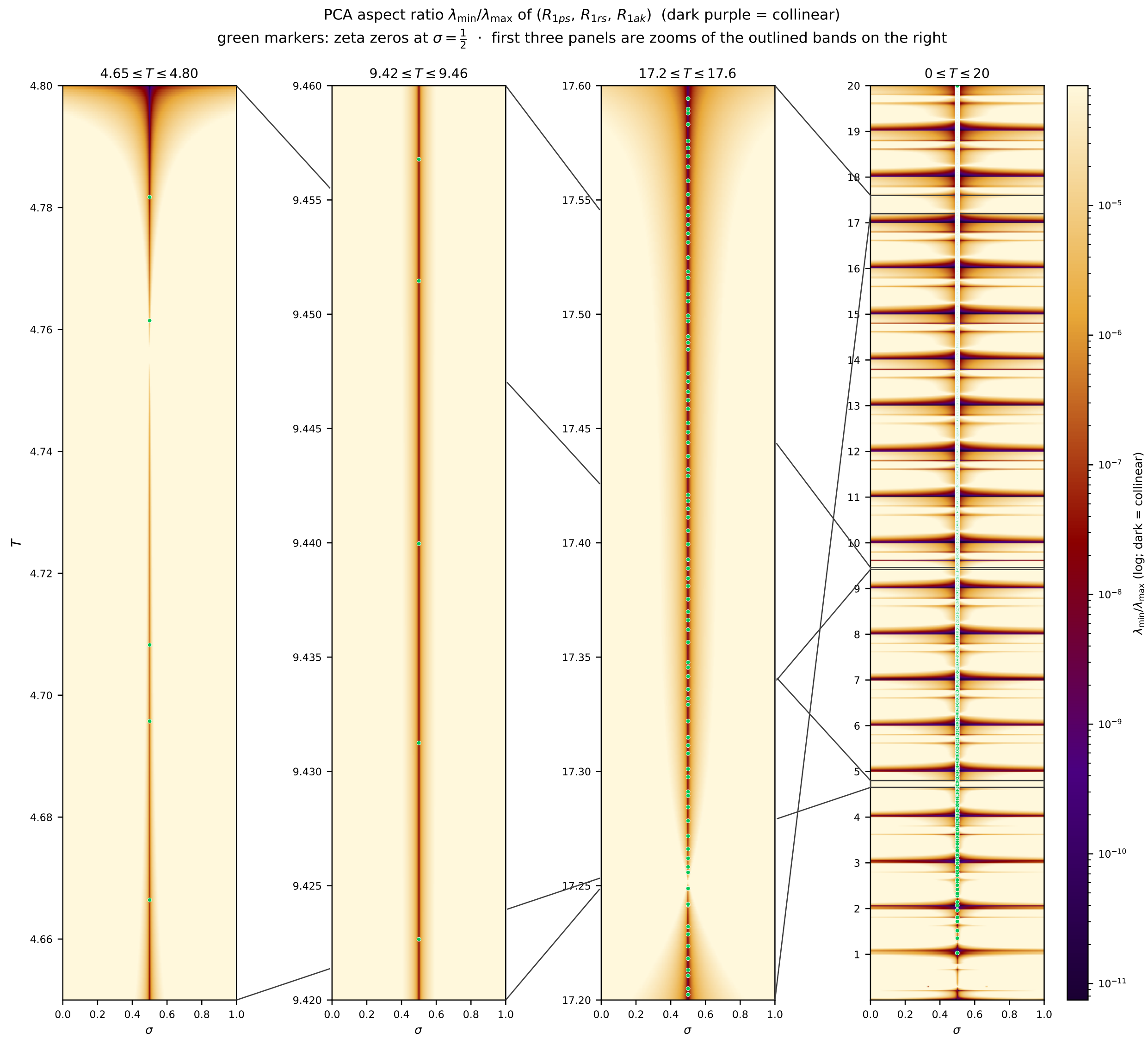

Figure 42: PCA aspect ratio $\lambda_{\min}/\lambda_{\max}$ of the three first-half remainders $(R_{1ps}, R_{1rs}, R_{1ak})$ (dark purple = collinear), in the four-panel layout of Figure 22. Horizontal axis $\sigma \in [0, 1]$; vertical axis the spiral index T. Color is on a log scale. Green markers are zeta zeros at $\sigma = \frac{1}{2}$. The rightmost panel is the full strip $0 \le T \le 20$; the first three panels are zooms of the outlined bands ($4.65 \le T \le 4.80$, $9.42 \le T \le 9.46$, $17.2 \le T \le 17.6$). The dark vertical band on $\sigma = \frac{1}{2}$ is the exact collinearity of the three first-half remainders on the critical line; the dark horizontal bands are the isolated off-line collinearity events. Generated by `plot_colinearity_four_panel.py`.

Why can this collinearity not be used like the two routes of §9.5? Because collinearity is *not* forced at a zero. For $\zeta \neq 0$ the equal-leg condition for a split says precisely that its head B_1 is equidistant from O and ζ, i.e. that B_1 lies on the perpendicular bisector of the chord from O to ζ; if all three splits had equal legs at a point where $\zeta \neq 0$, all three heads would lie on that single line and the three first-half remainders would be collinear. But at a zero the chord degenerates: $\zeta = O$, every point of the plane is equidistant from the two endpoints, and the (now automatic) equal-leg conditions carry no collinearity information at all. All three leg pairs are equal at a zero, as they must be, while the three points can remain in general position. So a zero implies equal legs for all three splits, but not collinearity, and the three points need not line up at an off-line zero.

A proof through collinearity therefore needs a bridge that the routes of §9.5 do not: first, prove that the exact collinearity along $\sigma = \frac{1}{2}$ is an identity of the three remainder formulas rather than a numerical coincidence; second, prove that off the line it fails outside isolated curves like the dark bands of Figure 41; and third, connect a hypothetical off-line zero to the collinearity structure, for instance through nearby off-line points with $\zeta \neq 0$ at which all three splits have equal legs, which by the perpendicular-bisector argument above would force a collinearity that the off-line geometry forbids. The first two steps would establish collinearity as an exact characterization of the critical line; only the third would turn it into a statement about zeros.

12.7 Joint angles

The joint angles of the forward chain can be read on their own, without the chain, and the Zest joint-angle view does exactly that: one dot per joint, the turning angle on the vertical axis, the joint index along the horizontal. The turning angle at joint n is the continuous joint angle (85) evaluated at an integer joint,

$$\theta_n \;=\; \operatorname{fold}\Big(-\,I(T)\,\log\tfrac{n}{n-1}\Big), \qquad \nu(n) \;=\; \frac{I(T)}{2\pi\, n(n-1)}, \tag{150}$$

where fold wraps to $[-\pi, \pi]$ and ν, the derivative of the unwrapped angle divided by 2π, is the local frequency of the strip in cycles per joint. Two features organize the picture, both consequences of $I(T)$. First, at the bisector joint $n = \lfloor T \rfloor + 1$ the frequency is $\nu = 1 + \frac{1}{6T(T+1)}$, one full turn per joint to nine decimals at $T = 13000$, and the angle there is $-\pi$ exactly: that is (86), the fold of §10.1, seen as the right edge of the strip. The joints just behind it inherit almost the same angle, 3.141109, 3.139659, 3.137242 at $n = T, T-1, T-2$, which is the same near-stationarity that makes the bisector frame of Figure 23 repeat. Second, since $\nu(n) \approx T^2/n^2$ falls from $\nu \sim 10^8$ at $n = 2$ to 1 at that fold, the strip is a sampled chirp, and it is stratified by the joints where ν is the reciprocal of a rational. Writing that rational as a Farey fraction $f = p/q$ in $(0, 1]$, its *caustic joint* is the solution of $n(n-1) = f\, I(T)/2\pi$, that is $\nu = 1/f$, so

$$n_f \;=\; \tfrac{1}{2}\Big(1 + \sqrt{1 + 2f\, I(T)/\pi}\Big) \;\approx\; T\sqrt{f}, \tag{151}$$

and there the joints fall into exactly p strands, the denominator of $\nu = q/p$, each a parabola of curvature $4\pi\nu/n$ per joint squared because the unwrapped angle is locally quadratic (exactly $|\widetilde{\theta}''(n_f)|$ of (153) below). Small denominators give the spines visible in Figure 43, drawn at the integer index $T = 13000$, where $t = I(T) = 1{,}061{,}939{,}999.369541$: there the fractions $\frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}$ sit at $n = 6501, 7506, 9193, 10615, 11259$, and $f = 1$ is the fold itself. The parabolic strands are the same local quadratic phase that produces the Cornu-like arcs of the curlicue literature of §13.2, here read off the joints rather than the partial sums.

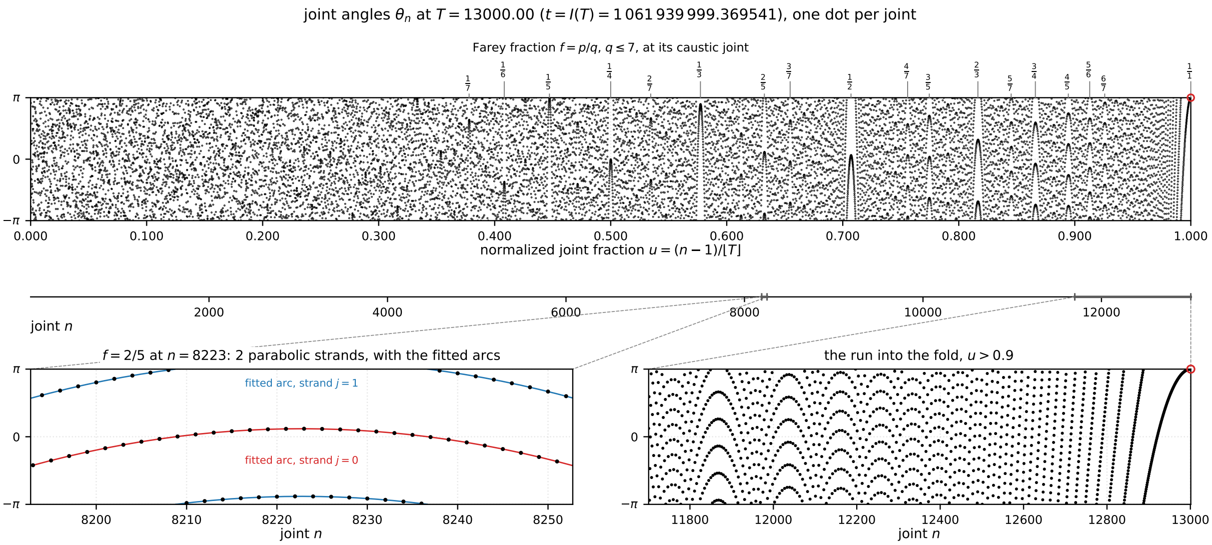

Figure 43: The joint-angle strip at $T = 13000$, where $t = I(T) = 1{,}061{,}939{,}999.369541$, one dot per joint of the forward chain, $n = 2, \ldots, \lfloor T \rfloor + 1$, with θ_n of (150) on the vertical axis. The two edges $\pm\pi$ are the same fold, so the strip is a cylinder. *Top:* all 13000 joints, carrying three horizontal scales: the Farey fractions $f = p/q$ with $q \le 7$ above, each at its caustic joint (151) and staggered in two rows where they crowd; the normalized joint fraction $u = (n-1)/\lfloor T \rfloor$ below, so the fold sits at $u = 1$; and the joint index n below that. The red circle at the right edge is the bisector joint $n = \lfloor T \rfloor + 1$, where $\theta = -\pi$ exactly. *Bottom left:* a window at $f = \frac{2}{5}$, showing its $p = 2$ parabolic strands as dots, each with its fitted arc (155) drawn through them; the two arcs sit half a turn apart, so strand 1 has its vertex wrapped to the opposite edge. *Bottom right:* the last tenth of the strip, where the strands steepen and run into that fold. Dashed lines carry each window back to the stretch of the joint-n scale it magnifies, marked there by a bracket. Generated by `fig_joint_angles.py`.

The fitted curve. The strands are not merely parabola-like; they can be written down. Let $\widetilde{\theta}(n) = -I(T) \log \frac{n}{n-1}$ be the unwrapped angle, so $\theta_n = \operatorname{fold} \widetilde{\theta}(n)$, and differentiate it three times:

$$\widetilde{\theta}'(n) = \frac{I(T)}{n(n-1)}, \qquad \widetilde{\theta}''(n) = -\frac{I(T)\,(2n-1)}{[\,n(n-1)\,]^2}, \qquad \widetilde{\theta}'''(n) = \frac{2\,I(T)\,(3n^2 - 3n + 1)}{[\,n(n-1)\,]^3}. \tag{152}$$

At a caustic these have closed forms in the fraction alone, since (151) fixes $n_f(n_f - 1) = f\,I(T)/2\pi$ and $2n_f - 1 = \sqrt{1 + 2f\,I(T)/\pi}$ exactly:

$$\begin{aligned} \widetilde{\theta}'(n_f) &= \frac{2\pi}{f} = \frac{2\pi q}{p}, \qquad \widetilde{\theta}''(n_f) = -\frac{4\pi^2}{f^2 I(T)} \sqrt{1 + \frac{2f\,I(T)}{\pi}} \;\approx\; -\frac{4\pi}{f^{3/2} T}, \\ \widetilde{\theta}'''(n_f) &= \frac{16\pi^3}{f^3 I(T)^2} \left(1 + \frac{3f\,I(T)}{2\pi}\right) \;\approx\; \frac{12\pi}{f^2 T^2}. \end{aligned} \tag{153}$$

Write $\delta = n - n_f$ for the offset in joints from the caustic. The linear term $\widetilde{\theta}'(n_f)\,\delta = (2\pi q/p)\,\delta$ is a carrier sweeping several radians per joint, and it is what interleaves the dots into strands in the first place. Along one strand, though, δ advances by p at a time, and $(2\pi q/p)\,pm = 2\pi q\,m$ folds away: the carrier contributes nothing but a constant. Absorbing it together with $\widetilde{\theta}(n_f)$, strand j carries the phase constant

$$C_j \;=\; \widetilde{\theta}(n_f) + \frac{2\pi q}{p}\big([n_f] + j - n_f\big), \qquad j = 0, \ldots, p-1, \tag{154}$$

with $[\,\cdot\,]$ the nearest integer, and the strand itself is the folded cubic

$$\rho_j(\delta) \;=\; \operatorname{fold}\Big(C_j + \tfrac{1}{2}\widetilde{\theta}''(n_f)\,\delta^2 + \tfrac{1}{6}\widetilde{\theta}'''(n_f)\,\delta^3\Big). \tag{155}$$

Because $\gcd(p, q) = 1$ the constants C_j run through all p residues of $2\pi/p$, so the p arcs are one parabola stacked at equal spacing around the cylinder. Third order is the right order: the second derivative supplies the parabola, the third its visible skew, and the fourth is smaller by a further $\delta/n_f \sim T^{-1/2}$.

The arcs drawn in Figure 43 are (155) at $f = \frac{2}{5}$ and $T = 13000$, where $n_f = 8222.74$, $\widetilde{\theta}''(n_f) = -3.8208 \times 10^{-3}$ and $\widetilde{\theta}'''(n_f) = 1.3941 \times 10^{-6}$ radians per joint squared and cubed, and the carrier is $2\pi q/p = 5\pi \equiv \pi$, which is why the two strands sit half a turn apart. Over the ± 30 joints shown they track the dots to 2.2×10^{-5} radians, that being the fourth-order term: despite the name it carries in the Zest joint-angle view, nothing is fitted in the statistical sense, and (155) is evaluated exactly. Two limits are worth noting. Toward small f the parabola stiffens like $f^{-3/2}$, which is why the left end of the strip looks like noise at this scale: consecutive joints there are many turns apart and only the caustics of very small f remain resolved. At the other end $f = 1$ closes the family, with $p = q = 1$: a single strand, a carrier of one full turn per joint, and $C_0 = -\pi$. That is the bisector joint of (86) again, so the fold this paper is built on is the last caustic of the sequence, and the right-hand panel of Figure 43 is the run of the high-denominator caustics into it.

12.8 Incremental change

The strip is a snapshot. Advance the index and every dot moves, though not by the same amount. Differentiating the unwrapped angle in T at a fixed joint,

$$\begin{aligned} \frac{\partial\widetilde{\theta}}{\partial T}(n) &= -\,I'(T)\log\frac{n}{n-1}, \qquad L = \log\Big(1 + \tfrac{1}{T}\Big), \\ I'(T) &= \frac{\pi}{L^2}\Big(2L + \frac{2T+1}{T(T+1)}\Big) = 4\pi T + 2\pi + O(T^{-1}), \end{aligned} \tag{156}$$

which in cycles is a drift rate

$$\mu(n) \;=\; \frac{I'(T)}{2\pi}\log\frac{n}{n-1} \;\approx\; \frac{2T+1}{n - \frac{1}{2}} \;\approx\; \frac{2}{u}, \qquad u = \frac{n-1}{\lfloor T \rceil}. \tag{157}$$

A value of μ is easiest to read as a lap count: $\mu(n) = 3$ means that dot laps the strip three times, falling from $+\pi$ to $-\pi$, wrapping, and repeating, while you advance T by one. Nothing here needs differencing: $I'(T)$ is available in closed form as $\pi\big(2L + (2T+1)/T(T+1)\big)/L^2$ with $L = \log(1 + 1/T)$, which is $4\pi T + 2\pi$ to leading order, about 163,369 at $T = 13000$.

The sign is the same everywhere, so the whole strip drifts downward; the magnitude is not, and that is the point. Since $\mu(n) \approx (2T+1)/(n - \frac{1}{2}) \approx 2/u$, the rate is essentially a $1/u$ hyperbola, equal to exactly 2 at the bisector joint $u = 1$ and growing leftward to 1.8×10^4 cycles per unit index at $n = 2$ — where, incidentally, the $2/u$ form is no longer any good, reading 26,000, since $\log\frac{n}{n-1}$ and $1/(n - \frac{1}{2})$ part company at the first few joints. The strip does not translate as the index advances, it shears. Read the other way, $1/\mu(n)$ is the index step that returns joint n to the same angle: a dot at $u = \frac{1}{2}$ comes back around every $\Delta T = 0.25$, one at $u = 0.1$ every $\Delta T = 0.05$, and one at the fold every half unit.

At the right edge the value is exact and already familiar: the middle expression of (157) at $n = \lfloor T \rfloor + 1$ is $(2T+1)/(T+\frac{1}{2}) = 2$, and the exact rate is 2.000000001 at $T = 13000$. Two cycles per unit index is one more than (86) reports for the joint angle, and the discrepancy is the joint itself moving: the fold sits at $n = \lfloor T \rfloor + 1$, which advances one joint per unit index, and one joint there is worth $\nu = 1$ turn, so an observer riding the fold sees $\mu - \nu = 1$ cycle, namely $\frac{d}{dT}\pi(2T+1) = 2\pi$. The two statements differ only in whether one rides the fold or stands at a fixed joint.

The pattern the dots drift through, by contrast, hardly moves at all. Differentiating (151) gives $dn_f/dT = f\, I'(T)\big/2\pi(2n_f - 1) \approx \sqrt{f}$, so a caustic creeps outward by a fraction of a joint per unit index, and its normalized position $u_f = (n_f - 1)/\lfloor T \rfloor \approx \sqrt{f}$ does not depend on T at all: the spines stand still while the joints stream through them. Their drift rate follows from pairing (157) with the cycles per joint $\nu = I(T)/2\pi n(n-1)$ of (150), which gives $\mu \approx 2\sqrt{\nu}$: at the caustic of Farey fraction f, where $\nu = 1/f$, the dots slide at $2/\sqrt{f}$ cycles per unit index, four at $f = \frac{1}{4}$ and two at $f = 1$. Only at an integer index does the frame change, when a new link is appended, a new joint appears at the right edge with $\theta = -\pi$, and u rescales by one part in $\lfloor T \rfloor$.

Figure 44 shows one step, $\Delta T = 0.02$, which advances t by $\Delta I = 3267.38$. The travel is $0.04/u$ cycles: 14.4° at the fold, 28.8° at $u = \frac{1}{2}$, 143.9° at $u = 0.1$, half a turn at $u = 0.0800$, joint $n = 1041$, and a full turn at $u = 0.0400$, joint $n = 521$. Since the motion is downward everywhere, the arrows in the figure are drawn downward and wrapped at the lower edge, which is where they land for joints left of $n = 1041$. Left of $n = 521$ the joint has gone round at least once, and rather than draw what is left over the arrow there sweeps the whole strip from π to $-\pi$; at $n = 2$ the travel is 2264 radians, some 360 turns. This is the sense in which the left end of the strip is a chirp in the index as well as in the joint number: it is not only that neighboring joints are many turns apart at fixed T, but that one joint sweeps many turns over a step in T too small to move the fold by a quarter of a radian. In the Zest joint-angle view μ is the blue cycles-per-T overlay, drawn against its own axis rescaled to whatever window is on screen; the bottom panel of Figure 44 is that overlay on a logarithmic axis, and the middle panel is a picture of the same quantity at $\Delta T = 0.02$.

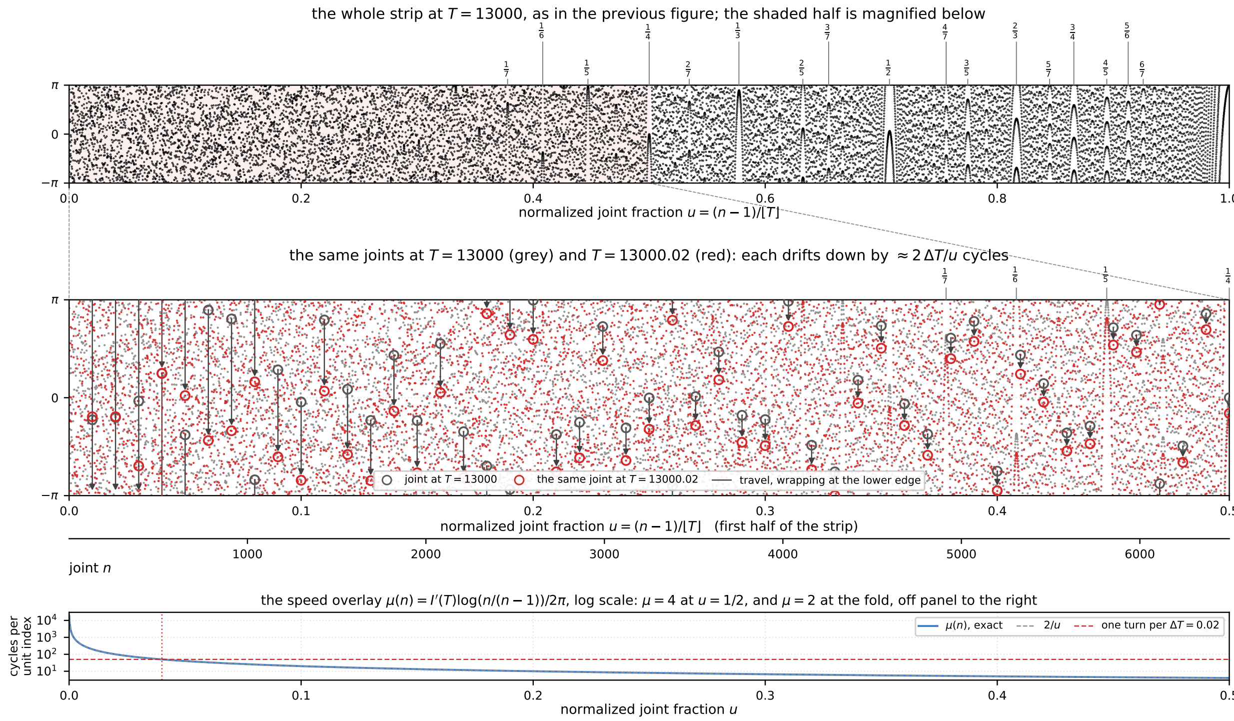

Figure 44: One increment of the index, at $T = 13000$. *Top:* the whole joint-angle strip as in Figure 43, with the shaded first half magnified below. *Middle:* that half twice over, grey at $T = 13000$ and red at $T = 13000.02$, every dot having drifted downward by $\mu(n)\,\Delta T \approx 0.04/u$ cycles of (157). Fifty joints spaced evenly across the panel are circled in both colours, with an arrow along the path travelled; since the drift is downward at every joint the arrows run downward and wrap at the lower edge, and where the travel exceeds a full turn the arrow sweeps the whole strip, from π to $-\pi$. The travel lengthens leftward like $1/u$, passing half a turn at $n = 1041$ and a full turn at $n = 521$, marked by the dotted vertical line. Both strips carry the Farey fractions $f = p/q$, $q \le 7$, each at its caustic joint (151); only $f \le \frac{1}{4}$ falls in the magnified half, since $u_f \approx \sqrt{f}$. *Bottom:* the drift rate itself on a logarithmic axis: exact (blue) against the $2/u$ form (grey), the two separating only at the first few joints, with the dashed line at one full turn per $\Delta T = 0.02$. Generated by `fig_incremental_change.py`.

12.9 Bearings of champions

The chain of this paper stops at the fold, but the chain of ζ itself runs much further, and one link far out along it carries a signature of the largest values of $|\zeta|$. Call a height a *champion* when $\left|\zeta(\frac{1}{2}+it)\right|$ exceeds every value it has taken below that height, a running record. We use a ground-truth list of 149 of them, from $t = 10.21$ to $t = 8.13 \times 10^{11}$, over which $|\zeta|$ climbs from 1.55 to 334.[6]

The link in question is the last one, the term at

$$n_{\text{last}} = \lfloor t/\pi \rfloor \;\approx\; 2T^2, \tag{158}$$

the index following from $\sqrt{t/2\pi} \approx T + \frac{1}{2}$ (§10). Its own heading is $\alpha = -t \ln n_{\text{last}}$, and the reading we want is the direction of the world origin $J_0 = 0$ as seen in the frame that lays this link along

[6] Provenance of the champion list: AIM workshop on AI and number theory.

the positive real axis: the change of frame of §10.1, applied at the far end of the chain instead of at the fold. Two features of that end of the chain make the reading cheap. First, the turning angle there is $-t\ln\frac{n}{n-1} \approx -t/n_{\mathrm{last}} = -\pi$ by the choice (158), so consecutive links are antiparallel and each added link reverses the frame; only the axial direction modulo 180° is meaningful, and the pattern drifts from joint to joint at the negligible rate $\nu = 1/4T^2$ cycles of (150). Second, the chain out there straddles its own limit. Taking $x = t/\pi$ in the approximate functional equation [20, Thm. 4.11] gives $\zeta(s) = \sum_{n\le x} n^{-s} + x^{1-s}/(s-1) + O(x^{-1/2})$, whose correction has modulus $1/\sqrt{\pi t}$, so the far joints lie within $O(t^{-1/2})$ of ζ; what they in fact do is bracket it, each end of the last link sitting half a link away, $|J-\zeta| = 2.806\times10^{-2}$ against $|v|/2 = 2.804\times10^{-2}$ at $t = 10^3$. That places ζ at the midpoint of the last link, to 4.8×10^{-5} there and to 1.4×10^{-7} at $t = 10^5$. Read at a Gram point, where ζ is real, that midpoint is therefore on the real axis, and the bearing collapses to the closed form

$$\beta \;=\; 180^\circ - \alpha \;=\; 180^\circ + t\ln\lfloor t/\pi\rfloor \pmod{180^\circ}, \tag{159}$$

a function of the height alone with no chain to sum, even though the chain in question has 2.6×10^{11} links at the top of the list.

Figure 45 draws the frame itself at six champions, one to a decade of the height, each of them the champion of its decade sitting closest to the mean axis. What the pictures show is that the chain does not spiral out to ζ so much as march. At $t = 6.73\times10^8$ its links spend $2\sqrt{n_{\mathrm{last}}} = 2.9\times10^4$ of arc length to cover a displacement of 124, a ratio of 236, and yet never stray more than 2.4 from the straight line joining its two ends: the arc length goes on loops the size of the links themselves, inside a thin tube whose heading is (159). Only the last stretch of the tube, where the logarithmic radius of the figure compresses least, shows those loops as loops.

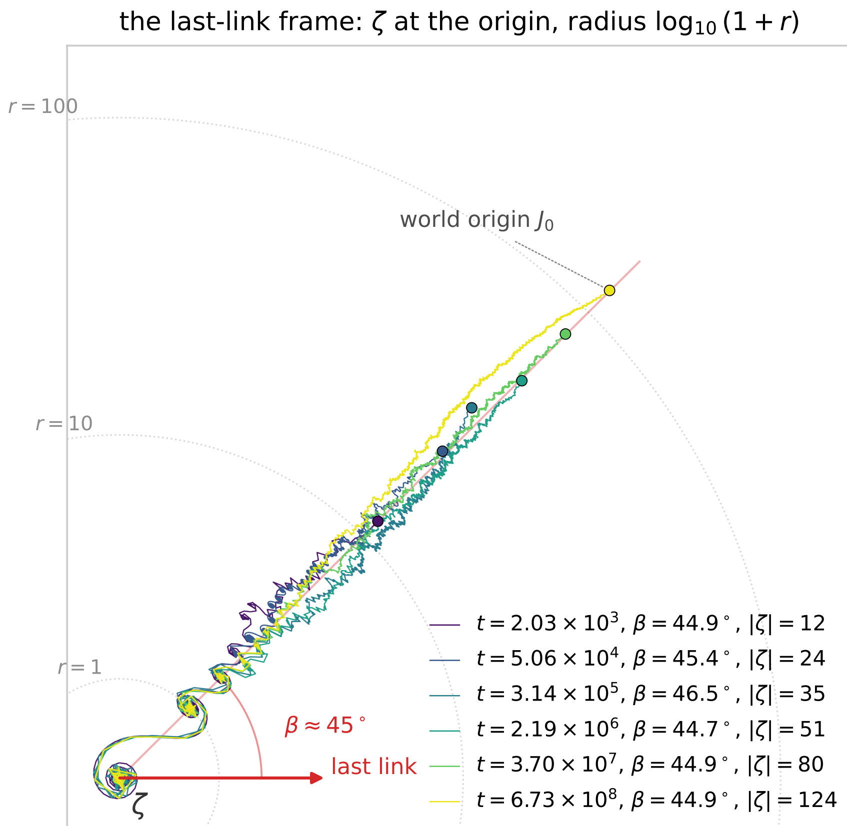

Figure 45: The chain of ζ in the frame of (159), at the champion of each decade of the height from 10^3 to 10^9 whose bearing lies closest to the mean axis. The frame is anchored at the far end of the chain: the last link lies along the positive real axis with its midpoint, which is ζ, at the origin, and each chain runs inward to it from the world origin $J_0 = 0$, marked by a dot at radius $|\zeta|$ in the direction β. The last link is drawn far longer than its true modulus $n_{\text{last}}^{-1/2}$, which is 6.8×10^{-5} at the outermost height against a picture 124 wide. The radius carries the same logarithmic compression as Figure 46: a point at distance r is drawn at $\log_{10}(1+r)$, which leaves every direction, and so the bearing, exact, while letting six decades of height share one frame; the compression is gentlest near ζ, which is why the coils of the final approach read as coils while the loops out near J_0 are flattened into the thread. The dotted rings are $r = 1, 10, 100$. Consecutive links out there being antiparallel, the frame is fixed only up to a half turn, and the representative drawn is the one that puts J_0 in the upper right. Chains are sampled at a fixed step of arc length, which keeps the 2.1×10^8 links of the outermost one plottable. Generated by `fig_last_link_frame.py`.

Read that at the Gram point nearest each champion and the bearings do not scatter. Over the 148 champions above $t = 17.8$, dropping only the first at $t = 10.21$ where the chain is three links long, the mean axis is $44.8°$ with Rayleigh concentration $R = \left|\text{mean } e^{2i\beta}\right| = 0.967$ against a noise floor of $1/\sqrt{148} = 0.082$, and 146 of the 148 lie between $25°$ and $65°$, the median distance from $45°$ being $4.3°$. The same count of ordinary Gram points, drawn log-uniformly over the same range of heights, gives $R = 0.052$ with 31 in that window, which is what a uniform bearing would give. So (159) is not manufacturing the direction; something about a record pins the last link of its chain to one axis. Written without any of the geometry, the observation is arithmetic:

$$\text{at a champion,} \qquad t \ln\lfloor t/\pi \rfloor \;\equiv\; 45° \quad (\text{mod } 180°). \tag{160}$$

What we did not expect is that the concentration tightens with height instead of fading. Splitting the list by decade of t:

Table 3: The last-link bearing by decade of the height: the mean axis never leaves 45° while the spread halves. R is the axial concentration and the last column is the median distance from 45°. Computed in `fig_champion_bearings.py`.

decade of t	champions	mean axis	R	median $\lvert\beta - 45°\rvert$
10^1–10^2	6	38.0°	0.852	12.9°
10^2–10^3	9	45.7°	0.885	8.5°
10^3–10^4	16	48.3°	0.949	7.7°
10^4–10^5	14	42.6°	0.975	5.7°
10^5–10^6	10	47.2°	0.982	3.5°
10^6–10^7	11	45.8°	0.984	5.5°
10^7–10^8	16	43.2°	0.985	3.7°
10^8–10^9	16	47.0°	0.987	3.7°
10^9–10^{10}	13	44.1°	0.989	2.8°
10^{10}–10^{11}	16	43.3°	0.987	3.5°
10^{11}–10^{12}	21	44.7°	0.993	3.2°

The ragged ring is the innermost one, the six champions below $t = 100$, whose bearings run 25.6°, 56.7°, 12.8°, 54.0°, 31.0° and 46.7°: the construction needs a chain of some length before it means anything, the first champion having three links.

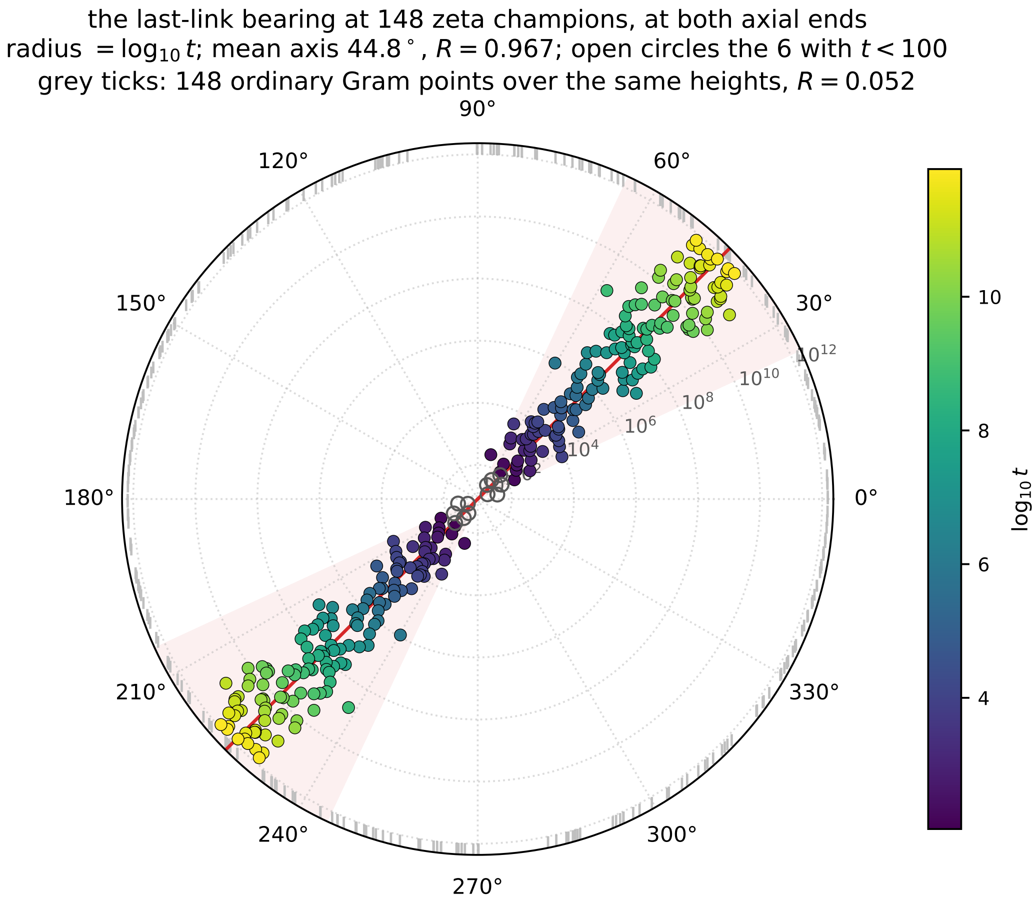

Figure 46: The last-link bearing (159) at the 148 champions above $t = 17.8$, read at the nearest Gram point. The bearing is axial, so each champion is drawn at both ends of its axis; the radius is $\log_{10} t$, the colour the height, the red line the circular mean and the shaded wedge $\pm 20^\circ$ about it. Open circles are the six champions below $t = 100$. The grey ticks around the rim are 148 ordinary Gram points over the same range of heights, whose bearings are uniform ($R = 0.052$). The beam narrows outward: the tightest decade is the outermost, $10^{11} \leq t < 10^{12}$, at $R = 0.993$. Generated by `fig_champion_bearings.py`.

Two cautions on what this is. It is a description of where records sit, not a way of finding them: (160) says nothing about which of the many heights satisfying it carries a large $|\zeta|$, and separating records from merely large values is the business of the alignment of Remark 12.6, not of a single phase. And the pinning itself is unexplained. The last link is one term of modulus $n_{\mathrm{last}}^{-1/2} \approx 1/\sqrt{2}\, T^{-1}$, far too small to matter to the size of ζ, so (160) is a coincidence of phases and not a mechanism; why a record height should fix the phase of that particular term to one axis, at every height reachable, we cannot say.

13 Prior literature

13.1 Nickel's Argand-diagram geometry

Nickel [14], with a condensed sequel [15], starts from the same object this paper starts from, the Argand diagram of the steps n^{-s}, and reaches a construction close enough to ours that the two are worth setting side by side. They agree in shape, and they differ in the one respect this paper is about.

Dictionary. His center of symmetry is the step $n_p = [\sqrt{t/2\pi}\,]$, with fractional part $p = \sqrt{t/2\pi} - n_p$. Since $\sqrt{t/2\pi} \approx T + \frac{1}{2}$ (§10.1), his index is $n_p \approx \lfloor T + \frac{1}{2} \rfloor$ and $p \approx \{T + \frac{1}{2}\}$, so it agrees with our $m = \lfloor T \rfloor$ on the first half of each unit interval and exceeds it by one on the second half: the same half-unit offset already recorded for Siegel's cutoff in §7.1. His factor $Q(s)$, defined by $\zeta(s) = Q(s)\zeta(1-s)$, is our $\chi(s)$; the displayed form $Q = n_p^{1-2s} e^{i(t+\pi/4)}$ is the familiar $e^{-2i\vartheta(t)}$ asymptotic, which reproduces χ to a relative 1.5×10^{-5} at $T = 20.3$ and 1.6×10^{-7} at $T = 200.3$ once the smooth $\sqrt{t/2\pi}$ is used in place of the rounded n_p. His symmetry angle Θ is our $\frac{\psi}{2} = \frac{1}{2}\arg\chi$.

What matches. Under that dictionary his Riemann–Siegel equation $\zeta = P(s) + Q(s)P(1-s)$ is our two-leg decomposition $\zeta = B_1 + B_2$ with $B_2 = \chi\,\overline{B_1}$ (Appendix B); his polar form $\zeta = 2\cos(\phi - \Theta)\,\mathsf{P}e^{i\Theta}$ is the shape of our $R = 2d_1\cos(\omega + \frac{\psi}{2})e^{i\psi/2}$; and his observation that the two magnitudes are equal exactly when $\sigma = \frac{1}{2}$ is Corollary 9.1. He also writes the classical remainder as $R = 2\mathsf{L}\cos(\theta_L - \Theta)$, where $L = \mathsf{L}e^{i\theta_L}$ runs from the joint at step n_p to his center: that is exactly the role played here by the pair $(d_1, -\omega)$. Where his index agrees with ours the two projected remainders converge, the ratio $2\mathsf{L}\cos(\theta_L - \Theta)\,/\,2d_1\cos(\omega + \frac{\psi}{2})$ falling through 2.24, 1.34, 1.11, 1.03, 1.01 at $T = 6.3$, 20.3, 60.3, 200.3, 600.3.

Two centers, one line. His $P(s)$ and our B_1 are not the same point, and they are not meant to be. The difference lies almost entirely along the direction in which ζ cannot see it: at $T = 600.3$ we find $|P - B_1| = 0.064$, of which only 1.2×10^{-4} lies along $e^{i\psi/2}$. That is his own remark that ζ is unchanged when $P(s)$ is translated along the symmetry axis. In the notation of §9.2 the admissible centers are the solutions X of $\zeta - X = \chi\overline{X}$, a line in the plane, and every one of them projects onto $\zeta/2$, which is Corollary 9.2. The freedom he notes and the projection we prove are the same fact.

The sequel. Two things in [15] touch this paper further. It traces $P(s)$ and $\zeta(s)$ as two curves over a range of t, joined at regularly spaced values of t to show the two segments, which is the figure the conveyor belt draws in Fig. 34 with his pendant center in place of B_1. And it counts zeros: it reads the count off Θ through the Gram points and tests that reading against the first 100,000 ordinates, finding one zero per Gram point on average with the zeros grouping between successive Gram points. The count there is the classical one, exact in the mean and never landing on an ordinate; the curves of §12.4 land on every one.

What differs. The correction itself. Nickel's is a lowest-order determination, carrying a factor $1/(2n_p^\sigma\cos 2\pi p)$ that grows without bound as $\cos 2\pi p \to 0$, and he is explicit that its error matters for the distribution of $P(s)$ even though it cancels in ζ; its direction, $e^{-i(t\ln n_p + 2\pi p)}$, is the step direction at n_p turned by the fractional offset $2\pi p$, not a summand direction of the chain. Our R_{1ps} and R_{2ps} are exact, and their directions are exactly those of the $(m+1)$st summand of each chain. That is what lets the serial chain of §6.2 close on ζ identically rather than asymptotically, and it is the content of the "one more partial summand" reading: the correction is not merely near the next link, it lies along it.

13.2 Spirals of exponential sums

For quadratic phases this geometry is rigorous ground. The partial sums $S_N(\tau) = \sum_{n\le N} e^{i\pi\tau n^2}$ draw what Berry and Goldberg [1] call curlicues: replacing the sum by an integral produces the Cornu spiral exactly, and a renormalization then carries the whole pattern onto a magnified and rotated copy of a shorter sum of the same kind. Coutsias and Kazarinoff [3, 4] work that picture the way this paper works ours. They give the polyline a discrete radius of curvature

$$R_N = \tfrac{1}{2}\bigl|\csc(\psi_N/2)\bigr|, \qquad \psi_N = \pi\tau(2N+1), \tag{161}$$

ψ_N being the turn between consecutive terms, so that inflections of the pattern sit where ψ_N is nearest a multiple of 2π and cusps where it turns around. They then cut the sum *at an inflection*, and replace each whole spiral of the first sum by a single vector carrying a scale and a phase, the Fresnel length $\sqrt{|\tau|}$ at $\pi/4$ to the spiral's mid-vector. What comes out is Hardy and Littlewood's approximate functional formula for the theta function, sharpened to an error uniform in τ and stated as a Diophantine inequality that contains Gauss's sum formula. Cutting at the fold and standing one link in for a whole spiral is the reading of §4; their $e^{i\pi/4}$ is the Gaussian integral's, the same factor that leaves the $-\pi/8$ in ϑ; and (161) is the bisector construction of §9.1 for unit links, the distance from a step to the meeting of the angle bisectors at its two ends, which is Nickel's L once the length $n^{-\sigma}$ is restored.

The arithmetic is what does not carry over. Their turn ψ_N is linear in N, so the second difference is the constant $2\pi\tau$, the arcs are Cornu spirals outright, and the theory turns on the continued fraction of τ: the pattern is a self-similar hierarchy of curlicues within curlicues. Our turn is $\theta_n = -t\log\bigl(1+\frac{1}{n}\bigr)$ and its second difference falls like t/n^2, passing 2π once. That is why the chain has a single distinguished fold, at $n \approx \sqrt{t/2\pi}$, instead of a hierarchy of them, and why its arcs are Cornu spirals only locally. The geometric machinery transfers; the renormalization does not.

Two further readings of the same diagram deserve mention. Lander's paired series [10, 11] reduce each pseudo-spiral to a principal-axis vector, and those axis vectors form a second chain running counter to the first, with matched arguments and with magnitudes that separate as σ leaves $\frac{1}{2}$: the counter-running chain of §11 and the equal lengths of Corollary 9.1 in embryo. Kapitonets [8] draws his axis of symmetry through the *midpoint* of the remainder vector at $\sigma = \frac{1}{2}$, observes that the remainder is then perpendicular to that axis, and concludes both that $\arg\zeta$ is the normal direction and that the two chains project onto the axis with cancelling sums. That is the half-split of §7.2 together with Corollary 9.2, read off the picture. His other construction, recovering ζ by repeatedly replacing a polygon of partial sums by the polygon of their midpoints, is unrelated to the bisector point: iterated midpointing is the binomial transform, and the limit is Cesàro summation of the Dirichlet series, as he says himself.

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- Freeman Dyson for his article *Birds and Frogs* (`https://pdodds.w3.uvm.edu/files/papers/others/2009/dyson2009a.pdf`), which inspired me to begin my journey to investigate one dimensional quasi-crystals and their possible connection to the Riemann Hypothesis nine years ago.
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A Lean formalization of $R = R_{1ps} + R_{2ps}$

This appendix reproduces the Lean file formalizing the algebraic content of Theorem 4.1. The file contains no `axiom` and no `sorry`: every step is proved, and the only axioms the final theorem depends on are Lean's own three (`propext`, `Classical.choice`, `Quot.sound`), as reported by `#print axioms`. It was checked against Lean 4.32.2 with Mathlib at commit `905b95818e`. The Lean code was written with the assistance of AI coding assistants (Anthropic's Claude).

A.1 Correspondence with the written proof

The file follows the proof of Theorem 4.1 step for step. The definitions `d_j1` and `d_j2` are the sine weights (19)–(20), and `R_1ps` and `R_2ps` are Step 1, transcribed from (17)–(18) in the $\lceil T\rceil^{\mp iI(T)}$ form rather than the $e^{\mp i\omega}$ form. Two small bridges convert them: `ceil_cpow` proves $\lceil T\rceil^{ir} = e^{ir\log\lceil T\rceil}$ for $T > 0$, which with $\omega(T) = I(T)\log\lceil T\rceil$ (the definition `omg`) is exactly $\lceil T\rceil^{iI(T)} = e^{i\omega}$; and `div_norm_eq_exp_arg` proves $z/|z| = e^{i\arg z}$ for $z \neq 0$, applied to χ to give $\chi/|\chi| = e^{i\psi}$. Then `factoring_lemma` is the regrouping of Step 2, pulling out the common factor $|R|/\sin(2\omega+\psi)$, and `key_algebraic_identity` is the content of Steps 3–5, namely

$$e^{-i\omega}\sin(\omega-\phi+\psi) + e^{i(\omega+\psi)}\sin(\omega+\phi) \;=\; e^{i\phi}\sin(2\omega+\psi); \tag{162}$$

it is proved by writing each sine as $(e^{i\theta} - e^{-i\theta})/2i$ (`sin_to_exp`, itself proved from Mathlib's $e^{i\theta} = \cos\theta + i\sin\theta$), clearing the denominator, merging the resulting products of exponentials into single exponentials, and letting `ring_nf` cancel the two copies of $e^{i(\psi-\phi)}$. The closing lines are Step 6: cancel $\sin(2\omega+\psi)$ and recognize $|R|e^{i\phi} = R$, which is Mathlib's `Complex.norm_mul_exp_arg_mul_I`.

A.2 What is proved and what is hypothesized

The theorem `R_1ps_plus_R_2ps_eq_R` carries three hypotheses. Two are needed for the objects to exist at all: $\chi(\sigma + iI(T)) \neq 0$, without which $\chi/|\chi|$ is meaningless, and $\sin(2\omega + \psi) \neq 0$, without which d_1 and d_2 of (19)–(20) are undefined. The third, $T > 0$, makes $\lceil T \rceil$ a positive base, so that its complex power can be written as an exponential. Nothing is assumed about whether T is an integer.

One convention has to be read carefully. Steps 2–6, that is `factoring_lemma` and `key_algebraic_identity`, hold for arbitrary real ω, ϕ, ψ and make no reference to T; the only T-dependent ingredient is the bridge `ceil_cpow`, which holds for every $T > 0$ with ω defined as $I(T) \log\lceil T \rceil$. That is the paper's ω whenever T is not an integer, since then $\lceil T \rceil = \lfloor T + 1 \rfloor$. At integer T the two part company, and the sign bookkeeping in the last paragraph of the proof of Theorem 4.1 turns on the factor $(-1)^{\lfloor T \rfloor}$ sitting inside Siegel's R; that factor is invisible here, because R enters as a free complex parameter, so what the file proves at integer T is the same identity read with $\lceil T \rceil$ throughout.

What remains outside the formalization is not algebra but analysis. The functional-equation factor χ, the index map I, and Siegel's remainder R enter as free parameters, so what is machine-checked is that any R written in polar form splits as $R_{1ps} + R_{2ps}$ with the weights (19)–(20), which is exactly Steps 1–6. That R is Siegel's remainder, and that I is the map of §10.1, are inputs to the theorem rather than consequences of it.

A.3 Source listing

```
/-
  Formalization of Theorem (R = R_1ps + R_2ps) from
  "Remainder Terms of the Riemann Zeta Function".

  Contains no ‘axiom‘ and no ‘sorry‘: every step is proved.
-/
import Mathlib.Analysis.SpecialFunctions.Complex.Circle
import Mathlib.Analysis.SpecialFunctions.Pow.Complex

open Complex Real

/-! ### Phases -/

/-- The phase of Siegel’s remainder, ‘φ = arg R‘. -/
noncomputable def phiR (R : ℂ) : ℝ := Complex.arg R

/-- The phase of the functional-equation factor, ‘ψ = arg χ(σ + i I(T))‘. -/
noncomputable def psi (χ : ℂ → ℂ) (σ T : ℝ) (If : ℝ → ℝ) : ℝ :=
  Complex.arg (χ (σ + If T * I))

/-- The turn angle ‘ω(T) = I(T) · log⌈T⌉‘, so that ‘⌈T⌉^{i I(T)} = e^{iω}‘. -/
noncomputable def omg (T : ℝ) (If : ℝ → ℝ) : ℝ := If T * Real.log (⌈T⌉ : ℝ)

/-! ### The two fractional link lengths -/

noncomputable def d_j1 (R : ℂ) (w f p : ℝ) : ℂ :=
  ((‖R‖ * (Real.sin (w - f + p) / Real.sin (2 * w + p)) : ℝ) : ℂ)

noncomputable def d_j2 (R : ℂ) (w f p : ℝ) : ℂ :=
  ((‖R‖ * (Real.sin (w + f) / Real.sin (2 * w + p)) : ℝ) : ℂ)

/-! ### The two partial-summand remainders, exactly as in the paper -/
```

```
noncomputable def R_1ps (σ T : ℝ) (R : ℂ) (χ : ℂ → ℂ) (If : ℝ → ℝ) : ℂ :=
  d_j1 R (omg T If) (phiR R) (psi χ σ T If) * ((⌈T⌉ : ℝ) : ℂ) ^ (-(I * (If T : ℂ)))

noncomputable def R_2ps (σ T : ℝ) (R : ℂ) (χ : ℂ → ℂ) (If : ℝ → ℝ) : ℂ :=
  d_j2 R (omg T If) (phiR R) (psi χ σ T If) * ((⌈T⌉ : ℝ) : ℂ) ^ (I * (If T : ℂ))
    * (χ (σ + If T * I) / (‖χ (σ + If T * I)‖ : ℂ))

/-! ### Step 0: elementary bridges -/

/-- `⌈T⌉^{i r} = e^{i r log⌈T⌉}` for `T > 0`, with no hypothesis on the
integrality of `T`. -/
lemma ceil_cpow (T r : ℝ) (hT : 0 < T) :
    (((⌈T⌉ : ℝ)) : ℂ) ^ (I * (r : ℂ))
      = Complex.exp (I * ((r * Real.log (⌈T⌉ : ℝ) : ℝ) : ℂ)) := by
  have hpos : (0 : ℝ) < (⌈T⌉ : ℝ) := lt_of_lt_of_le hT (Int.le_ceil T)
  have hne : (((⌈T⌉ : ℝ)) : ℂ) ≠ 0 := by
    simpa using (ne_of_gt hpos)
  rw [Complex.cpow_def_of_ne_zero hne, ← Complex.ofReal_log hpos.le]
  congr 1
  push_cast
  ring

/-- `z/‖z‖ = e^{i arg z}` for `z ≠ 0`. -/
lemma div_norm_eq_exp_arg (z : ℂ) (hz : z ≠ 0) :
    z / (‖z‖ : ℂ) = Complex.exp (I * (Complex.arg z : ℂ)) := by
  have hn : ((‖z‖ : ℝ) : ℂ) ≠ 0 := by
    simpa using norm_ne_zero_iff.mpr hz
  rw [div_eq_iff hn, mul_comm I ((Complex.arg z : ℝ) : ℂ),
    mul_comm (Complex.exp (((Complex.arg z : ℝ) : ℂ) * I)) ((‖z‖ : ℝ) : ℂ)]
  exact (Complex.norm_mul_exp_arg_mul_I z).symm

/-- Exponential form of the sine of a real argument, valued in `ℂ`. -/
lemma sin_to_exp (θ : ℝ) :
    (Real.sin θ : ℂ)
      = (Complex.exp (I * (θ : ℂ)) - Complex.exp (-I * (θ : ℂ))) / (2 * I) := by
  have h1 : Complex.exp (I * (θ : ℂ)) = Complex.cos (θ : ℂ) + Complex.sin (θ : ℂ) * I := by
    rw [mul_comm]; exact Complex.exp_mul_I _
  have h2 : Complex.exp (-I * (θ : ℂ)) = Complex.cos (θ : ℂ) - Complex.sin (θ : ℂ) * I := by
    have h : (-I * (θ : ℂ)) = (-(θ : ℂ)) * I := by ring
    rw [h, Complex.exp_mul_I, Complex.cos_neg, Complex.sin_neg]; ring
  rw [h1, h2, Complex.ofReal_sin]
  have hI : (2 * I : ℂ) ≠ 0 := by simp [Complex.I_ne_zero]
  field_simp
  ring

/-! ### Steps 3-5: the key algebraic identity -/

/-- The heart of the theorem: the two sine-weighted phasors combine into a
single phasor along `e^{iφ}`. -/
theorem key_algebraic_identity (w f p : ℝ) :
    Complex.exp (-I * (w : ℂ)) * (Real.sin (w - f + p) : ℂ) +
    Complex.exp (I * ((w + p : ℝ) : ℂ)) * (Real.sin (w + f) : ℂ) =
    Complex.exp (I * (f : ℂ)) * (Real.sin (2 * w + p) : ℂ) := by
  rw [sin_to_exp, sin_to_exp, sin_to_exp]
  have hI : (2 * I : ℂ) ≠ 0 := by simp [Complex.I_ne_zero]
  field_simp
  simp only [mul_sub, ← Complex.exp_add]
  push_cast
```

```
  ring_nf

/-! ### Step 2: factoring -/

/-- Pulling the common factor ‘|R|/sin(2ω+ψ)‘ out front. -/
lemma factoring_lemma (absR w f p : ℝ) :
    ((absR * (Real.sin (w - f + p) / Real.sin (2 * w + p)) : ℝ) : ℂ) *
        Complex.exp (-(I * (w : ℂ))) +
    ((absR * (Real.sin (w + f) / Real.sin (2 * w + p)) : ℝ) : ℂ) *
        Complex.exp (I * (w : ℂ)) * Complex.exp (I * (p : ℂ)) =
    ((absR : ℝ) : ℂ) / (Real.sin (2 * w + p) : ℂ) *
      (Complex.exp (-I * (w : ℂ)) * (Real.sin (w - f + p) : ℂ) +
       Complex.exp (I * ((w + p : ℝ) : ℂ)) * (Real.sin (w + f) : ℂ)) := by
  have hsplit : I * ((w + p : ℝ) : ℂ) = I * (w : ℂ) + I * (p : ℂ) := by
    push_cast; ring
  rw [hsplit, Complex.exp_add]
  push_cast
  ring

/-! ### The theorem -/

/-- **R = R₁ps + R₂ps.**  The only hypotheses are that the index is positive,
that ‘χ‘ does not vanish at the point in question, and that the denominator
‘sin(2ω+ψ)‘ is nonzero (without which ‘d₁, d₂‘ are undefined). In particular no
assumption is made about ‘T‘ being an integer or not. -/
theorem R_1ps_plus_R_2ps_eq_R (σ T : ℝ) (R : ℂ) (χ : ℂ → ℂ) (If : ℝ → ℝ)
    (hT : 0 < T)
    (hχ : χ (σ + If T * I) ≠ 0)
    (h_sin : Real.sin (2 * omg T If + psi χ σ T If) ≠ 0) :
    R_1ps σ T R χ If + R_2ps σ T R χ If = R := by
  have hsinC : (Real.sin (2 * omg T If + psi χ σ T If) : ℂ) ≠ 0 := by
    exact_mod_cast h_sin
  -- e^{iψ} for the normalized functional-equation factor
  have hchi : χ (σ + If T * I) / (‖χ (σ + If T * I)‖ : ℂ)
      = Complex.exp (I * (psi χ σ T If : ℂ)) := div_norm_eq_exp_arg _ hχ
  -- the two cpow factors become e^{∓iω}
  have hpow : (((⌈T⌉ : ℝ)) : ℂ) ^ (I * (If T : ℂ))
      = Complex.exp (I * (omg T If : ℂ)) := by
    rw [ceil_cpow T (If T) hT]; rfl
  have hpow' : (((⌈T⌉ : ℝ)) : ℂ) ^ (-(I * (If T : ℂ)))
      = Complex.exp (-(I * (omg T If : ℂ))) := by
    have h : -(I * (If T : ℂ)) = I * ((-(If T) : ℝ) : ℂ) := by push_cast; ring
    rw [h, ceil_cpow T (-(If T)) hT]
    congr 1
    simp only [omg]
    push_cast
    ring
  -- |R| e^{iφ} = R
  have hR : ((‖R‖ : ℝ) : ℂ) * Complex.exp (I * (phiR R : ℂ)) = R := by
    rw [mul_comm I ((phiR R : ℝ) : ℂ)]
    exact Complex.norm_mul_exp_arg_mul_I R
  unfold R_1ps R_2ps d_j1 d_j2
  rw [hchi, hpow, hpow', factoring_lemma, key_algebraic_identity]
  -- cancel sin(2ω+ψ)
  field_simp
  linear_combination hR
```

B Lean formalization of the critical line: $d_1 = d_2$, $|R_{1ps}| = |R_{2ps}|$, and equal legs

This appendix reproduces the Lean file establishing the three critical-line statements of the paper: Corollary 8.1 ($d_1 = d_2$ and $|R_{1ps}| = |R_{2ps}|$), Corollary 9.1 ($L_1 = L_2$), and Corollary 9.2 (the bisector point projects onto $\frac{\zeta}{2}$). As in Appendix A the file contains no `axiom` and no `sorry`, and each result depends only on Lean's own three axioms (`propext`, `Classical.choice`, `Quot.sound`) as reported by `#print axioms`; it was checked against Lean 4.32.2 with Mathlib at commit `905b95818e`. The Lean code was written with the assistance of AI coding assistants (Anthropic's Claude).

B.1 The chain of implications

All three statements descend from a single fact, and the file is organized around it.

Step 0: the two partial sums are conjugate. At $\sigma = \frac{1}{2}$ one has $\overline{-s} = s - 1$, so $n^{s-1} = \overline{n^{-s}}$ for every n; summing over $n \leq m$ gives $\Sigma_2 = \chi\,\overline{\Sigma_1}$. This is `partial_sum_reflect`, proved from Mathlib's `Complex.cpow_conj`.

Step 1: the rotated remainder is real. Write $\psi = \arg\chi$ and let $u = e^{i\psi/2}$, a square root of χ. With $R = \zeta - \Sigma_1 - \Sigma_2$,

$$e^{-i\psi/2}R \;=\; \underbrace{e^{-i\psi/2}\zeta}_{\text{real}} \;-\; \underbrace{\left(e^{-i\psi/2}\Sigma_1 + e^{i\psi/2}\overline{\Sigma_1}\right)}_{=\,2\Re\left(e^{-i\psi/2}\Sigma_1\right)} \;\in\; \mathbb{R}, \tag{163}$$

the first term being real because the functional equation at $\sigma = \frac{1}{2}$ reads $\zeta = \chi\overline{\zeta}$, so that $e^{-i\psi/2}\zeta = \overline{e^{-i\psi/2}\zeta}$. In the file this is `rotated_remainder_self_conj` (the algebra, from $u\bar{u} = 1$ and $u^2 = \chi$) and `rotated_remainder_im_eq_zero` (its imaginary part vanishes). It is the fact quoted as $\arg R = \frac{1}{2}\arg\chi$ in §8.1. Stated as the reality of $e^{-i\psi/2}R$ it is exact; stated through the argument it holds modulo π, which is visible in $R = 2d_1\cos(\omega + \frac{\psi}{2})e^{i\psi/2}$, whose argument is $\frac{\psi}{2} + \pi$ wherever the cosine is negative. The modulo-π version is all the sequel uses.

Step 2: $d_1 = d_2$. Since $\sin A - \sin B = 2\sin\frac{A-B}{2}\cos\frac{A+B}{2}$, the two numerators of (19)–(20) differ by

$$\sin(\omega - \phi + \psi) - \sin(\omega + \phi) \;=\; 2\,\sin\!\Big(\frac{\psi}{2} - \phi\Big)\cos\!\Big(\omega + \frac{\psi}{2}\Big), \tag{164}$$

and Step 1 kills the first factor: `sin_half_sub_arg_eq_zero` turns the reality of $e^{-i\psi/2}R$ into $\sin(\frac{\psi}{2} - \arg R) = 0$, using $\cos\arg R = \Re R/|R|$ and $\sin\arg R = \Im R/|R|$. Hence `d1_eq_d2_of_im_eq_zero` and its critical-line specialization `d1_eq_d2`. Note that ω is an arbitrary real throughout: the equality of the weights depends on the direction of R relative to χ and not at all on the turn angle.

Step 3: $|R_{1ps}| = |R_{2ps}|$. Both phases are unimodular, so this is $|d_1| = |d_2|$: `norm_R1ps_eq_norm_R2ps`. The sharper form (66), $R_{2ps} = \chi\,\overline{R_{1ps}}$, is `R2ps_eq_chi_conj_R1ps`, and it is what feeds Step 4; it needs only that d_1 is real and $d_1 = d_2$.

Step 4: the legs. Combining Steps 0 and 3, $B_2 = \Sigma_2 + R_{2ps} = \chi\,\overline{\Sigma_1 + R_{1ps}} = \chi\overline{B_1}$ (`leg2_eq_chi_conj_leg1`), and since $|\chi| = 1$ on the line the two legs have the same length (`legs_norm_eq`). Finally, from $\zeta = B_1 + \chi\overline{B_1}$,

$$e^{-i\psi/2}\zeta = e^{-i\psi/2}B_1 + \overline{e^{-i\psi/2}B_1} = 2\,\Re\big(e^{-i\psi/2}B_1\big), \tag{165}$$

which says that the perpendicular projection of the bisector point onto the line $\mathbb{R}\,e^{i\psi/2}$ carrying ζ lands exactly on $\frac{\zeta}{2}$: the apex of the isosceles triangle sits over the midpoint of its base. This is `proj_eq_half_zeta`, and it is the dotted bisector line drawn in Figures 13 and 15.

B.2 What is proved and what is hypothesized

Two analytic facts about χ on the critical line enter as hypotheses: $|\chi| = 1$, used in the form $\chi = e^{i \arg \chi}$ (eq_exp_arg_of_norm_one), and the functional equation $\zeta = \chi\bar{\zeta}$, which is $\zeta(s) = \chi(s)\zeta(1-s)$ together with $1 - s = \bar{s}$ at $\sigma = \frac{1}{2}$. Everything else above is derived, including the reflection $\Sigma_2 = \chi\overline{\Sigma_1}$ of Step 0, which is proved rather than assumed. The projection statement additionally takes $\zeta = B_1 + B_2$, which is Theorem 4.1 of Appendix A.

The remainders are written here in the form $R_{1ps} = d_1 e^{-i\omega}$, $R_{2ps} = d_2 e^{i(\omega+\psi)}$ of (24) rather than the $\lceil T \rceil^{\mp i I(T)}$ form of (17)–(18); the bridge between them is ceil_cpow of Appendix A. As there, ζ, χ, Σ_1 and ω are free parameters, so what is checked is the geometry of the critical line and not the identity of the functions involved.

B.3 Source listing

```
/-
  Critical-line consequences of the decomposition R = R1ps + R2ps:
    (1) e^{-iψ/2} R is real, i.e. arg R = ψ/2 modulo π,
    (2) d₁ = d₂,
    (3) |R1ps| = |R2ps|, indeed R2ps = χ · conj R1ps,
    (4) the two legs have equal length, and the bisector point projects onto ζ/2.

  Contains no ‘axiom‘ and no ‘sorry‘: every step is proved.
-/
import Mathlib.Analysis.SpecialFunctions.Complex.Circle
import Mathlib.Analysis.SpecialFunctions.Pow.Complex

open Complex Real ComplexConjugate

/-! ### Unit phases -/

/-- A purely imaginary exponent gives a unit phase. -/
lemma norm_exp_imag (x : ℝ) : ‖Complex.exp (I * (x : ℂ))‖ = 1 := by
  rw [Complex.norm_exp]
  simp

lemma norm_exp_neg_imag (x : ℝ) : ‖Complex.exp (-(I * (x : ℂ)))‖ = 1 := by
  rw [Complex.norm_exp]
  simp

/-- Conjugation flips the sign of a purely imaginary exponent. -/
lemma conj_exp_imag (x : ℝ) :
    conj (Complex.exp (I * (x : ℂ))) = Complex.exp (-(I * (x : ℂ))) := by
  rw [← Complex.exp_conj]
  congr 1
  simp [Complex.ext_iff]

lemma conj_exp_neg_imag (x : ℝ) :
    conj (Complex.exp (-(I * (x : ℂ)))) = Complex.exp (I * (x : ℂ)) := by
  rw [← Complex.exp_conj]
  congr 1
  simp [Complex.ext_iff]

/-- A number of unit modulus is the exponential of its argument: this is the
form in which ‘|χ| = 1‘ on the critical line will be used. -/
lemma eq_exp_arg_of_norm_one {χ : ℂ} (hχ : ‖χ‖ = 1) :
    χ = Complex.exp (I * ((Complex.arg χ : ℝ) : ℂ)) := by
```

```
  have h := Complex.norm_mul_exp_arg_mul_I χ
  rw [hχ] at h
  rw [mul_comm I _]
  simpa using h.symm

/-! ### Step 0: the second partial sum is the reflection of the first

At `σ = 1/2` we have `n^{s-1} = conj (n^{-s})` for every `n`, because
`conj(-s) = s - 1` there. Summing, `Σ₂ = χ · conj Σ₁`, which is the hypothesis
the results below are stated with. -/

lemma conj_natCast_cpow (n : ℕ) (z : ℂ) :
    conj ((n : ℂ) ^ z) = (n : ℂ) ^ (conj z) := by
  have harg : ((n : ℂ)).arg ≠ π := by
    rw [show ((n : ℂ)) = (((n : ℝ)) : ℂ) by push_cast; ring,
      Complex.arg_ofReal_of_nonneg (by positivity)]
    exact Ne.symm Real.pi_ne_zero
  rw [Complex.cpow_conj _ _ harg, Complex.conj_natCast]

/-- On the critical line the two Dirichlet partial sums are conjugate. -/
theorem partial_sum_reflect (m : ℕ) (t : ℝ) :
    ∑ n ∈ Finset.Icc 1 m, ((n : ℂ)) ^ (((1 / 2 : ℂ) + (t : ℂ) * I) - 1)
      = conj (∑ n ∈ Finset.Icc 1 m, ((n : ℂ)) ^ (-((1 / 2 : ℂ) + (t : ℂ) * I))) := by
  rw [map_sum]
  refine Finset.sum_congr rfl fun n _ => ?_
  rw [conj_natCast_cpow]
  congr 1
  simp [Complex.ext_iff]
  norm_num

/-! ### Step 1: the rotated remainder is real

On the critical line the functional equation reads `ζ = χ · conj ζ`, and the
second partial sum is the reflection of the first, `S₂ = χ · conj S₁`. With
`u = e^{iψ/2}` a square root of `χ`, the remainder `R = ζ - S₁ - S₂` satisfies
`conj (conj u * R) = conj u * R`, that is, `e^{-iψ/2} R` is real. -/

lemma rotated_remainder_self_conj (ζ S₁ χ u : ℂ)
    (hu : u * conj u = 1) (hu2 : u ^ 2 = χ)
    (hζ : ζ = χ * conj ζ) :
    conj (conj u * (ζ - S₁ - χ * conj S₁)) = conj u * (ζ - S₁ - χ * conj S₁) := by
  simp only [map_sub, map_mul, Complex.conj_conj]
  rw [← hu2] at hζ ⊢
  simp only [map_pow]
  linear_combination (-conj u) * hζ
    + (-u * conj ζ + u * conj S₁ - conj u * S₁) * hu

lemma rotated_remainder_im_eq_zero (ζ S₁ χ : ℂ) (p : ℝ)
    (hp : χ = Complex.exp (I * (p : ℂ)))
    (hζ : ζ = χ * conj ζ) :
    (Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) * (ζ - S₁ - χ * conj S₁)).im = 0 := by
  have hu : Complex.exp (I * ((p / 2 : ℝ) : ℂ)) * conj (Complex.exp (I * ((p / 2 : ℝ) : ℂ))) = 1
    := by
    rw [conj_exp_imag, ← Complex.exp_add]
    simp
  have hu2 : Complex.exp (I * ((p / 2 : ℝ) : ℂ)) ^ 2 = χ := by
    rw [hp, pow_two, ← Complex.exp_add]
    congr 1
    push_cast
```

```lean
    ring
  have hc : conj (Complex.exp (I * ((p / 2 : ℝ) : ℂ)))
      = Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) := conj_exp_imag _
  rw [← hc]
  exact Complex.conj_eq_iff_im.mp
    (rotated_remainder_self_conj ζ S₁ χ _ hu hu2 hζ)

/-! ### Step 2: reality of `e^{-ip/2} R` says `arg R = p/2` modulo `π` -/

lemma sin_half_sub_arg_eq_zero (R : ℂ) (p : ℝ) (hR : R ≠ 0)
    (him : (Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) * R).im = 0) :
    Real.sin (p / 2 - Complex.arg R) = 0 := by
  have hexp : Complex.exp (-(I * ((p / 2 : ℝ) : ℂ)))
      = ((Real.cos (p / 2) : ℝ) : ℂ) - ((Real.sin (p / 2) : ℝ) : ℂ) * I := by
    have h : -(I * ((p / 2 : ℝ) : ℂ)) = ((-(p / 2) : ℝ) : ℂ) * I := by
      push_cast; ring
    rw [h, Complex.exp_mul_I, ← Complex.ofReal_cos, ← Complex.ofReal_sin,
      Real.cos_neg, Real.sin_neg]
    push_cast
    ring
  have hnorm : ‖R‖ ≠ 0 := norm_ne_zero_iff.mpr hR
  rw [hexp] at him
  simp only [Complex.mul_im, Complex.sub_re, Complex.sub_im, Complex.mul_re,
    Complex.ofReal_re, Complex.ofReal_im, Complex.I_re, Complex.I_im] at him
  rw [Real.sin_sub, Complex.cos_arg hR, Complex.sin_arg]
  field_simp
  linear_combination -him

/-! ### The weights and the two fractional remainders -/

/-- `d₁ = |R| sin(ω - arg R + ψ)/sin(2ω + ψ)`. -/
noncomputable def d1 (R : ℂ) (w p : ℝ) : ℝ :=
  ‖R‖ * (Real.sin (w - Complex.arg R + p) / Real.sin (2 * w + p))

/-- `d₂ = |R| sin(ω + arg R)/sin(2ω + ψ)`. -/
noncomputable def d2 (R : ℂ) (w p : ℝ) : ℝ :=
  ‖R‖ * (Real.sin (w + Complex.arg R) / Real.sin (2 * w + p))

/-- `R1ps = d₁ e^{-iω}`. -/
noncomputable def R_1ps (R : ℂ) (w p : ℝ) : ℂ :=
  ((d1 R w p : ℝ) : ℂ) * Complex.exp (-(I * ((w : ℝ) : ℂ)))

/-- `R2ps = d₂ e^{i(ω+ψ)}`. -/
noncomputable def R_2ps (R : ℂ) (w p : ℝ) : ℂ :=
  ((d2 R w p : ℝ) : ℂ) * Complex.exp (I * ((w + p : ℝ) : ℂ))

/-! ### Result 1: the two weights are equal

Note that `ω` is arbitrary here: the equality of the weights depends only on the
direction of `R` relative to `χ`, not on the turn angle. -/

theorem d1_eq_d2_of_im_eq_zero (R : ℂ) (w p : ℝ)
    (him : (Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) * R).im = 0) :
    d1 R w p = d2 R w p := by
  rcases eq_or_ne R 0 with h | h
  · simp [d1, d2, h]
  · have hs : Real.sin (p / 2 - Complex.arg R) = 0 :=
      sin_half_sub_arg_eq_zero R p h him
    have hnum : Real.sin (w - Complex.arg R + p) = Real.sin (w + Complex.arg R) := by
```

```lean
      have hd := Real.sin_sub_sin (w - Complex.arg R + p) (w + Complex.arg R)
      have harg : (w - Complex.arg R + p - (w + Complex.arg R)) / 2
          = p / 2 - Complex.arg R := by ring
      rw [harg, hs, mul_zero, zero_mul] at hd
      linarith
    rw [d1, d2, hnum]

/-- **d₁ = d₂ on the critical line.** -/
theorem d1_eq_d2 (ζ S₁ χ : ℂ) (w p : ℝ)
    (hp : χ = Complex.exp (I * (p : ℂ)))
    (hζ : ζ = χ * conj ζ) :
    d1 (ζ - S₁ - χ * conj S₁) w p = d2 (ζ - S₁ - χ * conj S₁) w p :=
  d1_eq_d2_of_im_eq_zero _ w p (rotated_remainder_im_eq_zero ζ S₁ χ p hp hζ)

/-! ### Result 2: the two fractional remainders have equal length -/

/-- **|R1ps| = |R2ps|.** -/
theorem norm_R1ps_eq_norm_R2ps (R : ℂ) (w p : ℝ) (hd : d1 R w p = d2 R w p) :
    ‖R_1ps R w p‖ = ‖R_2ps R w p‖ := by
  rw [R_1ps, R_2ps, norm_mul, norm_mul, norm_exp_neg_imag, norm_exp_imag, hd]

/-- The sharper statement: `R2ps` is the mirror image of `R1ps` across the real
axis, followed by the rotation `χ`. -/
theorem R2ps_eq_chi_conj_R1ps (R χ : ℂ) (w p : ℝ)
    (hp : χ = Complex.exp (I * (p : ℂ)))
    (hd : d1 R w p = d2 R w p) :
    R_2ps R w p = χ * conj (R_1ps R w p) := by
  have hexp : Complex.exp (I * ((w + p : ℝ) : ℂ))
      = Complex.exp (I * (p : ℂ)) * Complex.exp (I * ((w : ℝ) : ℂ)) := by
    rw [← Complex.exp_add]
    congr 1
    push_cast
    ring
  rw [R_1ps, R_2ps, map_mul, Complex.conj_ofReal, conj_exp_neg_imag, hd, hexp, hp]
  ring

/-! ### Result 3: the two legs have equal length -/

/-- With `B₁ = S₁ + R1ps` and `B₂ = S₂ + R2ps = χ conj S₁ + R2ps`, the second leg
is the reflection of the first. -/
theorem leg2_eq_chi_conj_leg1 (S₁ R χ : ℂ) (w p : ℝ)
    (hp : χ = Complex.exp (I * (p : ℂ)))
    (hd : d1 R w p = d2 R w p) :
    χ * conj S₁ + R_2ps R w p = χ * conj (S₁ + R_1ps R w p) := by
  rw [map_add, mul_add, R2ps_eq_chi_conj_R1ps R χ w p hp hd]

/-- **L₁ = L₂.** -/
theorem legs_norm_eq (S₁ R χ : ℂ) (w p : ℝ) (hχ : ‖χ‖ = 1)
    (hp : χ = Complex.exp (I * (p : ℂ)))
    (hd : d1 R w p = d2 R w p) :
    ‖χ * conj S₁ + R_2ps R w p‖ = ‖S₁ + R_1ps R w p‖ := by
  rw [leg2_eq_chi_conj_leg1 S₁ R χ w p hp hd, norm_mul, hχ, one_mul,
    RCLike.norm_conj]

/-! ### Result 4: the bisector point projects onto ζ/2 -/

/-- If `B₂ = χ conj B₁` and `ζ = B₁ + B₂`, then the orthogonal projection of the
bisector point `B₁` onto the line `ℝ·e^{iψ/2}` through `ζ` is exactly `ζ/2`. The
apex of the isosceles triangle sits over the midpoint of its base. -/
```

```
theorem proj_eq_half_zeta (B₁ ζ χ : ℂ) (p : ℝ)
    (hp : χ = Complex.exp (I * (p : ℂ)))
    (hzeta : ζ = B₁ + χ * conj B₁) :
    (((Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) * B₁).re : ℝ) : ℂ) *
        Complex.exp (I * ((p / 2 : ℝ) : ℂ))
      = ζ / 2 := by
  set em := Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) with hem
  set ep := Complex.exp (I * ((p / 2 : ℝ) : ℂ)) with hep
  have hu : ep * em = 1 := by
    rw [hep, hem, ← Complex.exp_add]
    simp
  have hu2 : ep ^ 2 = χ := by
    rw [hep, hp, pow_two, ← Complex.exp_add]
    congr 1
    push_cast
    ring
  have hconj : conj em = ep := by rw [hem, hep, conj_exp_neg_imag]
  set z := em * B₁ with hz
  have key : ((z.re : ℝ) : ℂ) * 2 = z + conj z := by
    rw [Complex.add_conj]
    push_cast
    ring
  have hcz : conj z = ep * conj B₁ := by rw [hz, map_mul, hconj]
  rw [eq_div_iff (two_ne_zero : (2 : ℂ) ≠ 0), hzeta]
  linear_combination ep * key + ep * hz + B₁ * hu + ep * hcz + conj B₁ * hu2
```

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